Shape bias is a readout property: a developmental two-stream model, frozen networks and human cortex

Manuscript draft, compiled 2026-10-26 13:54 from `docs/PAPER_v3.md` at commit `a91d9de`.

Draft v3, 2026-10-19. Model of record: `fs2_dvd300_avg_w050_ls` (seeds 0–2), replacing the BA network `dkl_bcs` of PAPER_v2, which is kept as the comparison. Plan and claim-by-claim evidence: `docs/PAPER_v3_PLAN_dvd300.md`. Control seeds 1–2 and the channel swap (3 seeds) finished training; their behavioural (2026-10-23) and cortical (2026-10-24) numbers are filled. The nuisance-axis controls of Extended Data Fig. 3 and the estimator-bias simulation were rerun on the model (2026-10-24; Supplementary Note S3). Still marked ⟦PENDING⟧: the Supplementary rows for the model. Before submission: push branch `fusion-stage1`, which the code statement names and which is not yet on the public remote, and fill Author contributions. `tmp_analysis/verify_v3_claims.py` re-derives every number stated for the model and its controls (main text, Table 1, Extended Data Tables 1–3) and runs `tmp_analysis/verify_new_claims.py` as the comparison block for the BA network and the zoo. The abstract and the Discussion were re-led on the carrier × ramp conjunction and the per-image result, and the per-stream accuracies added to Results 2.2 (2026-10-24). Figures are regenerated on the model in `results/figures_v3/` (Fig. 5d shows the BA network; control seeds 0–2 and the channel swap are drawn, 2026-10-24; Fig. 6 draws the model in its two-stream row and as panel c's reference, 2026-10-24). The main text was cut to NMI length (2026-10-24); the paragraphs condensed are given in full in Supplementary Note S6. Supplementary Note S4 (which now says so) and the Supplementary Tables still describe the BA network; Extended Data Tables 1 and 2 are rewritten here by hand.

Abstract

Humans name objects by shape when shape and texture conflict, yet the ventral stream is well described by texture statistics. We model the subcortical split as two ResNet-18 trunks fed luminance and L−M under one objective. When the luminance input matures from infant to adult during training, its stream learns a shape bias; given the adult input from the start, or the same ramp on L−M, both streams lean to texture. Only this conjunction divides the cortical hierarchy of eight Natural Scenes Dataset subjects: the shape-biased stream carries V1, the stronger, texture-leaning stream extrastriate cortex. Only it tracks which images humans resolve (22% of the ceiling, against 8–9% for both controls on identical input). One readout weight moves its shape-bias rate from 0.48 to 0.85; in 38 frozen ImageNet networks, a gain on each network's shape and texture subspaces moves it by 0.08–0.29. The cue-conflict rate is a readout statistic.

1. Introduction

Humans presented with an image whose shape and texture point to different categories name the shape: on the cue-conflict images of Geirhos et al. (2018a), ten observers did so on 96% of decided trials, while ImageNet-trained convolutional networks named the texture. Closing that gap is a standard target for biologically motivated models (Hermann et al., 2020; Jang & Tong, 2021); training on degraded early input, as infancy supplies it (Vogelsang et al., 2018, 2024), has

been studied as a route to more human-like recognition (Avberšek et al., 2021; Jinsi et al., 2023; Yoshihara et al., 2023; Jang & Tong, 2024), and a developmental visual diet — input whose acuity, contrast and chromatic sensitivity follow the human developmental trajectory — is among the most effective routes to it (Lu et al., 2026).

A second body of evidence sits uneasily beside the first: ventral-stream representations are well described by texture statistics — category-selective human cortex fails to discriminate natural images from texture-matched scrambles yet still discriminates categories (Jagadeesh & Gardner, 2022), macaque V4 responses to natural textures are predicted by their image statistics (Okazawa et al., 2015), and in inferotemporal cortex texture is the more decodable cue (Jagadeesh & Livingstone, 2024; Komatsu & Goda, 2018). The argument needs only that the ventral description is substantially texture-like, not that shape is absent (the evidence is one human study and several macaque ones).

If behaviour is shape-biased and the ventral stream texture-biased, either the two measures disagree about what shape bias is, or behaviour is not a readout of one representation. The second horn is a hypothesis about architecture that single-trunk models cannot express, and two facts make it concrete: the visual system receives a luminance-dominated, coarse magnocellular signal and a chromatically opponent, finer parvocellular one, separated before cortex and later recombined (Livingstone & Hubel, 1987; Kaplan & Shapley, 1982; Derrington et al., 1984; Nassi & Callaway, 2009; Ferrera et al., 1994), and that input is not stationary over development. The two-stream hypothesis (Mishkin et al., 1983; Goodale & Milner, 1992) and the emergence of ventral- and dorsal-like streams in a two-pathway network trained on one objective (Bakhtiari et al., 2021) motivate an architecture whose streams are defined by their subcortical-like inputs.

Two-stream networks reproduce the dorsal–ventral division (Choi et al., 2023; Finzi et al., 2026; see also Dobs et al., 2022). Closest to our question, blurry, achromatic training before sharp, colour training made one network's classification more shape-biased, attributed to magnocellular-like units (Vogelsang et al., 2025); here the split is set by two streams' inputs, the input matures continuously to adulthood, and a control on identical test input, a channel swap and cortical encoding test the account.

This paper asks two questions, in the order the originating study posed them (Manocchio, 2026): can two streams fed magnocellular- and parvocellular-like inputs diverge while optimising a single objective and producing shape-biased behaviour, and does such a system explain how the ventral stream can be texture-like while behaviour is shape-biased? We build it — two ResNet-18 trunks fed the luminance and L−M channels of a cone-opponent image, the luminance channel under a developmental diet that matures from birth to adulthood over training (Fig. 1) — and examine its streams with probes and cue-conflict behaviour (Fig. 2), encoding models of eight subjects' 7T fMRI responses (Fig. 3) and the readout that combines them (Fig. 4). The streams diverge by different routes, and the shape-biased one only when its input is achromatic and matured during training: the same network shown the adult input from the first step, or given the ramp on its L−M channel instead, learns two texture-leaning streams. Only that conjunction divides the cortical hierarchy — the shape-biased stream carrying V1, the stronger, texture-leaning stream extrastriate cortex — and tracks which images human observers resolve. The integrated behaviour is shape-biased because a behavioural rate is a property of a readout over components, not a report of any one of them, in kind even in networks with no stream split (Fig. 6); robustness, error consistency and dimensionality come apart from the rate (Fig. 5).

2. Results

2.1 The model, and what the developmental trajectory costs

The network splits its input once, before any convolution (Fig. 1a): each image is converted to the DKL cone-opponent space, stream 1 receives the luminance channel and stream 2 the L−M channel, and each is a ResNet-18 trunk. Under the developmental diet the luminance channel passes through the acuity blur and contrast-sensitivity cut of Lu et al. (2026) at one age that rises linearly with the training step from birth to 300 months; every evaluation uses the endpoint, at which the magno stream sees nearly clean luminance (Methods 4.1). The trunks are initialised bit-identically, and a linear classifier reads out $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with $w = 0.5$ during training, under one cross-entropy on that output and no loss on either stream.

The decisive control holds the diet at its 300-month endpoint from the first step: the same architecture, recipe, seeds and test input, without the developmental trajectory. The originating study's network ("BA network") held a fixed mixture of ages instead (a $\sigma = 4$ px blur and a 48-month-old's contrast sensitivity) and is reported beside the model, with its no-diet arm; trained with our recipe, the same fixed diet separates the recipe from the trajectory. Networks were trained on MiniEcoset (112 classes; Thorat & Kietzmann, 2024) for 50 epochs, three seeds per condition (Methods 4.2).

The developmental trajectory costs accuracy, not stability (Fig. 1c): the model reaches 0.683 ± 0.002 validation top-1 at its final checkpoint, within 0.9 points of its best epoch in every seed, against 0.764 ± 0.003 for the control on the identical test input. It adds 1.8 points over the fixed BA diet under the same recipe (0.665 ± 0.003) and 0.067 of shape bias (training-time evaluation), and gives up 8.1 points against the network that learned the same input without the trajectory. A trajectory stopping at 48 months (one seed) is no better than the fixed diet (0.664 top-1, shape bias 0.637): the input has to reach adult acuity, not only start coarse. Every result for the model and its control below is from final checkpoints (Methods 4.7).

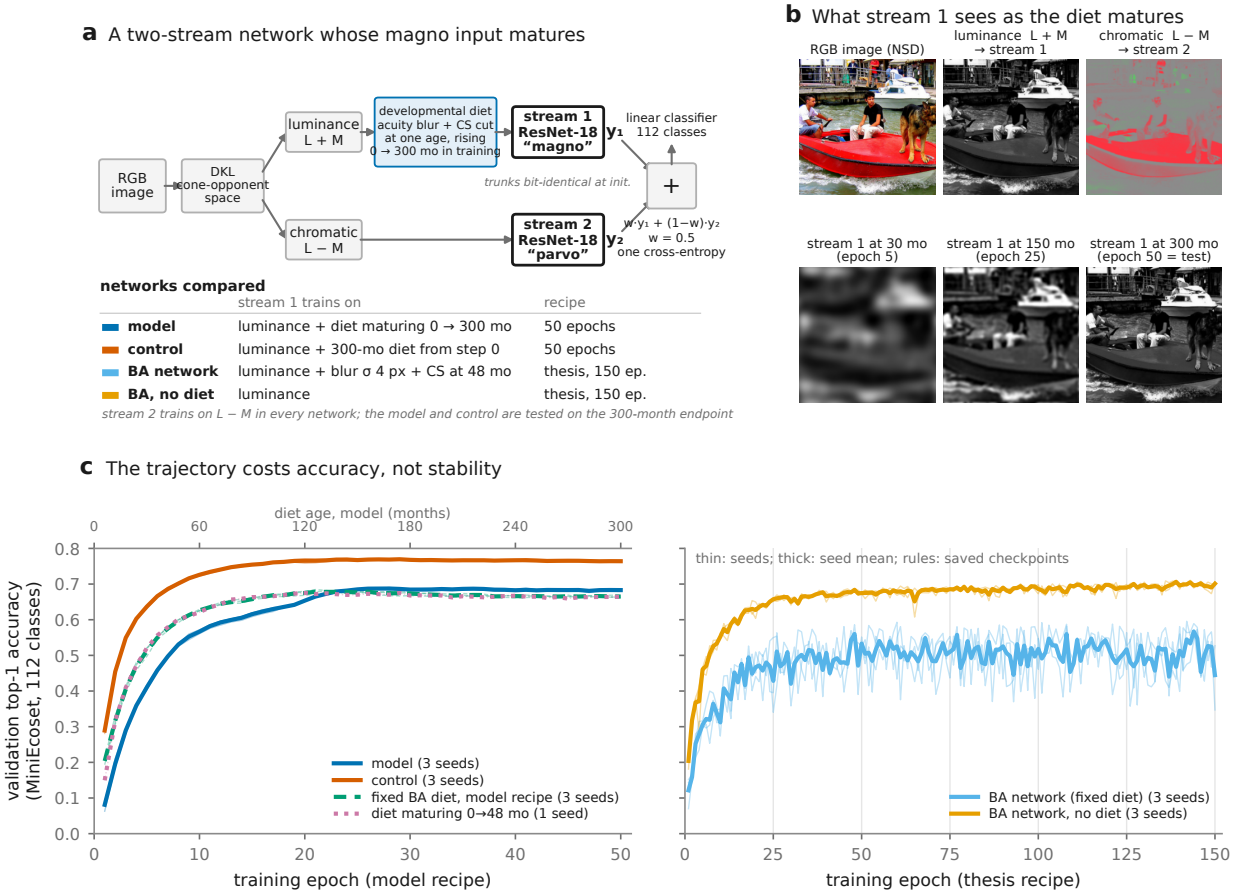


Figure 1 | A two-stream network fed magnocellular- and parvocellular-like inputs, with a developmental diet on the luminance channel that matures over training.

(**a**) The model. Each RGB image is converted to the DKL cone-opponent space and split once, before any convolution: stream 1 ("magno") receives the luminance channel, stream 2 ("parvo") the L−M chromatic channel. During training the luminance channel passes through the acuity blur and the contrast-sensitivity cut of the developmental visual diet of Lu et al. (2026), both evaluated at one age that rises linearly with the training step from birth to 300 months (Methods 4.1); every evaluation uses the 300-month endpoint. The two ResNet-18 trunks are initialised bit-identically, and a linear classifier reads out $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with $w = 0.5$ during training, under one cross-entropy and no loss on either stream. The table lists the networks compared: the model, its control (the 300-month diet from the first step: the same architecture, recipe, seeds and test input, without the trajectory), and the originating study's BA network (a fixed $\sigma = 4$ px blur and a 48-month contrast cut, thesis recipe) with its no-diet arm.

(**b**) What each stream sees, for one NSD image (id 15504, the most chromatic of a fixed candidate grid), rendered with the training front end. Top: the RGB image and its luminance and L−M channels (L−M contrast-stretched for display only). Bottom: stream 1's input at three ages of the trajectory, 30, 150 and 300 months (the ends of epochs 5, 25 and 50); the contrast cut leaves almost no fine detail until the second half of training, and the 300-month endpoint is nearly clean luminance.

(**c**) Validation top-1 accuracy (MiniEcoset, 112 classes) over training. Left, the model's recipe (50 epochs): the model reaches 0.683 ± 0.002 at its final checkpoint, within 0.9 points of its best epoch in every seed, against 0.764 ± 0.003 for the control; the BA diet under the same recipe reaches 0.665 ± 0.003, and a trajectory stopping at 48 months (one seed) 0.664. The top axis gives the model's diet age. Right, the BA network under the thesis's recipe (150 epochs): 0.556 at its best saved checkpoint, after which it degrades (final 0.445), against 0.70 without the diet. Thin lines: seeds; thick: seed mean; vertical rules: saved checkpoints.

Sources: run histories (`tmp_analysis/figs_v2/v3runs.py`); `tmp_analysis/figs_v2/fig1_model.py`.

2.2 The two streams specialise in opposite directions, by different routes

Read out separately with a linear probe on the cue-conflict images, the two streams end on opposite sides of the shape–texture boundary: on exemplars the probe was not trained on, shape − texture accuracy is +0.316 for the magno stream and −0.156 for the parvo stream, every seed on its side (Fig. 2a; Methods 4.3), as in the BA network (+0.337 and −0.168) at 0.683 top-1 against its 0.445. Read out alone, the magno stream's cue-conflict shape bias is 0.850 and the parvo stream's 0.477, against 0.682 for the integrated readout (Table 1).

Table 1. The two streams end on opposite sides of the boundary, by learning, and only with a maturing input. Final checkpoints (Methods 4.7). Probe values: shape − texture accuracy on held-out exemplars (Methods 4.3) / with exemplars shared. Untrained: seed-0 initialisation, which all trained networks share, on the model's test input. Control: the 300-month diet from the first step, the identical test input. Channel swap: the developmental ramp applied to the L−M channel, luminance untransformed, the identical test input.

	magno stream (luminance)	parvo stream (L−M)	integrated
linear probe, shape − texture (0 = sign flip), 3 seeds: exemplars held out / shared	**+0.316** / +0.578	**−0.156** / −0.454	—
cue-conflict shape bias (0.5 = parity), 3 seeds	**0.850**	**0.477**	**0.682**
linear probe, untrained	+0.087 / +0.429	−0.033 / −0.087	—
contribution of learning on the maturing input (trained minus untrained)	**+0.229** / +0.149	**−0.123** / −0.367	—
control, adult input from the first step, 3 seeds: linear probe	**−0.177** / −0.291	−0.069 / −0.413	—
control: cue-conflict shape bias	0.384	0.426	0.374
channel swap, ramp on L−M, 3 seeds: linear probe	−0.159 / −0.276	−0.016 / −0.276	—
channel swap: cue-conflict shape bias	0.397	0.390	0.389
BA network (fixed diet, thesis recipe), 3 seeds: probe held out / shared; cue-conflict	+0.337 / +0.819; 0.878	−0.168 / −0.518; 0.418	0.675
BA network, untrained on its own front end: probe held out / shared	+0.210 / +0.796	−0.033 / −0.087	—

Both components are built by learning, and the input's trajectory decides the sign of one (Fig. 2a, Table 1). Untrained, the magno stream on the model's test input sits where undieted luminance does (+0.087 against +0.090 on held-out exemplars), and the parvo stream begins nearly unbiased (−0.033). Training on the maturing input adds +0.229 to the magno stream — nearly twice what training added in the BA network (+0.127), whose fixed blur put +0.210 into the input before any learning — and −0.123 to the parvo stream. Training on the same adult input from the first step

drives the magno stream across the sign flip to −0.177, below its own parvo stream (−0.069): the magno stream's shape bias is a property not of the input it is tested on but of the input it learned from. The bias is present from the first evaluation, after one epoch at ages up to 6 months, and is kept as the input sharpens; whether the coarse-to-fine direction matters, or only exposure to coarse input, these runs cannot say. Nor is the stream a texture filter: on held-out exemplars it decodes texture at 0.148 (chance 0.0625) and shape at 0.464.

Which property of the input carries the effect? Crossing the carrier (achromatic luminance or chromatic L−M) with the developmental ramp gives four cells: achromatic with the ramp (+0.316), achromatic with the adult input from the start (−0.177), chromatic without the ramp (−0.156 in the model, −0.069 in the control) and chromatic with the ramp, the channel swap, which applies the same ramp to L−M with luminance untransformed (three seeds; −0.016). The ramp moves the achromatic stream by +0.49 and the chromatic stream by +0.05, leaving it below parity; the carrier × ramp interaction is +0.44 on held-out exemplars, and no seed of the conjunction cell (+0.30 to +0.33) overlaps a seed of any other cell. Only the ramp on an achromatic carrier crosses zero, as only the BA network's fixed transform on an achromatic carrier did (Supplementary Table S3). The swap reproduces its control on every other measure as well (0.760 top-1, an integrated rate of 0.389): applied to the chromatic channel, the trajectory buys neither the shape bias nor the accuracy cost. Blur-first training likewise helped a network's categorisation only with greyscale input (Jinsi et al., 2023).

The trajectory also decides which trunk carries the task. Read alone through readouts trained on the trunks' detached features (no gradient reaches the trunks; training-time evaluation), the model's luminance stream reaches 0.39 validation top-1 and its L−M stream 0.63, while in the control and the swap the luminance stream carries the task (0.75, against 0.25 and 0.17 for L−M); early in training the ramp all but silences the luminance trunk (0.07–0.09 at epoch 7, against 0.68–0.69 in the control and the swap). In each of the three networks the weaker trunk is the more shape-leaning one, but weakness does not cross parity: the swap's chromatic trunk is the weakest of all and stays below it (−0.016).

Two qualifications belong here. First, cue-conflict shape bias is computed only over the images a readout decides (426 to 580 of 1,200 in the model, 209 to 673 in the control; seed means); the probe, which classifies every image, and a forced two-way estimator preserve every ordering this paper draws a conclusion from, but the parvo stream sits within 0.03 of parity on both estimators, so its side is read from the probe. Second, the model is the less accurate network (0.683 against 0.764), but at the epoch where the control's top-1 matches the model's final 0.683 (epoch 7 of 50 in each of the three seeds) its magno stream's shape bias is 0.41–0.43, and at no epoch of any seed does it reach parity (highest 0.495; training-time evaluation).

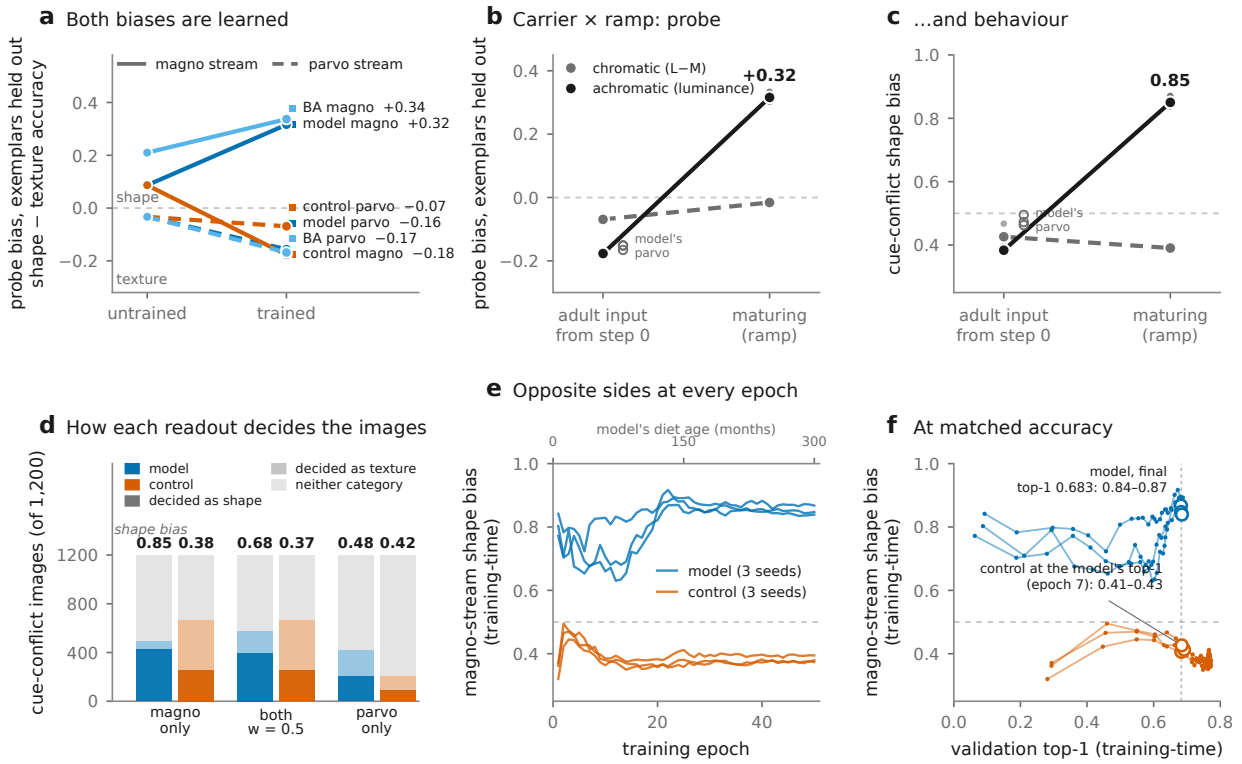


Figure 2 | The two streams specialise in opposite directions, both by learning, and the magno stream's shape bias only with a maturing input.

(**a**) Linear-probe shape − texture accuracy per stream on exemplars the probe was not trained on (exemplar-balanced, leave-one-exemplar-out; Methods 4.3), before and after training; solid: magno stream, dashed: parvo stream; bands: seed range. The model and its control start from one untrained network on one test input (magno +0.087, parvo −0.033). Training on the maturing input ends at +0.316 (magno) and −0.156 (parvo), means of three seeds; training on the adult input from the first step ends at −0.180 and −0.079 (control, seed 0). The BA network starts from its own blurred front end (+0.210, −0.033) and ends at +0.337 and −0.168.

(**b, c**) Carrier × ramp on the probe (b) and on cue-conflict shape bias (c). Achromatic carrier: the control's magno stream without the ramp, the model's with it (probe −0.180 → +0.316; behaviour 0.378 → 0.850). Chromatic carrier without the ramp: the control's parvo stream (probe −0.079; behaviour 0.388); open markers: the model's parvo stream, beside a ramped magno stream (−0.156; 0.477). The fourth cell, the ramp on L−M with luminance untransformed (`fs2_dvd300swap`), is training and is drawn when its results exist.

(**d**) How each readout decides the 1,200 cue-conflict images: decided as the shape category, as the texture category, or neither. Shape bias is computed over the decided images only, a count that varies across readouts (426 to 580 in the model, 209 to 673 in the control; seed means).

(**e**) Training-time evaluation (the shared head on the magno stream alone, three seeds each): the model's magno stream is shape-biased at every epoch evaluated (0.63–0.92) and the control's texture-biased (0.32–0.495); the top axis gives the model's diet age. (**f**) The same, against the network's validation top-1: at the model's mean final accuracy (0.683; open blue: its seeds), every control seed reaches it at epoch 7 of 50, with a magno-stream shape bias of 0.41–0.43 (open vermillion).

Sources: `results/figures_v2/exemplar_streams.json` (probe; `two_by_two_v3`), `results/cue_conflict/`, run histories; `tmp_analysis/figs_v2/fig2_streams.py`.

2.3 In cortex, the magno stream carries V1 and the texture-biased stream the rest of the hierarchy

The thesis's representational similarity analysis (Kriegeskorte et al., 2008) could not separate a general stream factor from regional structure, so we fitted voxel-wise banded-ridge encoding models for each of the eight NSD subjects, one band per stream, partitioning each voxel's cross-validated explained variance into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p) (Methods 4.5); Fig. 3a plots the per-voxel index (U_m − U_p)/(U_m + U_p), median over reliable voxels per ROI, across V1v → hV4.

The profile crosses zero, and only with the maturing input. In the model the index runs from +0.55 in V1v, magno-explained, through zero at V2v to −0.76 in hV4, parvo-explained (Fig. 3a; in V1, U_m 0.10–0.11 against U_p 0.03–0.04 in 8/8 subjects in every seed). The control, on the identical test input, is magno-explained everywhere and nearly flat (+0.81 in V1v, +0.73 in hV4), as are the channel swap (+0.85, +0.83) and the BA no-diet network (+0.91, +0.83); the BA network is parvo-explained from V1 on and descends without crossing (−0.00 to −0.84).

Quantified as the contrast hV4 − mean(V1v, V1d) of the per-ROI median (Fig. 3b; Extended Data Table 1), the gradient is −1.259 to −1.291 across the three seeds against −0.062 to −0.086 for the control's, each seed against the control seed of the same index at the exact Wilcoxon floor for eight subjects ($p = .0078$, 8/8), with the same separation on the raw statistic U_m − U_p (−0.186 to −0.193 against −0.003 to −0.007). The channel swap, the same ramp on the chromatic carrier, has no gradient either (−0.019 normalised, +0.002 raw, seed means), and the model's lies below it in 8/8 subjects ($p = .0078$): in cortex as in behaviour, only the ramp on the achromatic carrier departs from the control.

The trajectory does not create the gradient; it decides what training does to the one the input supplies. An untrained network on the same test input already descends (−0.480 normalised, −0.195 raw, 8/8 subjects, $p = .0078$ on both). Training on the maturing input steepens it on the normalised index (to −1.27) and holds it on the raw difference (−0.19), so under the rule that a cortical claim needs both statistics (Methods 4.7) we claim only that it preserves the gradient; training on the adult input from the first step erases it on both (−0.070; −0.004, seed means).

Pushed through models fitted on natural scenes only, the cue-conflict images give a predicted magno − parvo shape-bias difference of +1.32 to +1.34 in the model against +0.07 for the control and −0.01 for the channel swap (8/8 subjects; Fig. 3d).

The gradient continues past hV4 and does not turn. On NSD's full atlases (Extended Data Fig. 4), relative to V1, hV4 sits at −1.12 and every extrastriate group lower — ventral (VO, PHC) −1.25, lateral (LO, TO) −1.27, dorsal (V3A/B, IPS0–2) −1.22 — each in 8/8 subjects ($p = .0078$), and no contrast among the three extrastriate groups departs from zero; beyond V3 the magno band contributes about 0.013 unique variance against 0.149 for the parvo band. Nor is the gradient an eccentricity or receptive-field-size contrast in disguise: it is negative within each of NSD's five eccentricity bands (−0.58 to −1.33) and each pRF-size tertile (−1.26 to −1.32), below the control's in every one (Extended Data Fig. 3). Nor does the estimator manufacture it: with a true magno − parvo difference of zero everywhere the recovered contrast is small and of the wrong sign (+0.018, $p = .11$), and corrected by each arm's own floor the model's gradient is −1.277 against −0.127 for the control (Supplementary Note S3).

Three things this section does not settle. First, the trajectory buys the gradient at a small cost in BOLD prediction against its control (noise-ceiling-normalised median r^2 0.471 against 0.494, 8/8 subjects). Second, the gradient was not pre-specified: it is the survivor of seven regional hypotheses tested on the BA network, six of which failed, and is read as sign and ordering (Supplementary

Notes S1 and S3). Third, "texture-biased" names the stream, not the reason cortex prefers it: in every network the stream carrying extrastriate cortex is also the trunk that carries the task (Results 2.2), and in the model it is the chromatic one too, so the design separates neither task load nor the carrier from texture bias; what only the model has is a V1 carried by the other trunk.

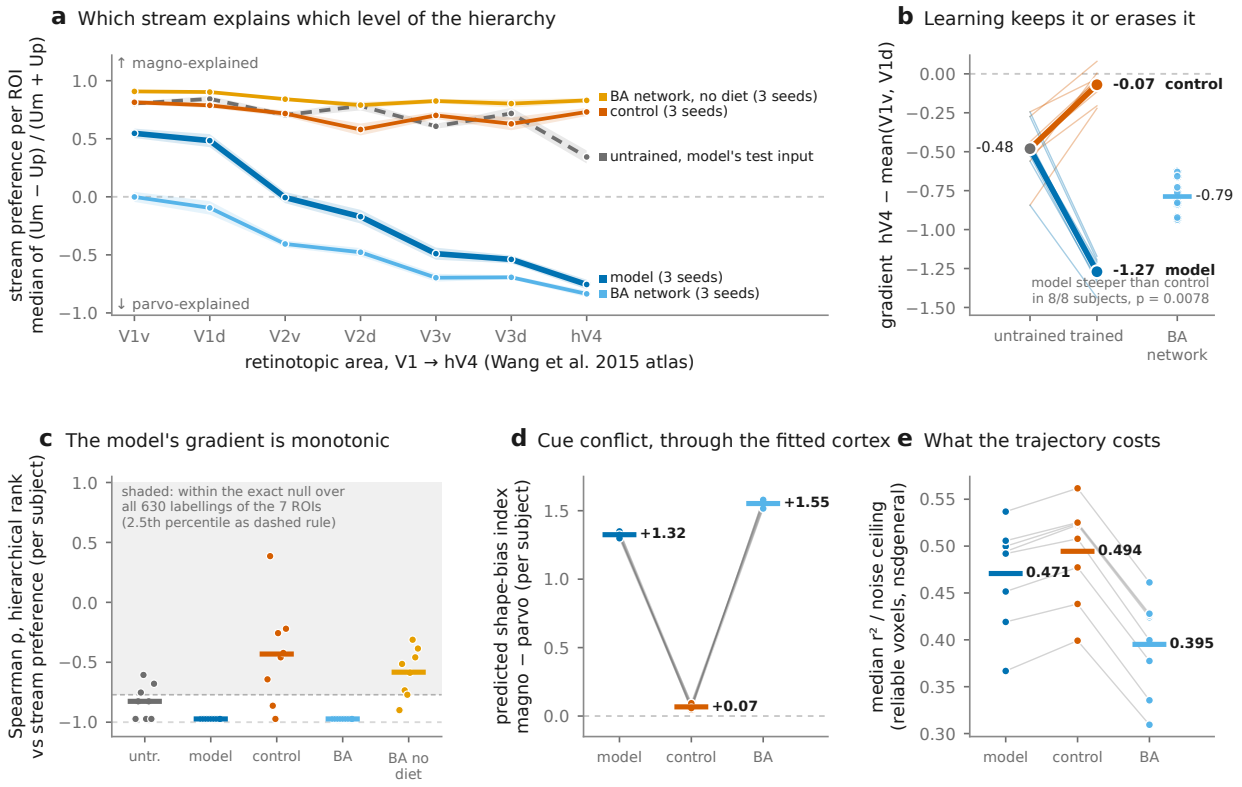


Figure 3 | In human visual cortex the magno stream carries V1 and the texture-biased stream the rest of the hierarchy, and only with a maturing input.

Voxel-wise banded-ridge encoding models of eight NSD subjects, one band per stream; each voxel's cross-validated explained variance is partitioned into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p) (Methods 4.5). Stream preference per ROI: median over reliable voxels (noise ceiling > 0.1, U_m + U_p $> 10^{-3}$, ≥ 20 voxels) of (U_m − U_p)/(U_m + U_p).

(**a**) Stream-preference profile from V1v to hV4 (Wang et al. 2015 atlas); line: mean over the eight subjects of each subject's seed mean; shading: ± s.e.m. The model's profile crosses zero, from +0.55 in V1v (magno-explained) to −0.76 in hV4 (parvo-explained). The control, on the identical test input, is magno-explained everywhere and nearly flat (+0.81 in V1v, +0.73 in hV4), as is the BA no-diet network (+0.91, +0.83); the BA network is parvo-explained from V1 on (−0.00 to −0.84). Dashed: the untrained network on the model's test input.

(**b**) The gradient hV4 − mean(V1v, V1d) per subject. One untrained network on one test input already descends (−0.48); training on the maturing input ends at −1.27 (seeds −1.259 to −1.291), training on the adult input from the first step at −0.07 (control, seeds −0.062 to −0.086); model steeper than control in 8/8 subjects (exact Wilcoxon p = .0078). Right: the BA network (seeds −0.745 to −0.834).

(**c**) Spearman ρ between hierarchical rank and stream preference per subject (seed-mean profile), against the exact null over all 630 labellings of the seven ROIs (shaded; dashed: its 2.5th percentile). Per-seed means: −0.964 to −0.973 for the model, −0.29 to −0.41 for the control.

(**d**) The cue-conflict images pushed through the encoding models fitted on natural scenes give a predicted magno − parvo shape-bias difference per subject: +1.32 for the model (seeds +1.32 to +1.34), +0.07 for the control, +1.55 for the BA network (seeds +1.53 to +1.57).

(**e**) Noise-ceiling-normalised median r² over reliable nsdgeneral voxels: 0.471 for the model against 0.496 for its control (8/8 subjects) and 0.395 for the BA network (8/8).

Sources: `docs/slides_2026-10-12_ba_vs_fusion/figures/fs2_cortex_summary.json` (`tmp_analysis/fs2_cortex_summary.py`), the untrained network's fits (`/scratch/byrong/mpk/encoding/results/2026-10-16_fs2`); `tmp_analysis/figs_v2/fig3_cortex.py`.

2.4 Behaviour reads out from both streams, and the rate is a property of the readout

The integrated readout's cue-conflict shape bias is 0.682 ± 0.020 (mean ± s.d. over three seeds; Table 1) against 0.374 for the control, 0.958 for the ten human observers and 0.172–0.429 for the CNNs of Geirhos et al. (2018a) on the identical images and statistic; integration costs 0.168 of the magno stream's 0.850 (Fig. 4c).

A readout of two oppositely biased components is nearly forced to land between them; what can be measured is how much of the rate it controls. Sweeping w while holding y_1 and y_2 exactly fixed (Fig. 4a) runs cue-conflict shape bias from 0.48 to 0.85 over 21 values of w and three seeds, and w = 0.5 reproduces each trained model exactly. The control's curve is flat (0.37–0.42), and so is the channel swap's (0.38–0.40): with both streams texture-leaning there is nothing for the weight to move. A two-layer head refitted on the frozen features at each w spans 106% of the frozen head's range (0.395 against 0.371) and 0.241 within 5 points of peak validation accuracy (Fig. 4b): the span belongs to the components, and its extremes are bought with accuracy.

The readout weight cortex prefers descends the hierarchy. With the readout's input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ as the design of a voxelwise ridge encoding model, the w that maximises an ROI's median r^2 is the weight that region prefers (Methods 4.5; Extended Data Fig. 5; Extended Data Table 2). In the model that weight falls from 0.65 in V1 to 0.30–0.35 in every extrastriate group, below V1 in 8/8 subjects ($\Delta = -0.323$ [−0.371, −0.269], exact p = .0078); in the control and the channel swap the gradient reverses (higher beyond hV4 in 8/8 subjects, and in 7/8 with one tie). The curves are shallow — every ROI is within 2% of its best r^2 at w = 0.5 — so this is an ordering of optima, not an r^2 difference, and the weight that best reproduces the observers' choices is not identified by these data (Supplementary Note S6).

Without a stream split, a gain on a network's own shape- and texture-predictive subspaces still moves the rate. The span of Fig. 4a is given by construction in a two-stream network, so we tested networks with no split, under plans committed before each analysis ran (Methods 4.8). In a frozen network's penultimate features, the fifteen discriminant directions of a 16-class shape and of a 16-class texture linear discriminant, made mutually orthogonal, span a shape- and a texture-predictive subspace; a gain of 2w on the first and $2(1 - w)$ on the second, the network's own head untouched, is the analogue of the mixing weight, every w is gated on top-1 accuracy on held-out natural images, and the subspaces are fitted on images sharing no exemplar with those the rate is scored on (Fig. 6; Methods 4.8). In the six ImageNet networks of Fig. 6 the rate rises monotonically with w by 0.14–0.20, 37–55% of the model's two-stream span (0.37); the pre-stated threshold ($\geq$ 0.20 in at least four of the six), met in all six with exemplars shared, is met in one (Fig. 6a, b; Extended Data Tables 3 and 4).

Across the field's zoo the result holds and the ranking does not. In 38 torchvision ImageNet classifiers the gated span is 0.08–0.29 (median 0.18) and unrelated to the trained rate ($\rho = +0.17$; Methods 4.8). Ranked by trained rate and by the rate at the shape end of the gate, the networks agree at Kendall $\tau = 0.63$; of the pairs the trained rate resolves, 39 of 490 reverse under image resampling (8.0%, against a floor of 0.33%) and 12 of 389 under texture-exemplar resampling (3.1%, against 0.47%), whereas the all-pairs figure (19%) is not decisive against the strictest resampling floor (13.6%). A gain on random 15-dimensional subspaces moves no network's rate by more than 0.08 (Supplementary Table S14).

The informative exception is a ResNet-50 trained on Stylized-ImageNet (Geirhos et al., 2018a): the highest trained rate of the 42 networks tested (0.81) and the only one whose probe leans to shape on exemplars it was not trained on (+0.23 [+0.13, +0.34]). Its bias sits in the representation, yet its span is not exceptional (0.11 on unseen exemplars, the 2nd smallest of 39); what sets it apart is which evidence the gain acts on: in a post hoc decomposition the gain moves 37 of the 38 ImageNet

networks toward shape mainly by removing texture evidence (median 89% of the effect) and the Stylized-ImageNet network mainly by adding shape evidence (texture share 11%; Supplementary Table S14).

Cortex loads on the texture-leaning units, not on the discriminant subspaces. With the two unit halves of the no-diet single trunk as encoding bands, the texture-leaning half carries every extrastriate group relative to V1 in 8/8 subjects ($\Delta = -0.32$, exact p = .0078; Fig. 6c, d; Supplementary Table S12). In ImageNet networks the texture-leaning half carries early visual cortex as well, so the V1-relative prediction fails there, and the discriminant subspaces that move the rate are nearly silent as unique cortical predictors (at most 1.3% of normalised unique variance each). The architecture is therefore not what makes the rate a readout property; it makes the components separable enough for one weight to move the rate from 0.48 to 0.85 (by 0.24 for a refitted head held within five points of peak accuracy), and for cortex to be carried almost entirely by one of them.

This is the mechanism we propose for the paradox: a system holding a shape-biased and a texture-biased component, whose behaviour reads out from both, will look shape-biased behaviourally and texture-biased wherever the texture-biased component dominates — here beyond V1, with the shape-biased component present only if its input matured during training. It is an existence proof, not a measurement of the human readout.

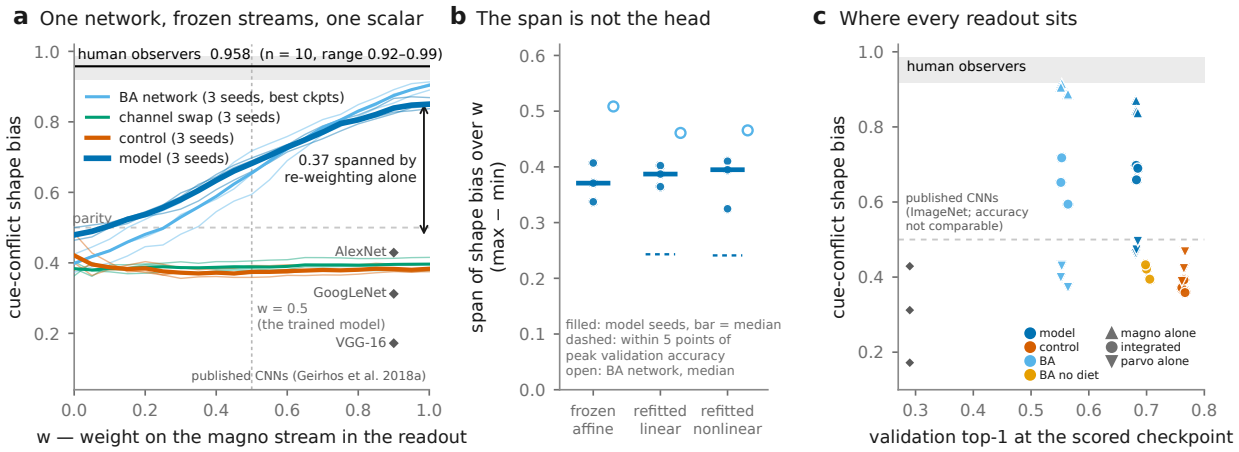


Figure 4 | Behaviour reads out from both streams, and the shape-bias rate is a property of that readout.

(**a**) Cue-conflict shape bias against the readout weight w on the magno stream, holding both streams' features fixed (21 values of w). The model (final checkpoints, three seeds; thin: seeds) runs from 0.48 at w = 0 to 0.85 at w = 1, and w = 0.5 reproduces each trained model; the control's curve is flat (0.37–0.42, three seeds), and so is the channel swap's (0.38–0.40, three seeds), the same ramp on L−M; the BA network (best saved checkpoints) for comparison. Grey band and line: the ten human observers (mean 0.958); diamonds: the CNNs of Geirhos et al. (2018a) on the identical images and statistic.

(**b**) The span of shape bias over w for the frozen affine head and for heads refitted on the frozen features at each w (linear; two-layer nonlinear), per seed of the model (bar: median). Median spans: 0.371 affine, 0.395 nonlinear (106%; weakest seed 96%). Dashed: the span over readouts within 5 points of peak validation accuracy (nonlinear median 0.241). Open circles: the BA network's medians.

(**c**) Shape bias against validation top-1 at the scored checkpoint, for each network's magno stream alone (▲), integrated readout (●) and parvo stream alone (▼); the model and its control at final checkpoints, the BA network at its best saved checkpoints, the BA no-diet arm at its final ones. Integration costs the model 0.168 of its magno stream's 0.850.

Sources: `results/fusion_sweep/`, `results/nonlinear_readout_summary.json`, `results/cue_conflict*/`, run histories; `tmp_analysis/figs_v2/fig4_readout.py`.

2.5 Properties that do not follow from shape bias

If shape bias is a property of a readout over components, quantities the field treats as travelling with it need not, and over identical test inputs three do not (Fig. 5).

Robustness. The standard reading of Geirhos et al. (2018a, 2021) is that shape bias buys out-of-distribution robustness. The BA study pre-registered that prediction for its diet and it failed (Supplementary Table S5); the model repeats the failure against a control that sees the same test images (Methods 4.6; Fig. 5a; Extended Data Fig. 1). It is less robust than its control on four of the five corruption families neither was trained on (uniform noise −0.135, high-pass −0.136, false colour −0.084, rotation −0.040; every model seed below every control seed, exact Mann–Whitney $p = 0.10$, the floor for three seeds against three), within 0.01 on phase scrambling, and more robust only to low-pass (+0.078), a family its trajectory trains in: robustness appears where it was trained in and nowhere else, as when stylised training was decomposed (Mummadi et al., 2021).

Agreement with human decisions. Error consistency — Cohen's κ on trial-by-trial agreement — subtracts the agreement two decision-makers' marginal rates force (Geirhos et al., 2020); across the ten observers the human–human ceiling is $\kappa = 0.234$. It does not follow shape bias: across the zoo it rises with the rate only as far as the bound the two marginals place on κ (Geirhos et al., 2020; Klein et al., 2025), at a matched rate it separates neither our seeds nor the zoo (Supplementary Table S13; Supplementary Note S6), and the BA study's pre-registered prediction, that the diet

would make the network more human-consistent than its control, is not supported (13.2% against 11.9% of the ceiling, seeds interleaved, exact Mann–Whitney $p = 1.0$). The measure does detect a correspondence when one is there, and here the trajectory supplies it. Asked not which cue the observers pick on an image but whether it resolves to either cue at all, the model's readout recovers 22.3% of the split-half ceiling on held-out images (seeds 19.6–24.3%), against 9.0% for the control (6.6–10.4%) and 8.2% for the channel swap (7.1–10.4%), every model seed above every seed of both (exact Mann–Whitney $p = 0.10$, the floor); on which cue is picked the three are interleaved (13.3%, 10.4% and 9.5%; $p \geq 0.70$; Fig. 5c). Unlike the BA network's version of this contrast, which crossing weights and test input removed (Fig. 5d), it belongs to the trajectory: the model and its control share their test input.

Representational dimensionality. The magno/parvo ratio of the streams' participation ratios on NSD images ends above parity in every subject and seed of the model (24/24; seed means 1.88–1.98) and below it in 23 of its control's 24 (seed means 0.89–0.95), from the same untrained starting point (1.63) and the same test input (Extended Data Fig. 2). The channel swap shows that dimensionality is not shape bias: it also ends above parity (24/24; seed means 1.27–2.15) with neither stream shape-biased, because its ramped chromatic stream contracts (participation ratio 52–86 against 116–124 in the control) where the model's ramped luminance stream expands (272–281).

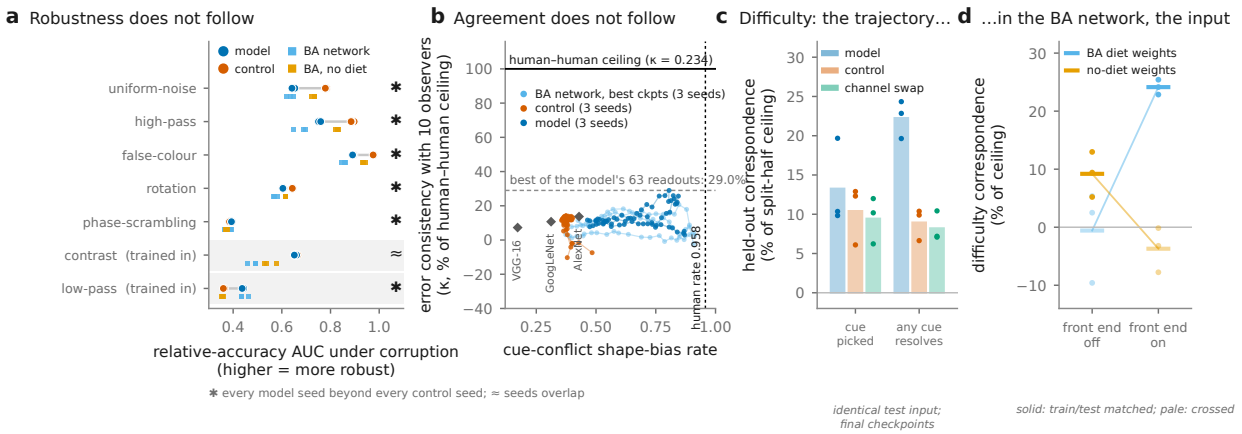


Figure 5 | Three properties the field treats as travelling with shape bias do not follow from it.

(**a**) Out-of-distribution robustness: relative-accuracy area under the curve (accuracy over clean accuracy, mean over severities) per corruption family, one marker per seed; the model against its control (final checkpoints, three seeds each, identical test input), the BA network's two arms as small squares (best saved checkpoints). The model is less robust than its control on uniform noise (-0.135), high-pass (-0.136), false colour (-0.084) and rotation (-0.040), every model seed below every control seed (exact Mann–Whitney $p = 0.10$, the floor for three against three); within 0.01 on phase scrambling ($+0.009$), level on contrast (-0.004, seeds interleaved) and more robust only to low-pass ($+0.078$); shaded: families the diets train in. $*$: every model seed beyond every control seed, whatever the size of the difference (phase scrambling separates by $+0.009$); $\approx$: seeds overlap.

(**b**) Error consistency with the ten observers (Cohen's κ as a percentage of the human–human ceiling, $\kappa = 0.234$) against the shape-bias rate, for every readout of the w sweep: the model (63 readouts; the best reaches 29.0% of the ceiling), its control and the BA network (best checkpoints), and the CNNs of Geirhos et al. (2020).

(**c**) Per-image correspondence with the ten observers on held-out images, as a percentage of each target's own split-half ceiling, after removing shape- and texture-category main effects: whether an image resolves to either cue at all (difficulty) and which cue is picked; bars are seed means, dots seeds. On difficulty the model reaches 22.3% (seeds 19.6–24.3%) against 9.0% for the control and 8.2% for the channel swap, every model seed above every seed of both (exact Mann–Whitney $p = 0.10$); on the cue picked the three are interleaved (13.3%, 10.4%, 9.5%). All three are tested on the identical input.

(**d**) The BA network's crossed front-end 2×2 on the same difficulty measure: its diet arm's 24.1% against 9.2% without the diet vanishes when weights and test input are crossed (-0.6% and -3.7%), so there it measured whether a network is tested on the input it was trained on, not the diet (solid: train/test matched, pale: crossed).

Sources: `results/ood/`, `results/ood_bestep/`, `results/sweep_error_consistency/`, `results/perimage_behavior/perimage_correspondence_v3.json` (`tmp_analysis/perimage_correspondence.py`); `tmp_analysis/figs_v2/fig5_dissociations.py`.

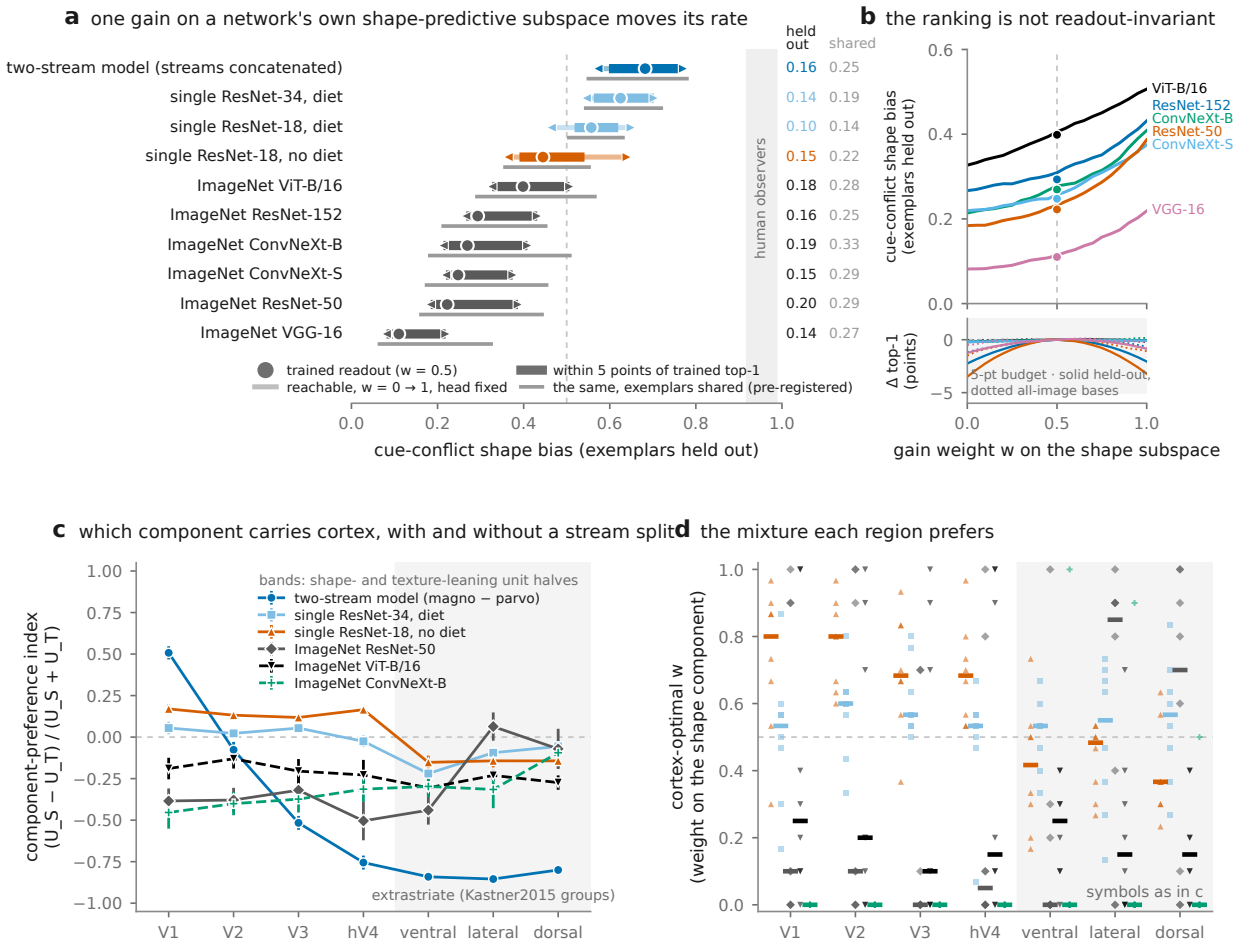


Figure 6 | The readout account beyond the two-stream architecture.

Every network here ends in logits = W·y + b for some penultimate feature vector y, and nothing is retrained. y is reduced by PCA and a 16-class shape and a 16-class texture linear discriminant give fifteen shape-predictive and fifteen texture-predictive directions, made mutually orthogonal by symmetric (Löwdin) orthogonalisation: bases B_S and B_T, with B_R the residual (Methods 4.8). Because the 1,200 cue-conflict images reuse 160 object and 48 texture exemplars, the curves drawn fit the subspaces on images whose object and texture exemplars both lie outside one of three exemplar folds and score the rate on the images whose exemplars both lie inside it (ten repetitions, mean curve; seeds averaged for our networks). The pre-registered estimator, fitted on one stratified half of the images and scored on the other, shares 98% of object and all texture exemplars between the halves; it is drawn as the grey brackets and gives spans about 1.6 times larger (Extended Data Tables 3 and 4).

(a) A reciprocal gain on a network's own shape- and texture-predictive subspaces moves its rate on exemplars the subspaces were not fitted on. logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b, the trained head untouched; w = 0.5 is the trained network (dot, its rate on all 1,200 images). Thin bars: the rate reachable as w runs from 0 to 1. Thick segments: the part of the sweep within 5 points of the trained top-1 on held-out natural images (MiniEcoset validation for our networks, a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks), with top-1 measured on bases fitted on all 1,200 images for our networks and, for the ImageNet networks, averaged over the held-out fold bases the curves use (Methods 4.8). Grey brackets beneath: the same gated range under the pre-registered estimator, whose halves share exemplars. The numbers at right are the gated spans of the seed-mean curves with exemplars held out and shared (the latter, and the ungated held-out spans, in Extended Data Table 3). Our networks by the paper's primary estimator, the ImageNet networks by the Geirhos 16-class rule (Methods 4.3). In the six ImageNet networks the whole sweep lies inside the gate (top-1 changes by at most 3.5 points on the held-out bases, 1.5 on bases fitted on all images) and the rate moves by 0.14–0.20 with exemplars held out (0.25–0.33 shared), monotonically ($\rho \geq 0.99$); the pre-registered 0.20 is reached in one of the six. In

our three single trunks the 16-class heads lose 7–15 points of top-1 at the ends of the sweep and the gated spans are 0.10–0.15 held out (0.14–0.22 shared). The pre-specified unit-level rule — a median split of the units by their shape-minus-texture η^2, the special case in which the two subspaces are disjoint sets of units, estimated with exemplars shared — reaches 0.07–0.16 within the same gates (Supplementary Table S11; no longer drawn). In the two-stream model the discriminant subspaces are not the streams (58–62% of the shape basis's energy lies on stream 1) and the subspace gain moves the rate by 0.20 with exemplars held out (mean over repetitions and seeds; the seed-mean curve runs 0.57 → 0.77, 0.16 within the gate) and by 0.29 with exemplars shared (0.52 → 0.80; 0.25 within the gate), where the streams themselves, which the unit-level rule recovers, give 0.38 and the readout weight of Fig. 4a 0.37. Grey band, the ten human observers (0.92–0.99).

(**b**) The held-out curves of the six ImageNet networks (top) and the change of top-1 along each sweep (bottom; solid, mean over the 30 held-out fold bases the curves use; dotted, bases fitted on all images; grey band, the five-point budget). Dots are the trained rates; the field's ranking is not invariant to the readout. With exemplars held out, four of the six networks can be moved past the trained rates of at least two of the other five within the budget, but the six can no longer be placed in the reverse of their trained order; with exemplars shared, five of the six pass the trained rates of four others (ViT-B/16, the highest, one) and the reverse order is reachable (Results 2.4).

(**c**) Which component carries cortex. For the single trunks the two unit halves of the pre-specified rule are the two bands of a banded ridge model of every NSD voxel (eight subjects, Methods 4.5); the index (U_S − U_T)/(U_S + U_T) is the median over reliable voxels per region, mean ± s.e.m. over subjects (seeds averaged per subject). The model's magno − parvo profile (Fig. 3, Extended Data Fig. 4) is drawn on the same axis under the single trunks' rule, voxels pooled per region: +0.51 in V1, −0.76 in hV4 and −0.80 to −0.85 beyond it. In the no-diet single trunk the texture-leaning half carries every extrastriate group relative to V1 (8/8 subjects, $\Delta = -0.32$, exact $p = .0078$) with no descent within V1–hV4; the diet-trained single ResNet-34 shows the V1-relative contrast (7/8, $p = .016$), a descent already within V1–hV4 (7/8, $p = .016$) and ventral cortex as its most texture-carried region (dorsal − ventral +0.16, 8/8); in ImageNet ResNet-50, whose units are almost all texture-leaning, that half carries V1 through hV4 and ventral cortex alike (−0.32 to −0.50) while lateral and dorsal cortex are balanced, so the V1-relative contrast reverses (+0.24, $p = .039$). In ConvNeXt-B and ViT-B/16, fitted after the pre-registered analysis, the texture-leaning half carries every region (index −0.09 to −0.45 and −0.13 to −0.31) with no V1-relative contrast (+0.22, 2/8 subjects negative; −0.08, 6/8, $p = .25$). Unique variances of the two halves in the single trunks and ConvNeXt-B are 0.006–0.04 (ViT-B/16 up to 0.09), against 0.10–0.25 for the parvo band with the split; most of a single trunk's explained variance is shared between its halves (Supplementary Table S12).

(**d**) The mixture each region prefers. The lossy sum $z(w) = w \cdot y_S + (1 - w) \cdot y_T$ (halves paired by rank) is the design of a ridge model at 11 values of w (0, 0.1, ..., 1), following Extended Data Fig. 5; points are subjects, bars medians, symbols as in (c). In the no-diet trunk the optimum falls from 0.80 in V1 to 0.37–0.48 beyond hV4 (7/8, $p = .023$); in the diet ResNet-34 the optimum sits near 0.55 in every region (the curves stay within 5% of their maximum over w in all three networks, as in Extended Data Fig. 5); in ResNet-50, V1–hV4 and ventral cortex prefer the texture-leaning half (0.0–0.1) while lateral and dorsal cortex prefer the shape-leaning one (0.7–0.85); ConvNeXt-B prefers the texture-leaning half in every region (median 0.0) and ViT-B/16 nearly so (0.10–0.25). Prediction P5 therefore holds in one of the three pre-specified networks and in neither of the two added after them.

Pre-specification and predictions: `docs/PLAN_GENERALITY_2026-09-14.md` (P1–P5, the unit-level rule) and `docs/PLAN_SUBSPACE_2026-09-15.md` (P-A, P-A′, P-B, P-C, P-D, the subspace rule). Panels c–d show the unit-level fits because the subspace rule's cortical predictions P-C1 and P-C3 failed and P-C2 held only in ConvNeXt-B: entered as three bands, the two 15-dimensional subspaces add $\leq 1.3\%$ unique cortical variance each and their index is at noise level (Supplementary Table S12, upper block); per-network values: Extended Data Table 3; Supplementary Tables S11 and S12, of which S11's last two columns hold a post hoc sensitivity analysis with the shape-leaning set defined as the units with $d_j > 0$ or as the top decile (Methods 4.8).

Sources: `results/figures_v3/subspace_behaviour.json`, `results/figures_v2/exemplar_zoo.json`, `results/figures_v2/exemplar_trunk_curves.json`, `results/figures_v2/exemplar_gate.json`, `results/figures_v2/generality_cortex.json` and `results/figures_v3/fig6_model_reference.json` (`tmp_analysis/figs_v2/fig6_model_reference.py`); `tmp_analysis/figs_v2/fig6_generality.py`. The PA-

PER_v2 version of this figure drew the BA network in the two-stream row and panel c.

3. Discussion

The two questions. Can two streams fed magno- and parvocellular-like inputs diverge under one objective while behaviour is shape-biased, and does such a system bear on why the ventral stream can be texture-like while behaviour is not? To the first the answer is yes, and the route is specific: both components are built by learning, and the luminance stream learns a shape bias only when its own input matures during training. The same adult input from the first step yields two texture-leaning streams, and the same ramp on the L−M channel leaves the network where its control is in behaviour, cortex and accuracy. To the second it is an existence proof, and it needs the same conjunction: only the network whose luminance input matured divides the cortical hierarchy, its shape-biased stream carrying V1 and its stronger, texture-leaning stream extrastriate cortex, and only it tracks which images the observers resolve (22.3% of the split-half ceiling, against 9.0% and 8.2% on identical input), while its behaviour, reading out from both, is shape-biased. Read as claims about one representation the two conflict; read as claims about a readout over components, they do not.

What the readout account licenses. A model matching the human shape-bias rate is routinely described as having human-like representations; here that inference fails twice. The same network holds components at +0.32 and −0.16 on held-out exemplars and behaves at 0.682, and one scalar over frozen representations moves the rate across most of the range the field's models occupy, so the rate does not identify a representation; and trial-by-trial agreement with the observers never exceeds 29% of the human–human ceiling in our networks, so the rate is at best a weak proxy for the decision process. Prior work established that a representation can hold more shape than its behaviour reveals (Hermann et al., 2020; Islam et al., 2021) and that the measured rate depends on the test design (Tartaglini et al., 2022), on a prompt (Gavrikov et al., 2024) and on a network's spatial-frequency channel (Subramanian et al., 2023), and retraining only the last layer reduces ImageNet networks' reliance on texture (Izmailov et al., 2022; Kirichenko et al., 2023); new here is one scalar over fixed components, the head untouched, its cortical grounding, and its persistence without a stream split (Fig. 6): the split amplifies the dissociation (a span of 0.37 between the streams against 0.20 for the same network's concatenated features with exemplars held out), it does not create it.

How to report shape bias. A shape-bias claim should carry three numbers — the rate at the trained readout, the span a reciprocal gain on the network's own shape- and texture-predictive subspaces reaches within an accuracy budget, and a probe's shape-minus-texture accuracy, the last two on held-out exemplars — because across 38 ImageNet networks they rank the zoo differently (τ = 0.63, and 0.43 for the probe); one command in the repository prints all three for any torchvision network (Methods 4.8).

What a developmental trajectory does, and what it costs. The intuitive account of low-pass developmental input is informational: coarse input shields the learner from texture, so it learns shape. Our results support a version of it in which the input's history matters and the information at test does not: the model and its control see the same images at test, and what differs is whether the magno stream's early learning happened on coarse, low-contrast input (cf. critical periods in deep networks; Achille et al., 2019). Its route runs through the competition between the trunks: the coarse early input hands the task to the chromatic trunk, and the luminance trunk, learning beside it from a world that sharpens, ends the weaker and shape-biased; the swap shows that being the weaker trunk is not enough (Results 2.2). The BA network's fixed 48-month transform instead put most of its magno stream's shape bias into the input (+0.210 of +0.337 before train-

ing), at a cost in accuracy (0.556) and BOLD prediction (0.395); the trajectory costs less (8.1 points of top-1 against its control, robustness to corruptions it was not trained on, 0.024 of noise-ceiling-normalised r^2). Its interest is as a model of development, and its prediction is developmental: a visual system whose early input is coarse should show a stream-level shape bias that a system given adult input from the start, on the same images, does not.

Limitations. The load-bearing cortical result rests on eight subjects from one dataset and is read as sign and ordering only. The mapping between our chromatic stream and texture-selective cortex is by explanatory role, not content, and in every network the trunk that carries extrastriate cortex also carries the task, so the cortical result cannot separate texture bias from task load; trunks matched in accuracy would. One trajectory shape was tested, and whether its direction matters is open: the control removes both the coarse-to-fine order and the exposure to coarse input, and a reversed or age-shuffled schedule would separate them. No single-stream network was trained with the trajectory. The model of record was changed after its first seed was seen; the rule for the change, and the control that could have blocked it, were fixed before seeds 1–2 and the control were trained (Methods 4.2). Seven plans were committed before their analyses ran, all on the BA network or the zoo, and every outcome, failures included, is reported (Supplementary Note S5); on the model, the robustness prediction fails again and the error-consistency one is not supported. Claims deliberately not made are listed in Supplementary Note S1, and the limitations in full in Supplementary Note S6.

4. Methods

4.1 Front end

RGB images are converted to the DKL cone-opponent space through RGB→XYZ (D65)→LMS (Hunt–Pointer–Estévez)→DKL at 256 × 256 px, each channel normalised to $[0, 1]$ with the training set's global range and quantised to 8 bits exactly as the training data were stored (Derrington et al., 1984). Stream 1 receives the luminance channel and stream 2 the L−M channel. Under the developmental diet the luminance channel passes through the acuity and contrast stages of the developmental visual diet of Lu et al. (2026) (colour stage off, the channel being achromatic), both evaluated at one age a: the acuity stage is a Gaussian blur whose σ follows the diet's acuity curve at a, scaled to the 256-px image (0.50 px at 48 months; inert, kernel size 1, beyond about six years), and the contrast stage is the scale-free contrast-sensitivity cut evaluated at a (`cs_logspan_start` $= 5 \times 10^{-3}$, decomposition mode "keep"; `src/analysis/encoding/dkl_frontend.py`, `apply_dvd_diet`). During training a rises linearly with the optimisation step from 0 to 300 months (6 months per epoch), as the diet's own age curve does; every evaluation — validation, cue-conflict, probes, NSD images and corruptions — uses a = 300 months. The conditions are: `fs2_dvd300` (model of record: luminance + ramp | L−M); `fs2_dvdfix300` (control: luminance + the 300-month diet from the first step | L−M), identical to the model at test; `fs2_dvd300swap` (luminance | L−M + ramp; training); `fs2_dvd48` (the ramp stopped at 48 months; one seed); and `fs2_bcs` (the BA network's fixed diet under the recipe of Methods 4.2). The BA network `dkl_bcs` (luminance + isotropic Gaussian blur, $\sigma = 4$ px, kernel 33, reflect padding + the same contrast cut at a fixed 48 months | L−M), its no-diet arm `dkl_base` and its two front-end controls `dkl_bcs_swap` and `dkl_bcs_parvo_highsf` are those of the originating study, retrained at three seeds with its recipe (PAPER_v2, Methods 4.1). Every analysis pipeline reads the front end from the checkpoint it evaluates and refuses a mismatch (`check_frontend`). Evaluation stimuli pass through the identical front end.

4.2 Architecture and training

Two ResNet-18 trunks (He et al., 2016), one per stream, initialised bit-identically, with a joint linear classification head over the weighted average of their penultimate features (w = 0.5), trained with one cross-entropy on the joint output (label smoothing 0.1) and no loss on either stream. Two per-stream heads are trained alongside on detached stream features to log each stream's accuracy during training; they send no gradient into the trunks (checked: zero gradient leak), and no measure in this paper uses them — the streams alone are read through the joint head (Methods 4.3). MiniEcoset at 256 px, 112 classes (Thorat & Kietzmann, 2024). 50 epochs, batch 256, AdamW (lr 10^{-3}, weight decay 0.05, none on biases and normalisation parameters; Loshchilov & Hutter, 2019), two warm-up epochs then cosine decay to zero, an exponential moving average of the weights (decay 0.9995) that is evaluated and saved as the final checkpoint, bf16 autocast, and the augmentation set of Lu et al. (2026) (horizontal flip, rotation $\pm$15°, colour jitter, random equalise, perspective, sharpness; random grayscale for RGB inputs only). Three seeds per condition. The recipe was fixed on the BA network's fixed diet before any trajectory was trained — eleven single-objective variants of that network (readout weight 0.5–0.9, label smoothing, gradient modulation, 25 or 50 epochs, norm-pinned features) all reached 0.64–0.67 top-1, and `fs2_bcs` was the best — and is unchanged for the trajectory and its controls. The model's seeds and the control's seed 0 trained on one NVIDIA B200 each (about 4.3 h per run), the control's seeds 1–2 and the swap on one RTX A5000 each (18.3–21.3 h per run; 99.9 GPU-hours for the five), with the same code and data. The rule for replacing `fs2_bcs` as the model of record — the trajectory's three-seed mean must exceed it on top-1 and on shape bias, and the adult-from-the-start control must fall short on shape bias, or the gain would be the adult input at test rather than the trajectory — was written after the trajectory's first seed and before seeds 1–2 and the control were trained; it passed on the training-time evaluation (+1.8 points, +0.067 shape bias; control −0.32 shape bias). The BA network's runs keep the thesis's recipe: 150 epochs, batch 256, Adam (lr 1×10^{-4}, $\beta_1 = 0.9$; Kingma & Ba, 2017), cross-entropy, the same augmentations, weights saved every 25 epochs; their compute accounting is that of PAPER_v2 (584 GPU-hours on RTX A5000s for 21 runs; 930 kWh and 325 kg CO_2e as an upper bound for the whole of that study). No hyperparameter search was run for any trajectory arm.

4.3 Behaviour and probe

Cue-conflict shape bias is computed on the Geirhos et al. (2018a) style-transfer set restricted to the 16 categories present in MiniEcoset and to images whose shape and texture categories differ (1,200 images). Logits are restricted to the 16 columns; the argmax is scored as a shape or texture response; images decided as neither are excluded from the denominator and their count is reported. Bootstrap 95% confidence intervals are over images. A second, forced two-way estimator keeps every image and scores whether the logit of the image's shape category exceeds that of its texture category; it is reported beside the primary estimator and computed from the same stored logits. The readout is full = classifier($w \cdot y_1 + (1-w) \cdot y_2$), magno_only = classifier($w \cdot y_1$), parvo_only = classifier($(1-w) \cdot y_2$), all through the same classifier and bias term; the w sweep evaluates full at 21 values of w on stored features. The refitted heads (Linear(512, 112) and Linear(512, 512)→ReLU→Linear(512, 112)) were trained on $w \cdot F_1 + (1-w) \cdot F_2$ of MiniEcoset training features at each w with an identical budget (Adam, lr 10^{-3}, batch 1024, 20 epochs) and the same initialisation seed at every w; the span within 5 points of peak validation accuracy is taken over the w at which the refitted head's validation top-1 is within 0.05 of its best (`tmp_analysis/nonlinear_readout_summary.py`). Human observers are the ten of the same release (1,280 trials each), scored by the identical rule; our recomputation reproduces the published 0.958 and the published AlexNet, GoogLeNet and VGG-16 values. Error consistency is Cohen's κ (Cohen,

1960) on the binary shape/texture choice over images both parties decided, with expected agreement p_m·p_h + (1−p_m)(1−p_h), reported as a percentage of the human–human κ over all 45 observer pairs (Geirhos et al., 2020). For the frozen networks of Methods 4.8 the same κ was computed at every gain w (`src/analysis/generality/kappa_sweep.py`), together with the largest κ attainable at the two marginals, κ_max = $(1 - |\text{p_m} - \text{p_h}| - \text{p_e})/(1 - \text{p_e})$, against which agreement is also reported as κ/κ_max, as a percentage of the same ratio for human pairs (0.353). Per-image correspondence uses the observers' per-image shape rate and cue-resolution rate (split-half reliabilities r = .658 and .794 after category removal), the models' shape-minus-texture logit margin, w chosen on half the images and scored on the held-out half, as a percentage of each target's own ceiling. The linear probe reduces each stream's features to 256 PCA components and fits multinomial logistic regression (C = 1.0) under 5-fold stratified cross-validation to shape and texture labels separately; bias is shape accuracy minus texture accuracy on the same folds. Because the 1,200 images are built from 160 object and 48 texture exemplars, stratified folds leave every test image's exemplars in the training folds, so the primary probe is exemplar-balanced: three exemplars per category for both labels (three of each shape category's ten, drawn at random; the texture set has three), texture images subsampled to the same per-category count, scored leave one exemplar out with PCA (min(256, n_train − 1) components) and scaling fitted inside each fold, mean of 20 draws with the 5–95% interval over draws (`src/analysis/generality/exemplar.py`; `docs/PLAN_EXEMPLAR_2026-09-21.md`). The stratified-fold probe is reported beside it as exemplar-shared. With about 250 training images per fold the balanced probe is sensitive to preprocessing: PCA fitted on all images and then standardised compresses every bias toward zero (median +0.03 over the 38 ImageNet networks), whereas fewer components or no standardisation agree in sign with the primary estimator (medians −0.14 and −0.39; $\rho = 0.94$ with the latter over the 42 zoo networks). The reporting standard (Discussion) fixes the primary preprocessing, and the spread over the five held-out variants is reported beside it (`tmp_analysis/figs_v2/exemplar_probe_spread.py`).

4.4 Effective dimensionality

Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ of the stream feature matrix on 1,000 NSD images per subject (Gao et al., 2017), computed Gram-side, reported as the magno/parvo ratio with parity at 1.0. Subjects are the replication unit within a seed; seeds are reported separately and never pooled. For the model and its control the stream features are those cached for the encoding fits (`tmp_analysis/fs2_pr_ratio.py`, `results/figures_v2/fs2_pr_ratio.json`), and the reference is the untrained network on the same front end; for the BA network the learning-free reference is a Gabor oriented-energy bank applied to each stream's own transformed input.

4.5 Encoding

NSD (Allen et al., 2022) `betas_fithrf_GLMdenoise_RR` single-trial betas, converted to percent signal change and z-scored per session per voxel, `nsdgeneral` mask, eight subjects. Splits are by 73k image id with the shared-515 images held out. Stream features for each subject's stimuli are reduced by PCA to 1,000 components per stream (fit on training images only) and entered as two bands into a banded ridge regression (himalaya; Dupré la Tour et al., 2022) with 3-fold GroupKFold by image id, per-voxel alphas and per-voxel band weights. Unique variance U_m (U_p) is the drop in cross-validated r^2 when the magno (parvo) band is removed. Noise ceilings are NSD's `ncsnr`; voxels below 0.1 are excluded and any ROI cell with fewer than 20 surviving voxels is reported as missing. ROIs are the `prf-visualrois` labels of the Wang et al. (2015) atlas as distributed with NSD (V1v, V1d, V2v, V2d, V3v, V3d, hV4), with hierarchical rank [1, 1, 2, 2, 3, 3, 4]. The model, its control and the untrained network on its front end are fitted with the

identical pipeline, reading each checkpoint's front end from the checkpoint (Methods 4.1). In the BA network the canonical-correlation spectrum between the two bands runs from ≈0 to 0.93, so each band retains directions the other cannot predict and unique variance is identifiable; a recovery simulation with the fitted feature geometry and a known across-ROI split returns the true ordering exactly ($\rho = +1.000$ in 8/8 subjects, both arms), which is why the gradient is reported as sign and ordering rather than as calibrated magnitude. The BA network's swap and `parvo_highsf` arms are fitted for all three seeds, as its diet and no-diet arms are, and their behavioural and probe measures use the same three seeds (Supplementary Table S3). Both are degenerate by construction and in every seed — the swapped network leaves the parvo band a median unique variance 0.005–0.010 of the magno band's, the parvo + high-SF network leaves the magno band 0.003–0.005 of the parvo band's — so they enter no cortical claim beyond the sign and size of their unique variances (PAPER_v2 Extended Data Table 1; Supplementary Table S10). The fits of this section were originally submitted at about 160 GB of memory per subject, which is not the size of the data: the peak is the group-ridge random search, which forms an (n_alphas × n_test × n_train) array per gamma sample and fold, about 30 GB in float64 with all 21 alphas at once. Batching that search seven alphas at a time changes no estimate — a refit of the diet seed-0 subject-01 fit reproduces the stored one at voxelwise r = 1.00000 on both unique variances, with the largest per-ROI index difference below 10^{-4} — and the same fits then peak at 32.6 GB (`slurm/encode_equiv_probe.sh`, `slurm/encode_cc_controls.sh`). Eccentricity bands are NSD's `prf-eccrois` labels (0–0.5°, 0.5–1°, 1–2°, 2–4°, > 4°); continuous pRF eccentricity, size and R^2 are NSD's population-receptive-field estimates in the same functional space, and the within-subject regressions of Supplementary Table S8 use z-scored log(eccentricity + 0.5) and log(size) over voxels with pRF $R^2 > 5\%$. The per-voxel predicted shape-bias index is obtained by pushing the cue-conflict set through each fitted encoding model. For Extended Data Fig. 4 and Supplementary Table S10 the same voxelwise fits were relabelled, without refitting, with the full Kastner2015 atlas as distributed with NSD (adding VO1/2, PHC1/2, LO1/2, TO1/2, V3A/B and IPS0–2; IPS3–5, SPL1 and FEF have no voxels inside `nsdgeneral`), with NSD's `streams` parcellation and with the fLoc face, place, body and word ROIs; regional means are unweighted means of the ROI medians, and region contrasts are tested per subject with the exact Wilcoxon test. For the cortex-optimal readout weight (Extended Data Fig. 5; Extended Data Table 2) the classifier's own input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$, where y_1 and y_2 are the 512-dimensional stream outputs the classifier mixes, was the design of a single-band ridge model of every `nsdgeneral` voxel for $w \in \{0, 0.05, \ldots, 1\}$, with the same split, grouped folds and alpha grid as the banded model and columns z-scored on the training trials; because $z(w)$ sums the streams dimension by dimension the ridge cannot undo the mixture, so w is a constraint on the design rather than a re-parameterisation of the two-band model. w*_cortex is the w that maximises the median test r^2 over an ROI's reliable voxels (seeds averaged per subject); the 1,024-dimensional concatenation $[y_1, y_2]$ is fitted as a reference and not as a bound, because although every mixture is a linear map of it the two models are regularised separately on a shared alpha grid, and in the model and its control the best mixture exceeds the fitted concatenation in 15 of the 112 subject × ROI × arm cells by a median 0.0023 r^2 — more than the 0.0012 the sweep itself buys from w = ½ to w, *so* w_cortex is read as the location of an optimum and never as an r^2 difference (Extended Data Table 2); and the behaviour-optimal weight is the trial-fit w of Fig. 4 (Methods 4.3) for each of the ten observers, reported with its per-fit bootstrap interval and permutation p and not compared with w*_cortex, because it is not identified (Results 2.4). The koniocellular control substitutes S−(L+M) for L−M in stream 2 of the untrained network at each of three seeds, each against the L−M network of its own seed, contrast-matched to L−M per image, with the magno band bit-identical across arms (gate $\max|\Delta r^2\text{_magno}| < 10^{-4}$, passed at 0.000).

4.6 Out-of-distribution robustness

Corruptions are applied to RGB before the front end and scored in the native 112-class label space. Five families were declared untrained-on before the BA study's run (uniform noise, high-pass, phase scrambling, rotation, false colour) and two as trained-in positive controls (low-pass, contrast); the declaration is kept for the model, whose trajectory trains in both low-pass and contrast. Reported as the area under the relative-accuracy curve across severities (accuracy over clean accuracy), at final checkpoints for the model and its control and at each seed's best saved checkpoint for the BA network. Two implementation differences from Lu et al. (2026) remain: our diet is a transform on one stream's input of a two-stream network rather than an augmentation on a single trunk, and we train on MiniEcoset and score corruptions in that label space, applied before the front end. The third difference of the BA network, a fixed rather than scheduled age, does not apply to the model.

4.7 Statistics

Within-subject comparisons use the paired exact Wilcoxon signed-rank test, n = 8 (exact two-sided floor p = .0078); seed-level comparisons use the exact Mann–Whitney U, 3 v 3 (floor p = 0.100, reached only on complete separation), and seeds are never pooled with subjects. Every cortical contrast is computed both on the raw per-voxel difference U_m − U_p and on the normalised index (U_m − U_p)/(U_m + U_p), each summarised by its median over reliable voxels per ROI, and a claim is made only where the two agree; disagreements are reported as unresolved. Equivalence uses two one-sided tests with bounds fixed before any interval was computed (Schuirmann, 1987; Lakens, 2017). No multiple-comparison correction is claimed: with an exact floor of .0078, Holm–Bonferroni over any family larger than six cannot yield a rejection, so claims rest on complete separation, agreement of both statistics, replication across seeds and effect size (dz = −14.3 for the model's gradient, seed means per subject; −6.9 for the BA network's and −0.59 for the control's by the same computation). The model and its control are evaluated at their final checkpoints throughout, which for this recipe are the exponential moving average of the weights and lie within 0.9 points of top-1 of the best epoch; the BA network's recipe declines late (final 0.445 against best 0.556), so its behavioural and robustness results use each seed's best saved checkpoint where stated, as in PAPER_v2, and its cortical fits the final checkpoint. Four predictions — the front-end 2×2, the inverted-front-end prediction, the corruption families and the error-consistency hypothesis — were written into a version-controlled plan and committed before the corresponding analysis was run on the BA network (internal pre-specification, not registry-based); two failed and are reported as failures. They were not re-registered for the model; its replacement rule (Methods 4.2) is the one decision fixed before its seeds 1–2 and control were trained.

4.8 Readout re-weighting in networks without a stream split

Pre-specified three times — in `docs/PLAN_GENERALITY_2026-09-14.md` (the unit-level rule; five directional predictions P1–P5), `docs/PLAN_SUBSPACE_2026-09-15.md` (the subspace rule; predictions P-A, P-A′, P-B, P-C and P-D) and `docs/PLAN_ZOO_2026-09-15.md` (the model zoo; P-Z1–P-Z4) — each committed before its analysis was run, with every outcome reported (Results 2.4; Extended Data Tables 3 and 4; Supplementary Tables S11 and S12). Both rules act on a frozen network with logits = W·y + b, y its D-dimensional penultimate features, and neither retrains anything. Subspace rule: on a fitting set of cue-conflict images y is centred and reduced by PCA to min(256, D) components; a 16-class linear discriminant analysis on the shape labels (scikit-learn, eigen solver, Ledoit–Wolf shrinkage) gives 15 shape-discriminant directions and one on the texture labels 15 texture-discriminant directions; mapped back to unit space and orthonormalised, the two bases

are made mutually orthogonal by symmetric (Löwdin) orthogonalisation, which treats shape and texture identically (sequential Gram–Schmidt in either order is the reported sensitivity; Extended Data Table 3), giving B_S, B_T and an orthonormal basis B_R of the residual, with projectors P_S, P_T and P_R. Fixed-head gain sweep: logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b, w ∈ {0, 0.05, …, 1}, so that w = 0.5 is the trained network exactly; the bases are fitted on one half of the 1,200 images, stratified by shape × texture pair, and the rate scored on the other, over 10 random splits in both directions. These halves share exemplars (98% of scored images share their object exemplar with the fitting half and all of them their texture exemplar), so every span is also computed on exemplar-disjoint folds (`docs/PLAN_EXEMPLAR_2026-09-21.md`, predictions P-X1–P-X16, committed before its code existed): in each of ten repetitions every texture category's three exemplars and every shape category's ten object exemplars are assigned at random to three folds, the bases are fitted on the images whose object and texture exemplars both lie outside fold k and the rate is scored on the images whose exemplars both lie inside it, and the three folds' scored images (about 400) form one curve; holding out only the texture or only the object exemplars gives the two components (median spans 0.195 and 0.293 over the 38 ImageNet networks, against 0.19 for both and 0.30 for neither). As for the exemplar-shared estimator, the reported span is max − min of the mean curve over the gated w; the pre-registered predictions were scored on the mean of the repetitions' own spans (median 0.189 against 0.187). The ranking's noise floor is the fraction of pairs reversing between two independent bootstrap resamples of the 1,200 images at the trained readouts (2,000 pairs of resamples, shared across networks). Unit-level rule, the special case in which the two subspaces are spanned by disjoint sets of units: each unit j receives d_j = η^2_shape(j) − η^2_texture(j), the one-way fraction of its variance over the images lying between the 16 shape or the 16 texture categories; the median split of d gives the shape-leaning half S and the texture-leaning half T (equal-sized, as the two streams are), and logits(w) = W·(g(w) ⊙ y) + b with g_j = 2w for j ∈ S and 2(1 − w) for j ∈ T, on the same image splits. At a referee's request, and not pre-specified, this rule was also run with S the absolute shape-leaning set (units with d_j > 0) or the top decile of units by d_j, T the remainder (Supplementary Table S11, last two columns; `slurm/generality_rules.sh`): both leave S small — 2–89 units under the absolute rule (a single unit, hence undefined, for ConvNeXt-S) and a tenth of the units (51–409) under the decile rule — so that w → 1 silences at least nine tenths of the representation; the ungated spans are 0.11–0.40 with positive mean ρ (0.29–0.99) but not every split is monotone in our trunks, and within 5 points of the trained top-1 — a gate that closes at w = 0.72–0.95 in every case but ConvNeXt-B under the decile rule, where the whole sweep passes — the reachable span is 0.02–0.21. Estimators: our networks by Methods 4.3; torchvision ImageNet networks (ResNet-50, ResNet-152, VGG-16, ViT-B/16, ConvNeXt-S/B, IMAGENET1K_V1 weights, model-vs-human preprocessing) by the Geirhos 16-class rule, mean softmax probability over each category's ImageNet classes and argmax over the 16, which assigns every image to one of the 16 categories; as in Methods 4.3, the rate is computed over the images assigned to their shape or their texture category (814 of 1,200 for ResNet-50; per-network counts in Extended Data Table 4). Accuracy gate: top-1 of the gained features, with the bases or the split defined on all 1,200 images, on held-out natural images — the MiniEcoset validation split for our networks and a stratified 10-per-class subset of the ILSVRC-2012 validation set for the ImageNet networks; the gated span is max − min of the seed-mean held-out curve over the w at which top-1 lies within 5 points of the trained value. For the exemplar-disjoint spans of the 42 zoo networks the gate is recomputed on the same fold bases the spans use (post hoc, `docs/PLAN_EXEMPLAR_2026-09-21.md` addendum C; `src/analysis/generality/exemplar_gate.py`, `results/figures_v2/exemplar_gate.json`): top-1 is averaged over the 30 fold bases at each w. Under those bases the largest top-1 change over the sweep is a median 1.5 times that on the all-image bases (up to 6.3 points against 4.4), the gate narrows in three networks (ResNet-18 to w = 0.10–0.95, MobileNet-V2 and the Stylized-ImageNet ResNet-50 to 0.05–0.95; four with ResNet-34 if every fold basis must

stay within budget), and the zoo summaries barely change (gated span 0.08–0.29, median 0.18 against 0.19, 11 of 38 ≥ 0.20; 19% of pairs reversed, $\tau = 0.63$; Spearman ρ with the trained rate +0.17 against +0.10); Results 2.4 gives these values. The six networks of Fig. 6 keep the whole sweep inside the gate (largest change 3.5 points, against 1.5 on the all-image bases). Our trunks' gate stays on the all-image bases, because their MiniEcoset validation features were not stored. Model zoo (Extended Data Fig. 6, Extended Data Table 4): the same pipeline on 42 networks — 38 torchvision `IMAGENET1K_V1` classifiers, ResNet-50 under torchvision's `IMAGENET1K_V2` recipe, and the three ResNet-50s of Geirhos et al. (2018a) trained on Stylized-ImageNet, on Stylized-ImageNet plus ImageNet, and on both then fine-tuned on ImageNet — with y the input of each network's last linear layer, captured on the network's own forward pass and checked against its logits; Resize 256 / CenterCrop 224 except Inception-v3 (342/299), EfficientNet-B3 (320/300) and EfficientNet-V2-S (384/384), which were not trained at 224. The probe bias is the linear probe of Methods 4.3 on the same 1,200 features, predicting the shape and the texture class in turn, under both of its estimators. Ranking statistics over the 38 ImageNet-trained networks are Kendall τ between rankings and the fraction of the 703 pairs whose order reverses, with no p-values, the networks being a fixed set rather than a sample. `python -m src.analysis.generality.report --model tv:<arch>` prints the three numbers for any torchvision classifier. Refit: because a refitted linear head can undo a per-direction gain, the refit test mixes the two components lossily, as the two-stream model does, and passes the rest of the representation unchanged — $z(w) = [w{\cdot}s + (1 - w){\cdot}t, B_R^T y]$ with $s = B_S^T y$ and $t = B_T^T y$ paired by discriminant rank (under the unit-level rule $z(w) = w{\cdot}y_S + (1 - w){\cdot}y_T$, the halves paired by rank of d) — and refits the linear and two-layer heads of Methods 4.3 at each w, with a head refitted on the unmixed y as the accuracy ceiling. Cortex: under the unit-level rule S and T (split on all 1,200 images) are the two bands of the banded ridge model of Methods 4.5 on the penultimate units of the single-trunk networks, giving U_S, U_T and the index $(U_S - U_T)/(U_S + U_T)$ per voxel; the cortex-optimal w of the lossy sum follows the protocol of Extended Data Fig. 5 on an 11-point grid $(0, 0.1, \ldots, 1)$. Under the subspace rule the bands are $s = B_S^T y$, $t = B_T^T y$ and the residual $r = B_R^T y$ (15, 15 and $D - 30$ dimensions; bases fitted on all 1,200 images), with $U_S = r^2_\{STR\} - r^2_\{TR\}$ and $U_T = r^2_\{STR\} - r^2_\{SR\}$ from the three-band model and the two two-band models (random search with 30 and 20 samples), and the w sweep on the 15-dimensional lossy sum $w{\cdot}s + (1 - w){\cdot}t$ alone; the five families are `rgb_tsn` and `dkl_bcs_single34` (three seeds each, averaged per subject), ResNet-50, and ViT-B/16 and ConvNeXt-B, whose NSD features were extracted for this test (predictions P-C1–P-C3; Supplementary Table S12, upper block). The `rgb_tsn` network — two trunks that receive the identical three-channel image and remain bit-identical — is one ResNet-18 on the untransformed image and is used as the no-diet single trunk. All of our networks enter this analysis at their final checkpoints, the policy of the cortical fits (Methods 4.7); the model's two-stream concatenation is run through the identical pipeline (`slurm/paper_fs2_generality.sh`, `slurm/paper_fs2_exemplar_zoo.sh`).

Evidence decomposition (post hoc; `docs/PLAN_SPAN_2026-09-18.md`, predictions S-1 to S-7 written before the analysis; Supplementary Table S14). For image i with shape category s and texture category t the Geirhos rule's margin is $\Delta_i = \log p_s - \log p_t$, p_c the mean softmax probability over c's ImageNet classes, and its gradient with respect to the logits is $u_i = q_s - q_t$, q_c the softmax restricted to c's classes. Because $\text{logits}(w) = \text{logits}(½) + (2w - 1){\cdot}W(P_S - P_T)y$ exactly, $e_S,i = u_i{\cdot}W\ P_S\ y_i$ and $e_T,i = u_i{\cdot}W\ P_T\ y_i$ give the first-order margin $\Delta_i + (2w - 1)(e_S,i - e_T,i)$ (model M0) and, with $z = \text{mean}\ \Delta\ /\ \text{sd}\ \Delta$ and $\lambda = \text{mean}(e_S - e_T)\ /\ \text{sd}\ \Delta$ over the 1,200 images, the two-number forced rate $\Phi(z + (2w - 1)\lambda)$ (M1); the shape term is mean e_S, the texture term $-$mean e_T and the texture share the second over their sum. All use the bases fitted on all 1,200 images and are compared with the forced-choice span in sample. Null: ten pairs of mutually orthogonal random 15-dimensional subspaces (seeds 0–9) inside each network's 256-dimensional PCA space of the cue-conflict features, under the same gain, estimator and gate (`src/`

`analysis/generality/span_model.py`, `slurm/zoo_span.sh`).

4.9 Use of large language models

No large language model is part of the scientific pipeline: no result, number, figure or statistical decision in this paper is produced by one. A large language model (Anthropic Claude, through the Claude Code command-line interface) was used as a coding and drafting assistant under the authors' direction — writing and refactoring analysis, figure and job scripts, drafting and revising manuscript text, and producing internal referee simulations (`docs/NMI_REVIEW_*.md`) used only to prioritise revisions. Its use is recorded commit by commit in the repository's `Co-Authored-By` trailers. Every pre-registration in `docs/PLAN_*.md` was written and committed before the corresponding analysis ran, every reported number is regenerated from committed result files by the scripts named in each caption and checked by `tmp_analysis/verify_new_claims.py`, and the authors take full responsibility for all content, including any text or code an assistant drafted.

Data availability

The Natural Scenes Dataset (Allen et al., 2022) is available at naturalscenesdataset.org under its own data-use agreement, which requires registration; we used the `betas_fithrf_GLMdenoise_RR` (b3) preparation, the `nsdgeneral` mask, the `prf-visualrois` and `prf-eccrois` labels and the pRF estimates as distributed with NSD. The Geirhos cue-conflict stimuli and the ten observers' trial-level responses are distributed at github.com/rgeirhos/texture-vs-shape (directory `raw-data/style-transfer-512-nomask-experiment/`) under CC BY 4.0; the same experiment is mirrored at github.com/bethgelab/model-vs-human, and the two mirrors were parsed by two independently written scripts that agree to four decimals on the human reference. MiniEcoset is at osf.io/msna2 (Thorat & Kietzmann, 2024). The accuracy gate of the ImageNet networks uses a stratified 10-per-class subset (10,000 images, drawn with seed 0 from the label file committed in the repository) of the ILSVRC-2012 validation set (Russakovsky et al., 2015), available from image-net.org under its terms of access, which require registration and non-commercial research use. The 39 torchvision networks use its published weights (38 `IMAGENET1K_V1`, one `IMAGENET1K_V2`), downloaded through `torchvision.models` (BSD-3-Clause). The three Stylized-ImageNet ResNet-50s are the published checkpoints of Geirhos et al. (2018a) as linked from github.com/rgeirhos/texture-vs-shape (bitbucket.org/robert_geirhos/texture-vs-shape-pretrained-models, fetched 2026-09-15; file names with their hash prefixes in `src/analysis/generality/models.py`). Derived result tables and fitted encoding-model summaries sufficient to reproduce every figure and Extended Data table are in the code repository, and will be deposited with a DOI on acceptance. Outputs derived from NSD at the level of voxels (the per-voxel encoding arrays that Fig. 3 reads) are not redistributed in the repository. They are deposited under the NSD data-use terms, so access requires the same NSD agreement. ROI-level summaries, which identify no participant and reproduce no NSD data, are in the repository.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article; a draft is `docs/REPORTING_SUMMARY.md` in the repository.

Code availability

Training, front-end, encoding, behavioural and analysis code is at github.com/LuZeking/Magno-Parvo-Konio-Net (branch `fusion-stage1`, commit as stamped on this manuscript's title page), under the BSD-3-Clause licence, with a pinned environment manifest (`requirements.txt`, Python 3.11) and the committed result files behind every figure; the figures regenerate with the scripts in `tmp_analysis/figs_v2/`, and the manuscript's numbers are locked by `tmp_analysis/verify_v3_claims.py`, which re-derives every number stated for the model and its controls from the committed result files and then runs `tmp_analysis/verify_new_claims.py` as its block for the BA network and the model zoo (Fig. 6, Extended Data Tables 3 and 4, Supplementary Tables S11 and S12). From a clean clone it runs 1,210 checks in under a minute and lists as skipped the nine blocks (four for the model, five for the BA network) that need the NSD-derived per-voxel encoding and readout-weight outputs. Those outputs are not redistributed in the repository; they are deposited alongside the derived tables under the same terms as the NSD data they derive from, and with the model's in place 1,515 checks run (the BA network's five per-voxel blocks run once its own deposited outputs are in `results/encoding_cc/`). The BA study's 21 trained networks (final and best saved checkpoints, $\approx$ 17 GB) and the model's and its controls' final checkpoints, which Figs. 4 and 6 and Extended Data Tables 2 and 3; Supplementary Tables S9–S12 read, are not in the repository; they will be deposited in the same DOI archive at acceptance and are available to referees through the private link. A private link is available to referees on request; a DOI-archived release accompanies acceptance.

Ethics and human participants

This study collected no new human data. Both human datasets are existing, de-identified public releases analysed under their own terms. NSD was collected under the approval of the University of Minnesota institutional review board with written informed consent from all eight participants (Allen et al., 2022); the behavioural data were collected under the approval described in Geirhos et al. (2018a). Secondary analysis of two de-identified public datasets required no additional approval under our institution's policy, and no participant-identifying information was accessed.

Sex and demographics. Participant sex was not under our control and is not a design factor; no analysis is disaggregated by sex, and at eight participants and ten observers none could be, because the exact tests used already sit at their resolution floor on the full samples. We do not restate sex or age: the Geirhos per-observer files carry only trial-level fields, so no per-observer demographic exists in the data we analysed, and for NSD we have not verified values against the released subject-information file and therefore point to the NSD release rather than transcribe from prose.

Constraints on generality. The neuroimaging sample is eight adults with normal or corrected-to-normal vision in the United States; the behavioural reference is ten adults tested in a German university laboratory. Both are small, Western, convenience-recruited adult samples. Claims about "human observers" should be read as claims about these samples; nothing here establishes cross-cultural, developmental or clinical generality, and the developmental motivation of the model's front end is a motivation for the model, not a claim about any developmental population we measured.

Author contributions

[To be completed at submission per Nature-portfolio policy: conceptualisation, model and train-

ing code, encoding analyses, behavioural analyses, figures, writing, supervision. The originating thesis (Manocchio, 2026) supplied the architecture, the front-end conditions and the first behavioural and RSA analyses; the seed replications, encoding models, controls and manuscript were developed subsequently.]

Competing interests

The authors declare no competing interests.

Extended Data tables

Extended Data Table 1 | The cortical gradient hV4 − mean(V1v, V1d), per seed and statistic. "normalised" is the per-voxel index (U_m − U_p)/(U_m + U_p), "raw" the per-voxel difference U_m − U_p, each summarised by its median over reliable voxels per ROI. Mean over 8 subjects of the per-subject contrast. "sep." is the number of subjects in which the model's seed is more negative than the control seed of the same index (against the control's seed mean it is also 8/8, p = .0078, for every seed and both statistics), and for the rows with no paired comparison the number in which the contrast is negative; p is the exact Wilcoxon, paired against the control and one-sample otherwise (floor .0078). The BA network's rows are those of PAPER_v2 (its no-diet arm: −0.078, −0.096 and −0.049 normalised; −0.027, −0.031 and −0.023 raw). The channel swap (the ramp on L−M, luminance untransformed) gives −0.015, −0.016 and −0.025 normalised (seeds 0–2) and +0.002 raw on the seed mean; alone, on the per-subject seed mean, 6/8 and 4/8 subjects below zero (p = .31 and .95). Sources: `tmp_analysis/fs2_cortex_summary.py` (normalised) and the per-voxel fits read with the verifier's `roi_profile` (raw), outside the repository.

statistic	seed	model (developmental order)	control (adult input from the start)	BA network (fixed diet)	sep.	p
normalised	0	−1.259	−0.062	−0.833	8/8	0.0078
	1	−1.291	−0.062	−0.785	8/8	0.0078
	2	−1.262	−0.086	−0.745	8/8	0.0078
raw	0	−0.186	−0.003	−0.166	8/8	0.0078
	1	−0.193	−0.003	−0.145	8/8	0.0078
	2	−0.190	−0.007	−0.153	8/8	0.0078
control alone (seed mean), normalised / raw	—		−0.070 / −0.004		6/8 / 4/8 < 0	0.11 / 0.64
untrained, the model's front end, normalised / raw	—	−0.480 / −0.195			8/8 < 0	0.0078
untrained, BA diet front end, normalised / raw	—			−0.544 / −0.120	8/8 < 0	0.0078

statistic	seed	model (developmental order)	control (adult input from the start)	BA network (fixed diet)	sep.	p
untrained, no-diet front end, normalised	—			−0.479	8/8 < 0	0.0078

Extended Data Table 2 | The readout weight cortex prefers against the one behaviour prefers. For each mixing weight w the classifier's own input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ (y_1 magno stream output, y_2 parvo stream output, 512-d each) is the design of a ridge encoding model of every nsdgeneral voxel (same split, folds and alpha grid as the banded model; Methods 4.5). w*_cortex is the w that maximises the median test R^2 over reliable voxels of the ROI (mean over seeds per subject; median [range] over 8 subjects). "R^2 at w = 0.5 / max" is the median R^2 at the trained readout weight relative to the best w. "concat" is the median R^2 of the 1,024-d concatenation [y_1, y_2], fitted as a reference and not as a bound (Methods 4.5); w*_cortex is read as the location of an optimum and never as an R^2 difference. What the table supports is the ordinal contrast: extrastriate below V1 in 8/8 subjects for the model ($\Delta = -0.323$ [−0.371, −0.269], 10,000-sample percentile bootstrap over subjects, exact Wilcoxon p = .0078) and reversed in the control (+0.308 [+0.194, +0.425], positive in 8/8 subjects, p = .0078) and in the channel swap (+0.358 [+0.221, +0.479], positive in 7/8 subjects and tied in one, p = .016, the tie dropped); the BA network gave 0.58 in V1 and 0.33–0.35 beyond it ($\Delta = -0.252$, 8/8; PAPER_v2). "Voxels magno / parvo" is the fraction of the ROI's voxels whose own best w is ≥ 0.75 / ≤ 0.25; "rate at w*_cortex" is the cue-conflict shape bias the network's own sweep gives at that weight. The behaviour-optimal weight (trial fit to each of 10 human observers' cue-conflict choices, model final checkpoints, 3 seeds) is given in the last row; it is not identified — the per-fit bootstrap intervals average 0.70 wide and 10 of 30 fits beat a label-permutation null — so no contrast with w*_cortex is tested (`results/fit_w_observers/`). Source: the per-subject readout-weight fits (outside the repository), reduced by the rule of `tmp_analysis/figs_v2/readout_w_cortex.py`.

arm	ROI	w*_cortex median [range]	R^2 at w = 0.5 / max	max R^2 (median)	concat R^2	voxels best w ≥ 0.75	voxels best w ≤ 0.25	rate at w*_cortex	n
model	V1	0.65 [0.50, 0.75]	0.989	0.1219	0.1313	0.41	0.11	0.75	8
model	V2	0.55 [0.45, 0.60]	0.997	0.0920	0.1018	0.26	0.20	0.70	8
model	V3	0.45 [0.35, 0.50]	0.998	0.0819	0.0877	0.17	0.29	0.66	8
model	hV4	0.33 [0.10, 0.55]	0.980	0.0912	0.0977	0.09	0.40	0.59	8
model	ventral (VO, PHC)	0.30 [0.20, 0.35]	0.985	0.1603	0.1642	0.03	0.41	0.58	8

arm	ROI	w*_cortex median [range]	R^2 at w = 0.5 / max	max R^2 (median)	concat R^2	voxels best w ≥ 0.75	voxels best w ≤ 0.25	rate at w*_cortex	n
model	lateral (LO, TO)	0.35 [0.20, 0.45]	0.980	0.1201	0.1234	0.05	0.44	0.61	8
model	dorsal (V3A/B, IPS0–2)	0.35 [0.25, 0.40]	0.990	0.0837	0.0861	0.07	0.39	0.61	8
control	V1	0.43 [0.35, 0.50]	0.996	0.1096	0.1185	0.21	0.23	0.37	8
control	V2	0.40 [0.20, 0.45]	0.982	0.0890	0.0990	0.18	0.32	0.37	8
control	V3	0.43 [0.30, 0.45]	0.993	0.0784	0.0863	0.20	0.28	0.37	8
control	hV4	0.50 [0.40, 0.60]	0.998	0.0920	0.0996	0.25	0.18	0.37	8
control	ventral (VO, PHC)	0.68 [0.60, 1.00]	0.990	0.1647	0.1654	0.50	0.04	0.38	8
control	lateral (LO, TO)	0.62 [0.45, 1.00]	0.986	0.1239	0.1273	0.50	0.07	0.38	8
control	dorsal (V3A/B, IPS0–2)	0.78 [0.50, 1.00]	0.984	0.0872	0.0869	0.50	0.08	0.38	8
channel swap	V1	0.38 [0.30, 0.45]	0.987	0.1085	0.1149	0.18	0.32	0.39	8
channel swap	V2	0.35 [0.25, 0.40]	0.978	0.0868	0.0951	0.16	0.38	0.39	8
channel swap	V3	0.28 [0.25, 0.45]	0.979	0.0805	0.0876	0.19	0.38	0.39	8
channel swap	hV4	0.40 [0.25, 0.55]	0.989	0.0910	0.0982	0.23	0.30	0.39	8
channel swap	ventral (VO, PHC)	0.65 [0.45, 1.00]	0.996	0.1612	0.1603	0.49	0.09	0.39	8
channel swap	lateral (LO, TO)	0.62 [0.30, 1.00]	0.996	0.1267	0.1272	0.46	0.15	0.39	8
channel swap	dorsal (V3A/B, IPS0–2)	1.00 [0.40, 1.00]	0.994	0.0839	0.0835	0.49	0.14	0.40	8

arm	ROI	w*_cortex median [range]	R^2 at w = 0.5 / max	max R^2 (median)	concat R^2	voxels best w ≥ 0.75	voxels best w ≤ 0.25	rate at w*_cortex	n
behaviour (model)	10 human observers, trial fit	0.79 [0.33, 0.97]	—	—	—	—	—	0.80	10

Extended Data Table 3 | The readout span without a stream split: one gain on a network's own shape- and texture-predictive subspaces, head fixed. For each frozen network with logits = W·y + b, the 15 discriminant directions of a 16-class shape LDA and of a 16-class texture LDA on the penultimate features y (PCA to 256 dimensions, Ledoit–Wolf shrinkage; fitted on one stratified half of the 1,200 cue-conflict images) are made mutually orthogonal by symmetric (Löwdin) orthogonalisation, giving orthonormal bases B_S and B_T (15 directions each) and the residual B_R. logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b with the network's own head untouched; w = 0.5 is the trained network. The rate is scored on the other half of the images (10 random splits, both directions; mean, with the range over splits and seeds in brackets), by the paper's primary estimator (112-class, decided images) for our networks and the Geirhos 16-class probability aggregation for the ImageNet networks; "forced-estimator span" is the same sweep on the forced two-way estimator. "gated span" is max − min of the seed-mean held-out curve over the w at which top-1 of the gained features (bases fitted on all 1,200 images) on held-out natural images (MiniEcoset validation for our networks; a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks) stays within 5 points of the trained value, followed by that w-range and the largest change of top-1 anywhere in the sweep. The two stratified halves share object and texture exemplars (98% and 100% of scored images; the 1,200 images reuse 160 object and 48 texture exemplars), so the span is also given with exemplars held out: fitted on images whose object and texture exemplars both lie outside one of three exemplar folds and scored on the images whose exemplars both lie inside it, ten repetitions, ungated (`docs/PLAN_EXEMPLAR_2026-09-21.md`; `results/figures_v2/exemplar_zoo.json`, `exemplar_streams.json`). "refit span" is the block-form lossy sum z(w) = [w·s + (1 − w)·t, $B_R^T y$], s = $B_S^T y$ and t = $B_T^T y$ paired by discriminant rank, with a linear and a two-layer head refitted at each w (Methods 4.3 budget), our networks only; in parentheses the largest shortfall of validation accuracy, over w, from a head refitted on the unmixed features. "variance in S / T" is the fraction of the feature variance on the fitting half lying in each subspace; "order sensitivity" replaces the symmetric orthogonalisation by sequential Gram–Schmidt with the shape basis or the texture basis first. "median-split gated span" is the gated span of the pre-specified unit-level rule of Supplementary Table S11 on the same networks and images. For the BA network's two-stream concatenation, which has no held-out natural-image features, the fraction of the shape basis's energy lying on stream-1 units (all-image bases; range over seeds) replaces the gate; for the model's, gated on MiniEcoset validation, it is given after the gate. Pre-specification: `docs/PLAN_SUBSPACE_2026-09-15.md` (predictions P-A, P-A′, P-B, P-D). Source: `results/figures_v2/subspace_behaviour.json`, aggregated by `tmp_analysis/figs_v2/subspace_behaviour.py` from the outputs of `src/analysis/generality/subspace_factor.py`, `gain_accuracy.py` and `refit_sum.py` (`slurm/subspace_{cc,refit}.sh`); the per-network files are outside the repository.

network	D	trained rate	w = 0	w = 1	span, exemplars shared [range]	span, exemplars held out [seed range]	ρ(w, rate)	gated span (w-range; largest change of top-1)	forced-estimator span	refit span linear / two-layer (largest shortfall of top-1)	variance in S / T	order sensitivity, S first / T first	median-split gated span	seeds
two-stream, BA diet, streams concatenated (validation)	1024	0.675	0.495	0.801	0.320 [0.18, 0.45]	0.249 [0.24, 0.27]	0.98	— (shape basis on stream 1: 0.63–0.70)	0.216	—	0.11 / 0.29	0.35 / 0.25	—	3
two-stream, model (developmental order), streams concatenated (validation)	1024	0.682	0.515	0.800	0.286 [0.25, 0.32] (seed range)	0.202 [0.17, 0.23]	0.99	0.25 (0.07–0.93; 7.0 pts; shape basis on stream 1: 0.58–0.62)	0.181	—	0.11 / 0.28	0.28 / 0.23	0.26	3
single ResNet-34, diet	512	0.625	0.516	0.735	0.226 [0.12, 0.37]	0.173 [0.14, 0.20]	0.98	0.19 (0.10–0.90; 11.9 pts)	0.163	0.08 / 0.10 (0.8 pts)	0.14 / 0.20	0.24 / 0.20	0.07	3
single ResNet-18, diet	512	0.557	0.433	0.674	0.249 [0.13, 0.41]	0.200 [0.18, 0.23]	0.98	0.14 (0.25–0.73; 16.7 pts)	0.189	0.08 / 0.10 (0.6 pts)	0.15 / 0.23	0.25 / 0.20	0.09	3

network	D	trained rate	w = 0	w = 1	span, exemplars shared [range]	span, exemplars held out [seed range]	ρ(w, rate)	gated span (w-range; largest change of top-1)	forced-estimator span	refit span linear / two-layer (largest shortfall of top-1)	variance in S / T	order sensitivity, S first / T first	median-split gated span	seeds
single ResNet-18, no diet (`rgb_tsn`)	512	0.445	0.311	0.651	0.344 [0.22, 0.47]	0.279 [0.23, 0.32]	0.99	0.22 (0.20–0.78; 13.4 pts)	0.242	0.13 / 0.15 (0.5 pts)	0.13 / 0.26	0.29 / 0.34	0.08	3
ImageNet ViT-B/16	768	0.399	0.287	0.570	0.282 [0.22, 0.34]	0.180	1.00	0.28 (0.00–1.00; 0.2 pts)	0.318	—	0.11 / 0.23	0.22 / 0.31	0.08	1
ImageNet ResNet-152	2048	0.293	0.209	0.455	0.246 [0.20, 0.33]	0.166	1.00	0.25 (0.00–1.00; 1.2 pts)	0.271	—	0.07 / 0.24	0.21 / 0.27	0.16	1
ImageNet ConvNeXt-B	1024	0.269	0.178	0.512	0.334 [0.29, 0.39]	0.196	1.00	0.33 (0.00–1.00; 0.2 pts)	0.371	—	0.08 / 0.33	0.31 / 0.35	0.11	1
ImageNet ConvNeXt-S	768	0.248	0.170	0.458	0.287 [0.24, 0.34]	0.156	1.00	0.29 (0.00–1.00; 0.1 pts)	0.353	—	0.08 / 0.33	0.27 / 0.30	0.12	1
ImageNet ResNet-50	2048	0.222	0.157	0.447	0.290 [0.21, 0.34]	0.208	1.00	0.29 (0.00–1.00; 1.5 pts)	0.310	—	0.07 / 0.28	0.28 / 0.29	0.13	1
ImageNet VGG-16	4096	0.110	0.061	0.329	0.268 [0.23, 0.31]	0.138	1.00	0.27 (0.00–1.00; 0.5 pts)	0.322	—	0.04 / 0.28	0.26 / 0.27	0.09	1

Extended Data Table 4 | The readout span across the field's model zoo. Frozen ImageNet networks under the subspace rule of Extended Data Table 3 (PCA to 256 dimensions, 16-class shape and texture LDA, 15 directions each, symmetric orthogonalisation; bases fitted on one stratified half of the 1,200 cue-conflict images and the rate scored on the other, 10 splits both directions),

each network's own head untouched, the Geirhos 16-class probability aggregation throughout; "images decided" is the number of the 1,200 assigned to their shape or their texture category at the trained readout, the denominator of the rate. "gate" is the w-range over which top-1 on a stratified 10-per-class ILSVRC-2012 validation subset (10,000 images; bases fitted on all cue-conflict images) stays within 5 points of the trained value, with the largest change of top-1 inside it; "texture end / shape end" are the held-out rates at the smallest and the largest gated w, and "gated span" is their difference, under the subspace rule and under the pre-specified unit-level median split of Supplementary Table S11. "probe" is the linear probe of Methods 4.3 (PCA to 256 dimensions, multinomial logistic regression, 5-fold stratified cross-validation) predicting the shape class and, separately, the texture class from the same features; bias is shape minus texture accuracy (chance 0.0625). The stratified halves and the probe's folds share object and texture exemplars (the 1,200 images reuse 160 object and 48 texture exemplars), so both are also given with exemplars held out (`docs/PLAN_EXEMPLAR_2026-09-21.md`; `results/figures_v2/exemplar_zoo.json`): "gated span, exemplars held out" fits the subspaces on images whose object and texture exemplars both lie outside one of three exemplar folds and scores the rate on the images whose exemplars both lie inside it (ten repetitions); "probe, exemplars held out" is the exemplar-balanced probe of Methods 4.3 (three exemplars per category for both labels, leave one exemplar out, PCA inside the folds, mean of 20 draws). With exemplars held out, over the 38 ImageNet-trained networks: gated span 0.08–0.29 (median 0.19; 0.18 with the gate recomputed on the held-out fold bases, as in Results 2.4 and Methods 4.8), ≥ 0.20 in 11; Spearman ρ with the trained rate +0.10 (+0.17 on the fold-basis gate); 18.6% of the 703 pairs reverse between the trained ranking and the shape-end ranking (forced two-way 16.2%), against 8.9% [6.0%, 12.2%] between two bootstrap resamples of the images at the trained readout; probe bias -0.52 to +0.01 (median -0.36), the 5–95% interval over draws below zero in 37. Preprocessing is Resize 256 / CenterCrop 224 except for Inception-v3 (342/299), EfficientNet-B3 (320/300) and EfficientNet-V2-S (384/384), which were not trained at 224. "κ with observers" is error consistency (Cohen's κ on trial-by-trial agreement, Results 2.5; Supplementary Table S6) with each of the ten human observers, mean over observers, as a percentage of the human–human ceiling $\kappa = 0.234$, at the trained readout and, in parentheses, at the texture end and the shape end of the gate under the same gain; computed under this pipeline's preprocessing, so the VGG-16 value differs from the published response file's 7.3% (Supplementary Table S6) as its trained rate does. Over the 38 ImageNet-trained networks: Kendall τ between the ranking by trained rate and by the shape-end rate 0.47 (by the texture-end rate 0.78); 188 of 703 pairs (27%) reverse order; gated span 0.20–0.50 (median 0.30), ≥ 0.20 in 97% and the whole sweep inside the gate in 100%; Spearman ρ between gated span and trained rate +0.04, and trained top-1 -0.22; error consistency at the trained readout 2.4–23.0% of the ceiling (median 7.9%), at most 36.7% anywhere inside a gate, changing by a median of +13.5 points from the texture end to the shape end; $\rho(\kappa$, trained rate) +0.80; Kendall τ between trained rate and probe bias 0.44. The three Stylized-ImageNet networks are the published checkpoints of Geirhos et al. (2018a), all ResNet-50: trained on Stylized-ImageNet (SIN), on SIN + IN, and on SIN + IN then fine-tuned on IN. Pre-specification: `docs/PLAN_ZOO_2026-09-15.md` (predictions P-Z1–P-Z4). Source: `results/figures_v2/zoo_behaviour.json`, aggregated by `tmp_analysis/figs_v2/zoo_behaviour.py` from the outputs of `slurm/zoo_cc.sh`; the per-network files are outside the repository.

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ImageNet-trained (torchvision IMAGENET1K_V1)													
ViT-B/32	768	0.758	0.617 (767)	0.00–1.00 (0.2 pts)	0.499 / 0.738	0.239	0.161	1.00	0.069	0.67 / 0.97 (-0.30)	0.46 / 0.46 (+0.01)	23.0 (12.9 → 34.0)	yes
ViT-B/16	768	0.815	0.399 (818)	0.00–1.00 (0.2 pts)	0.287 / 0.570	0.282	0.180	1.00	0.085	0.56 / 0.99 (-0.43)	0.38 / 0.57 (-0.19)	11.6 (7.2 → 19.2)	yes
ViT-L/16	1024	0.801	0.319 (885)	0.00–1.00 (0.2 pts)	0.207 / 0.545	0.337	0.193	1.00	0.078	0.57 / 0.99 (-0.43)	0.35 / 0.58 (-0.23)	11.5 (5.9 → 21.9)	yes
Inception v3	2048	0.759	0.305 (797)	0.00–1.00 (0.6 pts)	0.241 / 0.487	0.246	0.134	0.99	0.099	0.40 / 0.96 (-0.56)	0.29 / 0.63 (-0.35)	11.0 (5.5 → 22.5)	yes
MaxViT-T	512	0.842	0.295 (899)	0.00–1.00 (0.2 pts)	0.230 / 0.469	0.239	0.142	1.00	0.114	0.40 / 0.97 (-0.57)	0.22 / 0.60 (-0.38)	9.1 (4.5 → 20.7)	yes
ResNeXt-101 32x8d	2048	0.796	0.294 (732)	0.00–1.00 (0.6 pts)	0.228 / 0.511	0.283	0.184	0.99	0.220	0.47 / 1.00 (-0.53)	0.30 / 0.61 (-0.32)	7.9 (3.9 → 16.3)	yes
ConvNeXt-L	1536	0.848	0.293 (924)	0.00–1.00 (0.1 pts)	0.196 / 0.528	0.332	0.189	1.00	0.079	0.51 / 1.00 (-0.49)	0.28 / 0.66 (-0.38)	9.2 (4.4 → 18.1)	yes

network	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ResNet-152	0.785	0.293 (781)	0.00–1.00 (1.2 pts)	0.209 / 0.455	0.246	0.165	1.00	0.159	0.51 / 1.00 (-0.49)	0.31 / 0.64 (-0.33)	9.1 (2.1 → 16.0)	yes
EfficientNet-B3	0.818	0.291 (905)	0.00–1.00 (0.2 pts)	0.204 / 0.514	0.310	0.191	1.00	0.141	0.49 / 0.99 (-0.50)	0.34 / 0.74 (-0.39)	12.5 (6.8 → 26.0)	yes
ResNet-101	0.777	0.284 (770)	0.00–1.00 (1.4 pts)	0.216 / 0.493	0.276	0.206	1.00	0.143	0.48 / 1.00 (-0.52)	0.30 / 0.60 (-0.30)	8.7 (5.1 → 14.8)	yes
Wide-ResNet-101-2	0.790	0.279 (746)	0.00–1.00 (0.7 pts)	0.201 / 0.431	0.231	0.155	1.00	0.119	0.39 / 0.99 (-0.60)	0.26 / 0.57 (-0.31)	7.9 (3.4 → 18.5)	yes
DenseNet-201	0.764	0.273 (792)	0.00–1.00 (1.5 pts)	0.206 / 0.540	0.334	0.192	1.00	0.085	0.51 / 1.00 (-0.49)	0.31 / 0.59 (-0.28)	8.5 (3.0 → 18.2)	yes
EfficientNet-B0	0.777	0.271 (835)	0.00–1.00 (0.4 pts)	0.175 / 0.502	0.327	0.193	1.00	0.110	0.45 / 0.99 (-0.54)	0.31 / 0.67 (-0.36)	7.1 (3.6 → 19.0)	yes
ConvNeXt-B	0.843	0.269 (921)	0.00–1.00 (0.2 pts)	0.178 / 0.512	0.334	0.195	1.00	0.114	0.53 / 1.00 (-0.47)	0.25 / 0.70 (-0.45)	8.0 (2.1 → 21.6)	yes
MobileNet-V3-L	0.741	0.265 (796)	0.00–1.00 (1.2 pts)	0.156 / 0.591	0.435	0.263	1.00	0.102	0.53 / 1.00 (-0.47)	0.33 / 0.63 (-0.29)	10.8 (4.0 → 29.0)	yes

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
AlexNet	4096	0.552	0.261 (735)	0.00–1.00 (1.4 pts)	0.120 / 0.624	0.504	0.293	1.00	0.142	0.55 / 1.00 (-0.45)	0.34 / 0.58 (-0.24)	7.1 (3.0 → 14.3)	yes
RegNet-Y-16GF	3024	0.808	0.261 (797)	0.00–1.00 (0.6 pts)	0.180 / 0.480	0.300	0.174	1.00	0.208	0.50 / 1.00 (-0.50)	0.33 / 0.63 (-0.30)	7.7 (4.6 → 15.3)	yes
ResNet-34	512	0.732	0.259 (791)	0.00–1.00 (2.7 pts)	0.175 / 0.528	0.353	0.227	1.00	0.084	0.47 / 1.00 (-0.53)	0.24 / 0.56 (-0.31)	6.1 (3.5 → 15.9)	yes
GoogLeNet	1024	0.691	0.251 (876)	0.00–1.00 (1.4 pts)	0.187 / 0.483	0.296	0.189	1.00	0.056	0.47 / 0.99 (-0.52)	0.29 / 0.61 (-0.32)	9.5 (5.4 → 22.1)	yes
ConvNeXt-S	768	0.840	0.248 (901)	0.00–1.00 (0.1 pts)	0.170 / 0.458	0.287	0.155	1.00	0.124	0.47 / 1.00 (-0.53)	0.23 / 0.67 (-0.44)	8.1 (3.6 → 16.9)	yes
RegNet-Y-8GF	2016	0.802	0.244 (788)	0.00–1.00 (0.8 pts)	0.151 / 0.516	0.365	0.241	1.00	0.140	0.50 / 1.00 (-0.50)	0.31 / 0.67 (-0.36)	6.7 (1.8 → 15.9)	yes
ConvNeXt-T	768	0.830	0.241 (924)	0.00–1.00 (0.4 pts)	0.154 / 0.507	0.353	0.200	1.00	0.125	0.50 / 1.00 (-0.50)	0.29 / 0.71 (-0.42)	7.0 (2.4 → 22.3)	yes
Swin-S	768	0.836	0.236 (880)	0.00–1.00 (0.2 pts)	0.154 / 0.440	0.287	0.169	1.00	0.077	0.44 / 0.99 (-0.55)	0.25 / 0.68 (-0.42)	8.8 (5.3 → 16.6)	yes

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ShuffleNet V2 x1.0	1024	0.686	0.234 (781)	0.00–1.00 (2.5 pts)	0.146 / 0.527	0.381	0.245	1.00	0.034	0.57 / 1.00 (-0.43)	0.34 / 0.60 (-0.26)	9.2 (4.0 → 25.7)	yes
Swin-B	1024	0.839	0.233 (891)	0.00–1.00 (0.1 pts)	0.177 / 0.406	0.229	0.128	1.00	0.071	0.42 / 1.00 (-0.57)	0.23 / 0.63 (-0.40)	7.7 (4.0 → 17.0)	yes
ResNet-18	512	0.695	0.233 (849)	0.00–1.00 (3.8 pts)	0.163 / 0.479	0.316	0.195	1.00	0.050	0.47 / 0.99 (-0.53)	0.25 / 0.59 (-0.34)	6.3 (3.2 → 16.2)	yes
Wide-ResNet-50-2	2048	0.785	0.230 (792)	0.00–1.00 (0.6 pts)	0.166 / 0.427	0.261	0.177	0.99	0.105	0.38 / 0.99 (-0.62)	0.23 / 0.60 (-0.37)	5.0 (2.4 → 11.6)	yes
MNASNet 1.0	1280	0.734	0.226 (766)	0.00–1.00 (1.3 pts)	0.134 / 0.530	0.395	0.251	1.00	0.042	0.47 / 1.00 (-0.53)	0.29 / 0.66 (-0.37)	8.7 (3.7 → 19.7)	yes
Swin-T	768	0.820	0.225 (902)	0.00–1.00 (0.2 pts)	0.150 / 0.460	0.309	0.165	1.00	0.084	0.48 / 1.00 (-0.52)	0.29 / 0.68 (-0.39)	8.4 (4.2 → 18.4)	yes
DenseNet-121	1024	0.740	0.224 (839)	0.00–1.00 (1.9 pts)	0.147 / 0.494	0.347	0.211	1.00	0.071	0.45 / 1.00 (-0.55)	0.23 / 0.56 (-0.33)	5.7 (3.7 → 14.9)	yes
ResNet-50	2048	0.760	0.222 (814)	0.00–1.00 (1.5 pts)	0.157 / 0.447	0.290	0.203	1.00	0.130	0.43 / 1.00 (-0.56)	0.29 / 0.63 (-0.34)	5.1 (2.7 → 12.2)	yes

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ResNeXt-50 32x4d	2048	0.774	0.219 (776)	0.00–1.00 (0.9 pts)	0.145 / 0.461	0.316	0.201	1.00	0.193	0.41 / 0.99 (-0.58)	0.25 / 0.65 (-0.39)	5.6 (2.1 → 14.1)	yes
EfficientNet-V2-S	1280	0.832	0.184 (879)	0.00–1.00 (0.2 pts)	0.134 / 0.332	0.198	0.084	1.00	0.074	0.35 / 0.99 (-0.64)	0.22 / 0.74 (-0.52)	5.6 (3.6 → 8.1)	yes
MobileNet-V2	1280	0.715	0.184 (798)	0.00–1.00 (4.4 pts)	0.128 / 0.474	0.346	0.186	1.00	0.051	0.48 / 1.00 (-0.52)	0.27 / 0.65 (-0.38)	5.6 (1.9 → 18.0)	yes
VGG-19-BN	4096	0.743	0.150 (867)	0.00–1.00 (0.8 pts)	0.094 / 0.335	0.241	0.149	1.00	0.093	0.37 / 0.99 (-0.62)	0.27 / 0.70 (-0.43)	4.0 (2.2 → 6.8)	yes
VGG-16-BN	4096	0.733	0.115 (915)	0.00–1.00 (0.7 pts)	0.068 / 0.294	0.226	0.130	1.00	0.104	0.36 / 1.00 (-0.64)	0.24 / 0.70 (-0.45)	2.4 (2.9 → 5.2)	yes
VGG-16	4096	0.711	0.110 (918)	0.00–1.00 (0.5 pts)	0.061 / 0.329	0.268	0.137	1.00	0.093	0.33 / 0.99 (-0.66)	0.23 / 0.72 (-0.50)	3.4 (2.3 → 6.4)	yes
VGG-11-BN	4096	0.703	0.105 (927)	0.00–1.00 (1.0 pts)	0.059 / 0.294	0.235	0.110	1.00	0.092	0.42 / 1.00 (-0.57)	0.27 / 0.69 (-0.42)	4.2 (2.3 → 4.8)	yes

networkID	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
same architecture, different training recipe												
ResNet-204850 (V2 recipe)	0.814	0.185 (874)	0.00–1.00 (0.2 pts)	0.131 / 0.465	0.334	0.195	1.00	0.056	0.38 / 1.00 (-0.62)	0.26 / 0.72 (-0.47)	0.5 (1.9 → 7.8)	yes
Stylized-ImageNet controls (Geirhos et al., 2018a)												
ResNet-204850 SIN	0.582	0.810 (758)	0.00–1.00 (3.9 pts)	0.701 / 0.867	0.166	0.124	0.99	0.140	0.79 / 0.90 (-0.11)	0.54 / 0.30 (+0.23)	39.5 (28.7 → 49.7)	yes
ResNet-204850 SIN+IN	0.743	0.363 (802)	0.00–1.00 (2.4 pts)	0.256 / 0.606	0.350	0.238	1.00	0.170	0.59 / 0.99 (-0.40)	0.39 / 0.53 (-0.14)	11.7 (7.1 → 18.5)	yes
ResNet-204850 SIN+IN, ft IN	0.770	0.227 (797)	0.00–1.00 (1.9 pts)	0.170 / 0.447	0.277	0.175	1.00	0.154	0.41 / 0.99 (-0.58)	0.26 / 0.60 (-0.33)	5.9 (4.6 → 14.7)	yes

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> **Supplementary material in this draft.** Supplementary Notes S1–S5 and Tables S1–S14 below are PAPER_v2's and describe the BA network (`dkl_bcs`) and the zoo; the main text cites them for the BA network only. Their rows for the model of record are being regenerated ⟦PENDING⟧.

Supplementary Note S1 — claims deliberately not made

- That the two-stream split is what makes shape bias a readout property: networks with no split show the same dissociation — a reciprocal gain on their shape- and texture-predictive subspaces moves the rate by 0.14–0.20 in six ImageNet networks within 3.5 points of top-1 on exemplars the subspaces never saw (0.25–0.33 with exemplars shared), and in our trunks extrastriate cortex loads on the texture-leaning half relative to V1 — with the cortical part failing under the unit-level rule in all three ImageNet networks fitted (ResNet-50, ViT-B/16 and ConvNeXt-B, the last two post hoc) (Fig. 6; Results 2.4).
- That the two-stream split causes a shape-bias advantage over a single stream: at matched capacity the arms do not separate at n = 3, and capacity alone moves the rate by a comparable amount (Results 2.4).
- Any dorsal/ventral stream mapping. The thesis's framing "magno→dorsal, parvo→ventral" was tested rather than dropped: the pre-declared dorsal−ventral interaction is +0.0245 on the diet model and +0.0122 ($p = .195$) without the low-pass front end, so it does not survive its own control. Read directly on the Wang atlas, dorsal (V3A/B, IPS0–2) and ventral (VO1/2, PHC1/2) regions are indistinguishable under both arms (dorsal − ventral +0.015 with the diet, +0.005 without, $p > .1$; Extended Data Fig. 4c): the diet's magno stream contributes no unique variance anywhere beyond V3.

- hMT+ specificity: at n = 8 the two statistics disagree (p = .195 normalised, .039 raw) and the effect is dead in the blur control.
- A foveal/peripheral headline: the foveal band's sign flips when the blur is removed ($-0.057 \rightarrow +0.051$), so it is a front-end artefact.
- That the chromatic stream predicts colour-biased cortex: pooled VO is +0.0045 (p = .46), bounded within ± 0.017; hV4's parvo preference is weak (-0.012, 2/8). Substituting the koniocellular axis makes the prediction worse, and worse specifically in the colour set (hV4 -0.029, VO -0.024, each p = .0078, 0/8).
- Where in cortex the diet's shape-bias effect lands: a diet-specific hV4/LO step does not survive the full control set.
- That integration produces human-like shape bias, or that the diet improves neural predictivity: it does neither.
- That the diet makes the model's image-level difficulty profile more human-like: the crossed 2×2 shows train/test matching, not a weights effect.
- That human cortex uses this mechanism: a single-stream account in which texture-like ventral representations feed shape-biased behaviour directly is not excluded by these data.

Why the three regional claims above failed together, and what it implies for the survivor: without the low-pass front end essentially every ROI is magno-preferring, so the blur manufactures apparent regional specificity by handicapping the magno stream until the parvo stream can win somewhere — the same general-factor pathology that defeated the thesis's RSA, reappearing in encoding form. Any regional claim on these models must be shown to survive removal of the blur, and the ventral gradient is reported because it does.

Supplementary Note S2 — the koniocellular control

The thesis omitted the koniocellular pathway for simplicity. Because L−M fails to predict colour-biased cortex, two readings were live: that the chromatic stream does not predict colour cortex, or that L−M is the wrong chromatic axis. Swapping S−(L+M) into stream 2 of the untrained network, contrast-matched to L−M per image, with the luminance stream bit-identical, makes the ventral prediction worse and worse where colour lives (seed 0: hV4 -0.029, VO -0.024, the pre-declared colour set -0.026, each p = .0078 with 0/8 subjects positive on both statistics; colour set minus the other 17 ROIs -0.014, p = .0078), unchanged when the channel keeps its native 2.05× contrast, and replicated at two further untrained seeds, each against the L−M network of its own seed (colour set -0.027 and -0.020, colour set minus the other ROIs -0.016 and -0.014, every contrast p = .0078 with 0/8 subjects positive on both statistics; `results/konio_verdict/`). Both cone-opponent axes have now been tested against colour-biased cortex and neither predicts it. These fits are untrained, the native-contrast variant was run at seed 0 only, no third stream was added, and no blur was applied to the S channel; whether a three-band model beats a two-band one is a separate question requiring three-way variance partitioning.

Supplementary Note S3 — the cortical gradient against its nuisance axes and the instrument's bias

Results 2.3 states the outcome; this note carries the analyses. It reports them first on the model and its controls (Extended Data Fig. 3; `tmp_analysis/fs2_cortex_controls.py`, `results/figures_v3/cortex_controls.json`), then, as PAPER_v2 reported them, on the BA network (Supplementary Tables S7 and S8), where "diet" and "no diet" name the BA network's own two arms.

The model against the nuisance axes. Within NSD's five eccentricity bands the model's gradient is negative in every band (6/7 subjects in the foveal band, where subject 02 has too few hV4 voxels in every arm and subject 05 is positive; 8/8 in the other four) and grows with eccentricity, from −0.58 in the foveal band to −1.33 beyond 4°. The control stays between −0.07 and +0.16 in every band and the channel swap between −0.06 and +0.05, and the model lies below the control in every band (7/7, p = .016, in the foveal band; 8/8, p = .0078, in the other four). Within pRF-size tertiles the model sits at −1.28, −1.26 and −1.32 against −0.13, −0.13 and −0.09 for the control (6/6, p = .031, in the smallest tertile, where subjects 03 and 08 have too few voxels in the model; 8/8, p = .0078, in the other two). On the raw statistic the model-minus-control contrasts have the same counts in every band and tertile. On the continuous predictors (z-scored log(eccentricity + 0.5) and log size within subject, pRF $R^2 > 5\%$, mean VIF 1.64), the model's stream preference moves toward parvo with increasing pRF size at fixed eccentricity (partial slope −0.118, 8/8 subjects, p = .0078) and toward magno with increasing eccentricity at fixed size (+0.163, 8/8, p = .0078), on the raw statistic too (−0.391 and +0.444, 8/8 each). The control shows no eccentricity effect (−0.030, 6/8 negative, p = .25) and a size effect of the opposite sign (+0.047, 8/8 positive, p = .0078), and so does the channel swap (−0.025, p = .31; +0.048, 7/8 positive, p = .016); the model minus the control is +0.193 on the eccentricity partial and −0.165 on the size partial (8/8 each, p = .0078). The untrained network on the shared front end already carries the dissociation (+0.083, 7/8, p = .023; −0.040, 8/8, p = .0078), as the BA network's untrained networks do. Training on the maturing input keeps it and strengthens its size term on both statistics (model minus untrained −0.078 normalised, −0.102 raw, 8/8 each, p = .0078), while its eccentricity term grows on the normalised statistic only (+0.080, 7/8, p = .023; raw −0.027, 4/8, p = .64). Training on the adult input from the first step, or on the maturing input routed to the chromatic stream, removes the eccentricity term and reverses the size term (raw +0.265 and +0.270, 8/8 each). Within the model, then, the ventral gradient and the pRF-size dependence are the same fact seen on two axes, as they are in the BA network, and neither control shows either.

The estimator's bias on the model. The simulation of the BA network (below) was rerun on seed 0 of the model and of the control. It uses each network's fitted 1,000-component feature spaces from the encoding cache, the measured per-ROI noise ceilings (0.41 in V1v to 0.32 in hV4, identical across arms) and a true magno − parvo difference solved to zero (`slurm/vp_recovery_fs2.sh`). - **The floor.** On the model's geometry the recovered contrast is +0.018 (5/8 subjects positive, p = .11) and on the raw statistic −0.001 (p = .31), so the floor is indistinguishable from zero. On the control's geometry it is +0.066 (8/8, p = .0078), and +0.004 on the raw statistic (8/8, p = .0078). Both floors are of the wrong sign to manufacture a descent. - **Unlike the BA network's two arms,** the two floors differ (Mann–Whitney p = .0003), so each arm is corrected by its own. - **The displacement.** The real seed-0 contrast minus each subject's own floor is −1.277 for the model and −0.127 for the control, each non-zero in 8/8 subjects (p = .0078). The model's displacement is 10.1 times the control's and below it in every subject (−1.150, p = .0078). - **The control's non-zero value rests on the correction alone.** Uncorrected, its seed-0 gradient is −0.062 (5/8 negative, p = .25), as the BA network's no-diet arm rested on it. - **Fidelity.** The simulation recovers a known across-ROI split in the right order ($\rho = +1.000$ in 8/8 subjects, both geometries) but overstates its size by a factor of 1.23 on the model's geometry and 1.25 on the control's. As for the BA network, the corrected values are read as sign and ordering, not as calibrated effect sizes.

The two nuisance axes that co-vary with hierarchical rank. Receptive-field size and eccentricity coverage both rise from V1 to hV4, and the diet's magno stream is a low-pass instrument by construction, so a stream-preference gradient that merely tracked receptive-field size would be expected from the front end alone. We tested this three ways (Extended Data Fig. 3; Supplementary Table S8). First, the gradient computed within each of NSD's five eccentricity bands — V1 and hV4

voxels compared only with voxels of the same eccentricity — is negative under the diet in every band (6/7 subjects in the foveal band, 8/8 in the other four) while the no-diet arm stays between −0.10 and +0.07 in every band; the diet arm's gradient does grow with eccentricity (−0.26 in the foveal band, −0.93 beyond 4°), so eccentricity modulates it, but the contrast is present at matched eccentricity and is therefore not an eccentricity contrast in disguise. Second, stratifying the same voxels by pRF size gives the same result within size tertiles (−0.71, −0.77 and −0.90 under the diet against −0.08 in each tertile without it). Third, regressing the per-voxel index on log pRF eccentricity and log pRF size jointly within subject (NSD's pRF estimates, voxels with pRF $R^2 > 5\%$; the two predictors are only modestly collinear, VIF 1.66) shows that under the diet, stream preference moves toward parvo with increasing pRF size at fixed eccentricity (partial slope −0.086, 8/8 subjects, $p = .0078$) and toward magno with increasing eccentricity at fixed size (+0.119, 8/8, $p = .0078$), while the no-diet arm shows neither the size effect in that direction (+0.035, opposite sign) nor any eccentricity effect (−0.004, $p = 1.00$); the arm interaction on the eccentricity partial is +0.134 (8/8, $p = .0078$). A learning-free Gabor bank on the diet front end carries no such structure (+0.035, $p = .15$), so it is not a property of the input filter alone; an untrained network carries it on either front end, however (diet front end +0.083, 7/8, $p = .016$ on the eccentricity partial and −0.038, 6/8, $p = .039$ on the size partial; undieted front end +0.089, 7/8, $p = .023$ and −0.038, 8/8, $p = .0078$), and training with the diet leaves it where random weights put it — trained minus untrained is +0.036 (6/8, $p = .11$) on the normalised statistic and −0.077 (2/8, $p = .11$) on the raw one, opposite in sign and neither significant — while the network trained without the diet carries none of it on either statistic (+0.028 raw, −0.004 normalised; the paired decrease from the untrained network is significant on the raw statistic only, −0.374, 0/8, $p = .0078$, against −0.087, 2/8, $p = .15$). The diet preserves a dissociation that random convolutional weights already produce; it neither installs nor amplifies it (Supplementary Table S8). Within the diet arm, then, the ventral gradient and the pRF-size dependence are the same fact seen on two axes — larger receptive fields, further along the hierarchy, are increasingly explained by the texture-biased stream — and the network trained without the diet shows neither.

The estimator's own bias points the wrong way. The mean noise ceiling declines from V1v to hV4 ($\rho = -0.89$ against rank), a nuisance falling along the axis of the effect, so we simulated the instrument: synthetic voxels from each arm's fitted feature spaces, the measured noise ceilings imposed, the true magno − parvo difference set to zero everywhere. The recovered contrast is a small, positive floor (+0.038 with the diet's feature geometry, +0.051 without, 8/8 subjects each), flat across ROIs (+0.025 to +0.027 per ROI), and indistinguishable from zero on the raw statistic — opposite in sign to the observed effect and without its gradient. Because the floor is a signed bias, the right comparison is a paired displacement: the real contrast minus each subject's own floor is −0.871 with the diet and −0.130 without, each non-zero in every subject, the diet arm's 6.7 times larger (Supplementary Table S7). The floor subtraction relocates the recovery simulation's residual misfit rather than removing it, and we do not correct that residual, because the correction is under-specified: six variants swing the no-diet arm between 4/8 and 6/8 subjects and between $p = .055$ and $p = 1.00$, while the diet arm is non-zero in all eight subjects at $p = .0078$ under all six (Supplementary Table S7). The no-diet arm's non-zero gradient therefore rests on the floor correction alone, and since the simulation is not magnitude-faithful (it recovers the true contrast with a slope of 1.21 to 1.28), we read −0.834 as a sign-and-ordering result, not a calibrated effect size. The raw statistic per ROI is drawn beside the normalised one in Extended Data Fig. 3a; the diet arm's normalised index approaches −0.84 at hV4, near the index's bound, and the raw profile carries the same ordering without that bound.

Supplementary Note S4 — passages condensed in the main text

The main text was shortened to the journal's length. Every paragraph it condensed is given here in its full, pre-condensation wording, under the section it came from, so no number, control or failed prediction was dropped. These passages are PAPER_v2's and describe the BA network: "the diet" in them is its fixed 48-month transform under the thesis's recipe, not the model's trajectory. Where they differ from the main text, the main text is the current statement. Tables are not repeated.

Introduction.

Humans presented with an image whose shape and texture point to different categories name the shape: on the cue-conflict images of Geirhos et al. (2018a), ten observers did so on 96% of decided trials, while ImageNet-trained convolutional networks named the texture. Closing that gap has become a standard target for biologically motivated models (Hermann et al., 2020; Jang & Tong, 2021), and a developmental visual diet — training on input whose acuity, contrast sensitivity and chromatic sensitivity follow the human developmental trajectory — is among the most effective routes to it (Lu et al., 2026).

A second body of evidence sits uneasily beside the first: along the ventral stream, object representations are well described by texture statistics — category-selective human cortex fails to discriminate natural images from texture-matched scrambles yet still discriminates categories (Jagadeesh & Gardner, 2022), macaque V4 responses to natural textures are predicted by their image statistics (Okazawa et al., 2015), and in inferotemporal cortex texture is the more decodable cue (Jagadeesh & Livingstone, 2024; Komatsu & Goda, 2018). The premise is scoped — one human study with one manipulation, the rest macaque, and "texture-like" meaning that a texture-statistics description suffices, not that shape is absent — and the argument needs only that the ventral description is substantially texture-like.

If behaviour is shape-biased and the ventral stream texture-biased, either the two measures disagree about what shape bias is, or behaviour is not a readout of one representation. The second horn is a hypothesis about architecture that single-trunk models of the ventral stream cannot express, and two facts make it concrete: the visual system receives not one image but a luminance-dominated, spatially coarse magnocellular signal and a chromatically opponent, finer parvocellular one, separated before cortex and later recombined (Livingstone & Hubel, 1987; Kaplan & Shapley, 1982; Derrington et al., 1984; Nassi & Callaway, 2009; Ferrera et al., 1994), and that input is not stationary over development, so the system first learns from a low-passed world. The two-stream hypothesis (Mishkin et al., 1983; Goodale & Milner, 1992) and the emergence of ventral- and dorsal-like streams in a two-pathway network trained on one objective (Bakhtiari et al., 2021) motivate an architecture whose streams are defined by their subcortical-like inputs rather than by their cortical targets.

This paper asks two questions, in the order the originating study posed them (Manocchio, 2026): can two streams fed magnocellular- and parvocellular-like inputs diverge while optimising a single objective and producing shape-biased behaviour, and does such a system explain how the ventral stream can be texture-like while behaviour is shape-biased? We build it — two ResNet-18 trunks receiving the luminance and L−M channels of a cone-opponent image, with a developmental diet on the luminance channel (Fig. 1) — and examine it with linear probes and cue-conflict behaviour on the streams (Fig. 2), voxel-wise encoding models of eight subjects' 7T fMRI responses (Fig. 3) and the behavioural readout that combines the streams (Fig. 4). The streams diverge by different routes; the texture-biased one carries extrastriate cortex; and the integrated behaviour is shape-biased because a behavioural rate is a property of a readout over components, not a report of any one of them — a property that holds, in kind, in networks with no stream split at all (Fig. 6).

Because the account predicts that properties bundled with shape bias need not travel with it, we test that too: robustness, agreement with human decisions and representational dimensionality all come apart from the rate (Fig. 5).

Relative to the originating thesis, every condition is trained with three seeds and every claim states its replication unit; the representational-similarity analysis that had found no stream-to-brain mapping is replaced by variance partitioning of encoding models, which removes the global stream preference that had defeated it; and the stream weighting the thesis fixed by hand is swept, so that the dependence of behaviour on the readout is measured rather than assumed.

Results 2.1.

The network splits its input once, before any convolution (Fig. 1a): each image is converted to the DKL cone-opponent space, stream 1 receives the luminance channel and stream 2 the L−M channel, and each is a ResNet-18 trunk. Under the developmental visual diet the luminance channel passes through a Gaussian blur ($\sigma = 4$ px) and the scale-free contrast-sensitivity cut of Lu et al. (2026) at a fixed age of 48 months, separating the transform from the trajectory (Methods 4.1). The trunks are initialised bit-identically — a control in which both streams receive the same image leaves all 120 tensor pairs identical after 150 epochs — so every stream difference below is attributable to the inputs, and a linear classifier reads out $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with $w = 0.5$ during training.

The diet costs accuracy (Fig. 1c): the no-diet conditions converge to 0.70 validation top-1, the diet arm reaches 0.556 at its best saved checkpoint against 0.701 and, uniquely, degrades after its peak, ending at 0.445 (range 0.347–0.527), so behavioural and out-of-distribution results use each seed's best saved checkpoint, stated wherever the policy matters (Methods 4.7). Two single-stream controls trained with the identical diet — a ResNet-18 (11.2M parameters) and a ResNet-34 parameter-matched to the two-stream model (21.3M against 22.4M) — reach 0.599 and 0.623, so the accuracy cost of the split is not a capacity artefact (Fig. 4c).

Results 2.2.

Read out separately with a linear probe on the cue-conflict images, the two streams of the diet-trained network end on opposite sides of the shape–texture boundary: scored on exemplars the probe was not trained on, shape − texture accuracy is +0.337 for the magno stream and −0.168 for the parvo stream, means of three seeds, every seed on its side (Fig. 2a; +0.819 and −0.518 with exemplars shared; Methods 4.3); the magno stream decodes a new exemplar's texture at chance (0.08, against 0.0625). Behaviour agrees: read out alone at the final checkpoint, the policy of the cortical fits, the magno stream's cue-conflict shape bias is 0.878 and the parvo stream's 0.418, against 0.675 for the integrated readout (Table 1; best-checkpoint values in Supplementary Table S9) — the divergence the thesis reported on a single model per condition, now replicated across seeds.

The decomposition arises asymmetrically (Fig. 2a, Table 1). Scored on the untrained network with the same front end, most of the magno stream's shape bias is present before any weight is updated — the low-pass and contrast-sensitivity transform does it — at +0.210 of the trained +0.337 on held-out exemplars; the parvo stream begins nearly unbiased (−0.033) and acquires its texture preference over training. The shape-biased component starts in the input, the texture-biased component is built by learning, and what ordinary training does to the input's contribution is the decisive contrast: with the transform removed, the untrained luminance stream starts lower (+0.090) and training drives it across the sign flip to −0.122 — it does not merely fail to become shape-biased, it becomes texture-leaning — while the parvo stream ends texture-leaning whether or not the other stream is dieted (−0.168 against −0.118). Nor does the dieted input keep its bias

alone: a single ResNet-18 trained on the magno stream's exact input and recipe ends at a rate of 0.557 and a penultimate-layer probe of −0.06 to −0.10 on held-out exemplars, where the same input as one of two streams ends at 0.878 and +0.26 to +0.27 — the shape-biased component survives training beside a second stream.

Which property of the input supplies the shape bias? Crossing the carrier a stream receives (achromatic luminance or chromatic L−M) with whether the developmental transform is applied to it gives four cells, two of which the swap and `parvo_highsf` conditions provide (Fig. 2b, c): on the probe, scored on held-out exemplars, the chromatic carrier gives −0.118 without the transform and −0.002 with it, the achromatic carrier −0.139 without and +0.337 with, so only the conjunction crosses zero, with an interaction of +0.36 over the additive prediction of the other three cells (−0.02) and no seed overlap between the conjunction cell (+0.333 to +0.340) and any other (at most +0.008); with exemplars shared the interaction is +0.73, and cue-conflict shape bias on the same four cells replicates it (Supplementary Table S3). The shape bias requires the developmental transform applied specifically to an achromatic channel — a relational fact about what a stream receives, not a property of a stream called "magno".

Two qualifications belong here. First, cue-conflict shape bias is computed only over images a readout decides as the shape or the texture category, a count that varies four-fold across cells (165 to 598 of 1,200; Fig. 2d), so the three non-conjunction behavioural cells are read as "no strong preference"; the probe, which classifies every image, gives the same pattern, and so does a forced two-way estimator over every image, which compresses values toward 0.5 (the four cells read 0.44, 0.51, 0.35 and 0.76) while preserving every ordering this paper draws a conclusion from, the two no-preference cells trading places (Supplementary Table S9). Second, the diet-trained models are less accurate: we pre-registered the fatal outcome — a no-diet model at the diet's accuracy with a shape-biased magno stream — and retrained the no-diet network with per-epoch checkpointing, selecting by rule, in each of three seeds, the epoch nearest the diet's accuracy (0.546–0.554 against 0.556); matching accuracy closes 4.3% of the probe gap on held-out exemplars (1.5–5.9% across seeds; 7.2% with exemplars shared) and 6.1% of the behavioural gap (2.2–9.1%), and the matched magno stream stays on the texture side in every seed (−0.10 to −0.12; Fig. 2e, f).

Results 2.3.

The thesis had asked whether the streams align with different cortical streams by representational similarity analysis (Kriegeskorte et al., 2008) against seven regions of the Natural Scenes Dataset, and found that whichever stream was the more predictive correlated better with every region — a general factor that a correlation cannot separate from regional structure. We replaced the instrument with voxel-wise banded-ridge encoding models for each of the eight NSD subjects, one band per stream, partitioning every voxel's cross-validated explained variance into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p) (Methods 4.5); the per-voxel index (U_m − U_p)/(U_m + U_p), median over reliable voxels per ROI, across V1v → hV4 is Fig. 3a.

The two trained models sit on opposite sides of the global stream-preference axis — without the diet every retinotopic area is magno-explained, with it every area is parvo-explained — and that global difference, the general factor that defeated the correlational analysis, is not the result. The result is the shape of the profile: only the diet arm descends steeply from V1 to hV4, early visual cortex relatively more magno-explained and hV4 relatively more parvo-explained, the direction a luminance/low-spatial-frequency versus chromatic split predicts, and the descent belongs to the model whose global preference points the other way, which a model simply preferring one stream everywhere could not produce.

Quantified as the contrast hV4 − mean(V1v, V1d) of the per-ROI median (Fig. 3b; Extended Data

Table 1), the gradient is −0.745 to −0.834 across the three diet-trained seeds against −0.049 to −0.096 without the diet, with the same separation on the raw statistic U_m − U_p, and all six comparisons reach the exact Wilcoxon floor for eight subjects ($p = .0078$) with complete seed separation. The per-subject Spearman ρ between hierarchical rank and stream preference is −0.964 to −0.973 with the diet and −0.46 to −0.66 without (Fig. 3c; Supplementary Table S1); the rank vector $[1, 1, 2, 2, 3, 3, 4]$ admits exactly 630 labellings of the seven ROIs, and against that enumerable null — which calibrates the extremity of a result that was not pre-specified rather than testing a prediction — the diet arm's ρ is the single most extreme in every seed and on both statistics ($p = 1/630 = .0016$) while the no-diet arm is not ($p = .019$ to $.092$), so we claim nothing there.

The diet does not create the gradient; it decides what training does to the one the input supplies. An untrained network on the diet front end already sits at −0.544 (8/8 subjects, $p = .0078$), and on the no-diet front end at almost the same value (−0.479), so the two front ends alone impose no different gradient; training with the diet steepens it and training without it erodes it to near zero (Fig. 3b). Correcting each arm for the signed bias of the estimator itself — a simulated floor of the wrong sign, +0.04 with the diet and +0.05 without — leaves a paired displacement of −0.871 with the diet against −0.130 without, each non-zero in every subject, and an independently trained checkpoint from the thesis's own pipeline, with its progressive developmental schedule, gives −0.793.

Pushed through models fitted on natural scenes only, the cue-conflict images yield a per-voxel predicted shape-bias index whose magno − parvo difference is +1.53 to +1.57 with the diet against +0.08 to +0.11 without (8/8 subjects, $p = .0078$ in every seed; Fig. 3d; Supplementary Table S2) — a test that needs neither probe nor readout.

The gradient survives its nuisance axes and the instrument's own bias. Receptive-field size, eccentricity coverage and the noise ceiling all change from V1 to hV4, and the diet's magno stream is a low-pass instrument by construction; yet the diet gradient is negative within every one of NSD's five eccentricity bands and every pRF-size tertile while the no-diet arm stays near zero in each, it holds on continuous pRF eccentricity and size within subject, and the estimator's own simulated bias is small, flat across ROIs and of the wrong sign (Extended Data Fig. 3; Supplementary Tables S7 and S8; Supplementary Note S3).

The gradient does not stop at hV4, and it does not turn. Relabelling the stored fits with NSD's full atlases (Extended Data Fig. 4; Supplementary Table S10), the diet index runs from −0.05 in V1v through −0.81 in hV4 and then sits between −0.81 and −0.94 in every extrastriate region — ventral, lateral and dorsal alike, in the Kastner2015 atlas, NSD's stream parcels and the fLoc category regions — with every region-minus-V1 contrast negative in 8/8 subjects ($p = .0078$) and no region-minus-region contrast departing from zero under either arm (all $p > .05$). Beyond V3 the diet's magno band contributes about 0.01 unique variance while the parvo band's rises from 0.10 in V1v to 0.26 in PHC2 (Extended Data Fig. 4b): the texture-biased stream carries extrastriate cortex as a whole, and the thesis's magno→dorsal, parvo→ventral mapping fails for want of any magno variance to map.

Three things this section does not settle. The diet predicts BOLD worse, not better (median r^2 normalised by the noise ceiling 0.405 against 0.464 without the diet, in 8/8 subjects; Fig. 3e), so its case rests on stream differentiation and the gradient, not on prediction accuracy. The two front-end controls are degenerate on this instrument (swap: U_p = 0.006; `parvo_highsf`: U_m = 0.007; degenerate in every one of their three seeds, Methods 4.5), so only the diet on luminance yields two streams that both carry substantial unique cortical variance and a gradient between them (Fig. 3a). And the gradient was not pre-specified: it is the survivor of seven regional hypotheses, six of which failed — among them the thesis's magno→dorsal mapping, an hMT+ specificity and a

foveal/peripheral contrast whose sign flipped when the blur was removed — reported because it alone survives removal of that manipulation, and read as sign and ordering rather than as a calibrated effect (Supplementary Notes S1 and S3; Extended Data Table 1 for the subject-01 influence check).

Results 2.4.

The two streams are oppositely biased and the texture-biased one carries extrastriate cortex; what remains is what the behavioural readout does with them. The integrated readout's cue-conflict shape bias is 0.675 ± 0.025 with the diet against 0.416 ± 0.020 without (mean ± s.d. over three seeds, final checkpoints, as in Table 1; the thesis's checkpoint reproduces at 0.697 against its reported 0.677). On the identical images and statistic the ten human observers give 0.958 ± 0.019 (range 0.919–0.987) and the Geirhos et al. (2018a) CNNs 0.429, 0.312 and 0.172, values our pipeline reproduces. The diet-trained magno stream, at 0.878, closes most of the distance from the CNN regime toward the observers without matching the lowest of them, and integration costs 0.203 of shape bias at this checkpoint policy (0.878 → 0.675; 0.248 at the best saved checkpoints; Fig. 4c, Supplementary Tables S4 and S9).

The integrated readout lands between its two components — an outcome two oppositely biased components nearly force, so not a prediction — and what can be measured is how much of the rate the readout controls. Sweeping w while holding y_1 and y_2 exactly fixed (Fig. 4a) runs cue-conflict shape bias from 0.40 to 0.90 over 21 values of w, three seeds and both checkpoint policies; $w = 0.5$ reproduces each trained model exactly, and the two streams alone sit at 0.41 and 0.78 on the forced two-way estimator, so the span is not an artefact of the decided-image denominator (Supplementary Table S9). Because the affine sweep is exact interpolation between two measured endpoints, under a rule fixed before the test we refitted a two-layer nonlinear head, and a linear head as a control for the refit itself, on the frozen features at each w (Fig. 4b): the nonlinear head's median span over the three best-checkpoint seeds is 0.466 against 0.508 for the frozen affine head (92% of the pre-registered baseline, 86% in the weakest seed, threshold 80%), the refitted-linear and nonlinear medians (0.461 and 0.466) agree within 0.01, and restricted to readouts within 5 points of peak validation accuracy the nonlinear span is 0.283 (0.26–0.31 across seeds): the span is a property of two oppositely biased components under any readout, and the extreme rates near $w = 1$ are bought with accuracy.

On the ladder of Fig. 4c the two streams alone bracket the integrated readout (0.903 and 0.400 around 0.655) and the single-stream controls trained with the identical diet lie between (0.567 and 0.625), inside the 0.058 that capacity alone moves the rate, so the ordering among these three models is attributed to capacity and no shape-bias advantage is claimed for the split.

Nothing here fits w to humans, and it could not be: fitted to each of the ten observers individually on a 0.001 grid, no observer's rate is reachable in any run (0 of 90 observer-runs, the closest approach missing the lowest observer by 0.027), and a weight fitted to an observer's trial-by-trial choices does not cross-validate (held-out agreement +0.026 against +0.023 for an arbitrary weight, better in 45 of 90). The rate is a free parameter of the readout, and nothing in this family of readouts selects the human operating point.

The readout weight cortex prefers descends the hierarchy. The readout's input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ is a feature vector like any other, so for each w it was the design of a ridge encoding model of every voxel in NSD's general visual mask, and the w that maximises an ROI's median r^2 is the weight that region prefers (Methods 4.5; Extended Data Fig. 5; Extended Data Table 2). Under the diet that weight descends the hierarchy — 0.58 in V1 to 0.33–0.35 in every extrastriate group, ventral, lateral and dorsal alike, below V1 in 8/8 subjects ($\Delta = -0.252$ [−0.283, −0.219], exact $p = .0078$) — and at the weight extrastriate cortex prefers the readout's rate is 0.56, below the trained

0.675 and far from the observers' 0.958. The weight that best reproduces the observers' trial-by-trial choices is not identified by these data: its median over observers is 0.67 (range 0.49–0.89), but each observer-run's bootstrap interval averages 0.69 wide on $[0, 1]$ and covers the extrastriate optimum in 23 of 30 fits, only 2 of 30 fits beat a label-permutation null, and the fitted weight does not cross-validate (above), so we do not claim that cortex and behaviour prefer different weights (Extended Data Table 2). Without the diet the cortical gradient reverses (0.65 in V1 to 0.75–1.00 beyond hV4; $\Delta = +0.188$ [+0.077, +0.285], 7/8). The curves are shallow (every ROI within 3% of its best r^2 at $w = 0.5$ with the diet and within 6% without) and their heights sit at the ridge's own reproducibility — the 1,024-dimensional concatenation, which every mixture is a linear map of, is itself exceeded by the best mixture in 36 of 112 subject × ROI × arm cells by a median 0.0019 r^2, more than the 0.0016 the sweep buys — so this is an ordering of optima and never an r^2 difference: the dissociation of Fig. 3 in the readout's own currency.

Without a stream split, a gain on a network's own shape- and texture-predictive subspaces still moves the rate. The span of Fig. 4a is given by construction in a network built around two streams, so we tested networks with none, under plans committed before each analysis was run (Methods 4.8). In any frozen network the penultimate features hold a shape-predictive and a texture-predictive subspace — the fifteen discriminant directions of a 16-class shape and of a 16-class texture linear discriminant, made mutually orthogonal — and a gain of 2w on the first and $2(1 - w)$ on the second, everything else including the network's own head untouched, is the analogue of the mixing weight; every w is gated on top-1 accuracy on held-out natural images (Fig. 6). Because the 1,200 cue-conflict images reuse 160 object and 48 texture exemplars, the subspaces are fitted on images sharing no exemplar with those the rate is scored on (Methods 4.8); the pre-registered split shared exemplars, and its values are given as exemplar-shared. In the six ImageNet networks of Fig. 6 the rate rises monotonically with w ($\rho \geq 0.999$ on the mean curve) by 0.14–0.20 with top-1 within 1.5 points of the trained value over the whole sweep (3.5 points on the held-out bases) — about a third of the two-stream span — so the pre-stated threshold (≥ 0.20 in at least four of the six), met in all six with exemplars shared (0.25–0.33), is met in one; four of the six can still be moved past the trained rates of at least two others, but the reverse of their trained order, reachable with exemplars shared, no longer is (Fig. 6a, b; Extended Data Tables 3 and 4). Across the field's zoo the result holds and the ranking does not: in 38 torchvision ImageNet classifiers the whole sweep stays within the five-point gate (top-1 changes by at most 4.4 points on the all-image bases; on the held-out bases the gate narrows in three networks, Methods 4.8), the reachable span is 0.08–0.29 (median 0.19; 0.20–0.50 with exemplars shared, the difference coming mostly through the texture exemplars; Methods 4.8) and unrelated to the trained rate ($\rho = +0.10$), the rankings by trained rate and by the rate at the shape end of the gate agree at Kendall $\tau = 0.63$ with 19% of the 703 pairs reversing (16% on the forced estimator; 27% and 21% with exemplars shared), against 8.9% [6.0%, 12.2%] for the trained ranking re-estimated on resampled images, and the ranking by trained rate agrees with that by a linear probe's shape-minus-texture accuracy on held-out exemplars only at $\tau = 0.43$ (Extended Data Fig. 6; Extended Data Table 4). The span needs the labelled directions: a gain on pairs of random 15-dimensional subspaces of the same features moves no network's rate by more than 0.08 (median 0.01 over ten draws each; Supplementary Table S14), below every network's span on unseen exemplars. Error consistency with the ten observers (Results 2.5), added after the pre-registered analysis, is 2–23% of the human–human ceiling at the trained readouts (median 8%); it rises with the trained rate across networks ($\rho = 0.80$) and, within every network, from the texture end to the shape end of its gate (38 of 38; median +14 points of the ceiling), yet no gated readout of an ImageNet-trained network exceeds 37% of the ceiling — a rise that is mostly the measure's own bound moving with the rate (Results 2.5; Supplementary Table S13). The informative exception is a ResNet-50 trained on Stylized-ImageNet (Geirhos et al., 2018a): the highest trained rate of the 42 networks tested (0.81), the only one whose probe leans

to shape on exemplars it was not trained on (+0.23 [+0.13, +0.34], against −0.52 to +0.01 for the ImageNet-trained networks, 37 of 38 below zero; −0.11 against −0.30 to −0.66 with exemplars shared), and the highest agreement with the observers (40% of the ceiling at its trained readout, 53% at its shape end) — its bias sits in the representation, which the rate alone could not have shown. Its span is among the smallest (0.12 on unseen exemplars, the 3rd smallest of 39; 0.17 and the smallest with exemplars shared, short of its pre-registered 0.20 threshold, P-Z1–P-Z4, the other three predictions met), but that ranking belongs to the estimator — in sample it is the 9th smallest of 39, and on the forced estimator the 16th — and what sets the network apart is which evidence the gain acts on (below). Our own single trunks move by 0.23–0.34 ungated (0.17–0.28 on unseen exemplars) but lose 7–15 points of MiniEcoset top-1 at the ends of the sweep (seed means; up to 12–17 points in single seeds, Extended Data Table 3), 0.14–0.22 within the gate (0.10–0.15 on unseen exemplars, Fig. 6a), and a head refitted at each w on the lossy sum of the two subspaces, the residual passed unchanged, keeps validation accuracy within 1.2 points of its ceiling in every seed (0.5–0.8 points as family means) while the rate still moves by 0.10–0.15 (two-layer head; 0.08–0.13 linear). The information the gain uses is distributed — the two subspaces hold 4–15% (shape) and 20–33% (texture) of the feature variance — and the pre-specified unit-level rule, a median split of the units by their shape-minus-texture η^2, reaches gated spans of only 0.07–0.16 (Supplementary Table S11); on the two-stream network the data-driven subspaces are not the streams (63–70% of the shape basis's energy on stream 1, against the pre-stated 90%) and the subspace gain moves the rate by 0.32 (0.25 on unseen exemplars) where the streams themselves give 0.46 and the readout weight of Fig. 4a 0.50, so the architectural split remains that network's most effective factorisation.

What the gain acts on. Post hoc, under a fifth plan committed before the analysis ran (`docs/PLAN_SPAN_2026-09-18.md`), we decomposed each network's decisions at its trained readout. Because the gain adds $(2w - 1)\cdot W(P_S - P_T)y$ to the logits, its first-order effect on each image's shape-minus-texture margin is a shape term, the evidence the head receives through the shape subspace, plus a texture term, the texture evidence it receives through the texture subspace, the first added and the second removed as w rises (Methods 4.8). This account reproduces every network's forced-choice curve (median largest error 0.009) and, reduced to two numbers per network, ranks the forced-choice spans at $\rho = 0.95$, but it ranks the reported, held-out spans only at $\rho = 0.59$, and of seven predictions two were met, four failed and one held in half (Supplementary Table S14). What it does show is which evidence the gain moves. In 37 of the 38 ImageNet networks the texture term is the larger (median 89% of the effect; ViT-B/32 is the exception, at 45%): the gain moves an ImageNet network toward shape by removing texture evidence more than by adding shape evidence. The texture share tracks the probe's representational bias about as closely as it tracks the trained rate when the probe is scored on held-out exemplars ($\rho = -0.58$ [−0.78, −0.29] against +0.59), and more closely only with exemplars shared (−0.81 [−0.91, −0.62]); the Stylized-ImageNet network is its extreme at 11%, its texture term below every ImageNet network's (0.31 against 0.93–3.95 margin units, as predicted), the share returning to 75% and 91% as ImageNet is added to its training and then fine-tuned on.

Cortex loads on the texture-leaning units, not on the discriminant subspaces. With the two unit halves of the no-diet single trunk as encoding bands, the texture-leaning half carries every extrastriate group relative to V1 in 8/8 subjects ($\Delta = -0.32$, exact p = .0078), and the cortex-optimal weight of the lossy sum falls from 0.80 in V1 to 0.37–0.48 beyond hV4 (7/8, p = .023) while the behavioural rate rises toward w = 1 (Fig. 6c, d; Supplementary Table S12); the diet-trained single ResNet-34 shows the same contrast (7/8, p = .016). In the ImageNet networks the texture-leaning half carries early visual cortex as well — V1 through hV4 and ventral cortex in ResNet-50, whose units are almost all texture-leaning, and every region in ConvNeXt-B (index −0.09 to −0.45) and ViT-B/16 (−0.13 to −0.31), two networks fitted after the pre-registered analysis — so the V1-

relative prediction fails there (extrastriate − V1 = +0.24, p = .039; +0.22, 2/8; −0.08, 6/8, p = .25). These unique variances are small (≤ 0.04 for the texture-leaning half against 0.10–0.25 for the parvo band with the split, 80–96% of the explained variance being shared between the halves against 45–63% between the streams; ViT-B/16 is the least redundant, at up to 0.09 and 64–72%), and under the subspace rule they all but vanish: entered as three bands with the residual, the two 15-dimensional subspaces each add at most 1.3% of noise-ceiling-normalised unique variance in any region against 23–63% carried by the shared and residual variance, their index sits at noise level, and the pre-registered cortical predictions fail — the extrastriate − V1 contrast is negative in 2/8 subjects each in the no-diet trunk, the diet ResNet-34 and ResNet-50, in 7/8 in ConvNeXt-B ($\Delta = -0.44$, p = .016) and in 3/8 in ViT-B/16, and the 15-dimensional lossy sum's cortex-optimal weight is lower beyond V1 in no family at the pre-registered 7/8. The directions that move the rate are nearly silent as unique cortical predictors — which component cortex loads on is decided by how the representation is partitioned — so the cortical claim is made under the unit rule only. The architecture is therefore not what makes the rate a readout property, nor, in the networks trained on our images, what makes extrastriate cortex load on the texture-biased component; it is what makes the components separable enough for one weight to move the rate across the whole range models occupy, and for cortex to be carried almost entirely by one of them.

Results 2.5.

If shape bias is a property of a readout over components rather than of a representation, quantities the field treats as travelling with it need not. Over identical inputs, three do not (Fig. 5).

Robustness. The standard reading of Geirhos et al. (2018a, 2021) is that shape bias buys out-of-distribution robustness. We pre-registered that prediction and tested it on corruptions applied before the front end, scored in the models' own label space at each seed's best saved checkpoint (Fig. 5a; Extended Data Fig. 1; Supplementary Table S5). The diet-trained model is less robust on four of the five corruption families it was not trained on — uniform noise, high-pass, false colour and rotation (relative-accuracy AUC 0.634 against 0.729, 0.678 against 0.827, 0.854 against 0.938 and 0.574 against 0.615), with complete seed separation on each — equivalent within ± 0.033 on phase scrambling, more robust only to low-pass (0.444 against 0.356), declared trained-in before the run because the diet is a low-pass blur, and not to contrast reduction (0.481 against 0.549) although its front end includes a contrast-sensitivity cut. Robustness appears exactly where it was trained in and nowhere else, the strongest available evidence that the nulls are real rather than a broken measurement. This does not contradict Geirhos et al., whose stylisation route yields shape bias together with robustness — it shows the coupling belongs to that route — and it differs from the source of the diet, whose title reports robust, shape-based vision: we reproduce the shape-based half (their released model 0.822, our diet-trained magno stream 0.878) and not the robust half, and Methods 4.6 lists three implementation differences any of which could carry the divergence.

Agreement with human decisions. Two decision-makers with the same marginal rate can agree on no individual image beyond what those rates force; error consistency — Cohen's κ on trial-by-trial agreement — subtracts that forced agreement (Geirhos et al., 2020), and across the ten observers the human–human ceiling is $\kappa = 0.234$. We pre-registered that the diet-trained model would be more human-consistent than its control; it is not, on either readout, and the sharpest true statement is stronger: our most shape-biased model, the diet-trained magno stream at 0.878, has, averaged over seeds, the lowest error consistency of the seven models compared ($\kappa = 0.0082$, 3.5% of the ceiling, seeds spanning −6.1% to 15.5%, against 7.3% for VGG-16 at shape bias 0.172; Supplementary Table S6); across all 189 readouts of the w sweep, with w chosen post hoc per run to maximise agreement, the best κ never exceeds 20.7% of the ceiling (Fig. 5b), and equivalence intervals fixed in advance exclude a κ change larger than 0.038 across the 0.71 span of shape bias

the readouts cover; across the field's zoo the same measure does rise with the rate, to at most 37% of the ceiling (Results 2.4) — and that rise is the measure's own bound. Cohen's κ cannot exceed a value fixed by the two marginals, which with the observers at 0.958 runs from 0.006 to 0.219 across the 798 gated readouts of the 38 networks; one quadratic in the rate accounts for the agreement ($R^2 = 0.90$) and, normalised by the bound, its rank dependence on the rate disappears (0.001 against 0.891). Agreement is thus readout-controlled as the rate is: over the 33 networks that reach a common rate of 0.40, ranking by agreement at the trained readouts against at that rate gives $\tau = 0.33$ with 34% of pairs reversing, the 1.9-fold residual spread is unrelated to the probe ($\rho = +0.01$), and the Stylized-ImageNet network is not separable from ViT-B/32 (135% against 120% of the human-pair value). Four of six predictions of a fourth plan failed in this way and a fifth was untestable as specified (`docs/PLAN_KAPPA_2026-09-17.md`, post hoc here; Supplementary Table S13). The measure does detect a correspondence when one is there — the same machinery recovers 24.1% of the split-half ceiling on whether an image resolves to either cue at all, against 8.7% on which cue is picked (Fig. 5c) — a positive control that invites a diet reading we do not make, because the crossed front-end 2×2 shows a train/test-matching pattern rather than a weights effect (Fig. 5d).

Representational dimensionality. The participation ratio of each stream's features on 1,000 NSD images per subject gives a magno/parvo ratio whose ordering across the four front ends is reproduced by a learning-free Gabor bank, so it tracks the bandwidth of the carrier a stream is handed; what the diet decides is which side of parity training ends on — every diet-trained seed ends magno-dominant (24/24 subject-seeds in each of three diet arms), including the `parvo_highsf` front end, which starts inverted in 8/8 subjects and which we had pre-registered would stay inverted, while every no-diet seed collapses to parity (Extended Data Fig. 2). No level is claimed, because the within-arm seed range exceeds the gaps between arms; and `parvo_highsf`'s second stream becomes the more dimensional stream while remaining texture-leaning (probe −0.139 on held-out exemplars, −0.271 shared), which separates dimensionality from shape bias.

Discussion.

The two questions. The thesis asked whether magnocellular- and parvocellular-driven streams inside one network could diverge while the network behaves shape-biased, and whether such a system bears on why the ventral stream can be texture-like while behaviour is not. To the first the answer is yes, replicated across seeds and with the mechanism located: divergence requires the developmental transform on an achromatic carrier, the shape-biased component starts in the input and is kept, and extended, by training only under the diet and beside a second stream, and the texture-biased component is built by learning. To the second the answer is an existence proof the thesis could not reach with its instruments: fitted to human cortex, the texture-biased stream's unique variance carries the extrastriate end of the hierarchy — ventral, lateral and dorsal alike — while behaviour, reading out from both, is shape-biased. Read as competing claims about a single representation the two cannot both be right; read as claims about a readout over components, they are not in competition.

What the readout account licenses. A model matching the human shape-bias rate is routinely described as having human-like representations; that inference fails twice over here. The same network holds components at +0.34 and −0.17 on held-out exemplars (+0.82 and −0.52 with exemplars shared) and behaves at 0.675, and one scalar over frozen representations moves the rate across most of the range the field's models occupy, so the rate does not identify a representation; and across readouts spanning a five-fold range of shape bias, trial-by-trial agreement with ten observers never exceeds 21% of the human–human ceiling in our networks and is lowest, 3.5%, for our most shape-biased stream; across the zoo the two do move together ($\rho = 0.80$ over 38 networks, every network's agreement rising from the texture to the shape end of its gate), yet no gated readout exceeds 37% of the ceiling and that rise is the bound on κ moving with the rate (Supple-

mentary Table S13), so the rate is at best a weak proxy for the decision process — as is agreement, read as the field reads it. Prior work established that a representation can hold more shape than its behaviour reveals (Hermann et al., 2020; Islam et al., 2021), that the measured rate depends on the test design (Tartaglini et al., 2022), that a prompt — a readout-level manipulation — steers it in vision–language models (Gavrikov et al., 2024), and that it tracks a network's spatial-frequency channel (Subramanian et al., 2023), the quantity the diet constrains; new here is the form of the dissociation — both representations held fixed, the rate moved by the readout alone, a refitted nonlinear head preserving the span — and its cortical grounding: the same mixing weight, fitted to cortex, prefers the magno-heavy end in V1 and the parvo-heavy end in every region beyond it (Extended Data Fig. 5); where human choices place it is not identified by these data. Nor does the account need the split: in 38 frozen ImageNet networks with none, a reciprocal gain on the network's own shape- and texture-predictive subspaces moves the rate by 0.08–0.29 on exemplars those subspaces never saw, within five points of top-1, re-ordering 19% of network pairs (8% of the pairs the trained rate resolves, against a resampling floor of 0.33%), and in our single trunks the texture-leaning units again carry extrastriate cortex relative to V1 (Fig. 6); the split amplifies the dissociation (0.50 against 0.32 on the same network, 0.25 on unseen exemplars), it does not create it.

What a developmental constraint does (the BA network's fixed diet). The intuitive account of low-pass developmental input is informational — coarse input shields the learner from texture, so it learns shape — and for the BA network's fixed transform our results do not support it: the transform removes bandwidth rather than adding structure, most of the shape bias it produces is present before training, and what the diet decides is whether learning builds on that bias or destroys it (on held-out exemplars, training drives the luminance stream from +0.09 to −0.12 without it and from +0.21 to +0.34 with it, and a second instrument replicates the side of parity rather than a level). This is a constraint on the solution learning converges to, not a better input representation, consistent with the thesis's observation that the transformed magno stream becomes less predictive of the category as the parvo stream becomes more so. The model's trajectory supports a version of the informational account in which the input's history matters and the information at test does not (Discussion).

What comes apart, and what the diet costs. Shape bias, robustness, dimensionality and human-likeness are discussed as a package; over identical inputs this system separates them three times, and a readout account says why it should: none of the three is a property of a readout. The diet is also expensive — less accurate (0.556 against 0.701), less stable late in training, less robust on four of five untrained-on corruptions, a worse predictor of BOLD — so nothing here recommends it as a training method; its interest is as a model of development, and the thesis's central mechanistic reading, that divergence and shape bias are bought by degrading the magno stream's evidence, survives in every instrument.

How to report shape bias. A shape-bias claim should carry three numbers, each answering a different question: the rate at the trained readout (an operating point); the span the same frozen features reach under a reciprocal gain on their own shape- and texture-predictive subspaces within an accuracy budget (how much of the rate the readout controls); and a probe's shape-minus-texture accuracy (what the representation holds) — the last two scored on exemplars the subspaces and probe were not fitted on, since the standard cue-conflict set reuses 48 texture images across 1,200 stimuli and exemplar-shared estimates inflate both. Across 38 ImageNet networks the three rank the zoo differently ($\tau = 0.63$ and 0.43), and the Stylized-ImageNet network shows what their agreement looks like when a bias really is representational — seen from the readout, the difference is not how far the gain moves it but which evidence the gain moves (Results 2.4); error consistency does not make a fourth: read at each network's own readout it is largely a re-expression of the

rate, because κ is bounded by the mismatch with the observers' 0.958, and at a matched rate it separates neither the zoo nor the Stylized-ImageNet network from it, so where human trial data exist it should be read at a matched rate (Supplementary Table S13). None of these numbers alone licenses a claim about representations. One command in the repository prints all three, with the exemplar-shared values beside them, for any torchvision network (Methods 4.8).

Limitations (sentences condensed). Beyond hV4 the index saturates near its bound in every atlas, so the extension establishes that the effect is hierarchy-wide but cannot order extrastriate regions against one another. The `parvo_highsf` condition varies architecture alongside the front end. The diet is a fixed 48-month front end rather than a trajectory, though a schedule-trained checkpoint behaves like it on every instrument we tried. The single-trunk subspaces are linear discriminants of one cue-conflict set, and their spans are lower bounds on what a readout can reach, not measurements of the best one.

Supplementary Note S5 — the pre-registration ledgers

The generality test was pre-registered three times and all three ledgers are complete (Methods 4.8; outcomes in the plan files): of the unit-level plan's five predictions, the behavioural thresholds were met by direction but not by magnitude, the stream-recovery check was met, the V1-relative cortical contrast held in our two trunks and reversed in ImageNet ResNet-50, and the mixing-weight ordering held in one network of three; of the subspace plan's, the magnitude threshold was met in all six ImageNet networks but in one of our three trunks, the stream-recovery check failed (63–70% of the shape basis on stream 1, against 90%), the subspace span exceeded the unit-level span in every network (Extended Data Tables 3 and 4; Supplementary Tables S11 and S12), and of its three cortical predictions only the ConvNeXt-B contrast held, the two discriminant subspaces carrying almost no unique cortical variance; of the zoo plan's four predictions three were met and the Stylized-ImageNet span fell short of its threshold, read, as pre-stated, as a bias held in the representation rather than in the readout — a reading of the span that a fifth plan does not support (Results 2.4). A fourth plan, committed before its own analysis ran, predicted that agreement with the observers would separate representations at a matched rate; four of its six predictions failed, for the reason given in Results 2.5. A fifth, on what sets the span, met two of its seven predictions; which evidence the gain moves is a post hoc reading of it (Results 2.4). A sixth, re-estimating spans and probes with exemplars held out, is post hoc with respect to the five before it, written after a referee simulation found the exemplar reuse; it met 15 of 17 predictions; the two failures concern the probe, whose ImageNet networks stay texture-leaning where we predicted they would not (`docs/PLAN_EXEMPLAR_2026-09-21.md`). The held-out accuracy gate, the reversal floors by resampling unit and the probe's preprocessing spread were added to that plan after a later referee simulation and are post hoc too.

Supplementary Note S6 — the model's passages condensed in the main text

The main text was cut to the journal's length on 2026-10-24. Each paragraph it condensed is given here in full, as it stood before the cut, under the section it came from; the condensed paragraph keeps the claim, and every number it gives is taken from the passage here (`tmp_analysis/verify_v3_claims.py` checks both). Unlike Supplementary Note S4, these passages describe the model of record.

From 1. Introduction.

If behaviour is shape-biased and the ventral stream texture-biased, either the two measures disagree about what shape bias is, or behaviour is not a readout of one representation. The second

horn is a hypothesis about architecture that single-trunk models cannot express, and two facts make it concrete: the visual system receives a luminance-dominated, coarse magnocellular signal and a chromatically opponent, finer parvocellular one, separated before cortex and later recombined (Livingstone & Hubel, 1987; Kaplan & Shapley, 1982; Derrington et al., 1984; Nassi & Callaway, 2009; Ferrera et al., 1994), and that input is not stationary over development, so the system first learns from a low-passed world. The two-stream hypothesis (Mishkin et al., 1983; Goodale & Milner, 1992) and the emergence of ventral- and dorsal-like streams in a two-pathway network trained on one objective (Bakhtiari et al., 2021) motivate an architecture whose streams are defined by their subcortical-like inputs rather than by their cortical targets.

This paper asks two questions, in the order the originating study posed them (Manocchio, 2026): can two streams fed magnocellular- and parvocellular-like inputs diverge while optimising a single objective and producing shape-biased behaviour, and does such a system explain how the ventral stream can be texture-like while behaviour is shape-biased? We build it — two ResNet-18 trunks fed the luminance and L−M channels of a cone-opponent image, with a developmental diet on the luminance channel whose acuity and contrast sensitivity mature from birth to adulthood over training (Fig. 1) — and examine its streams with probes and cue-conflict behaviour (Fig. 2), encoding models of eight subjects' 7T fMRI responses (Fig. 3) and the readout that combines them (Fig. 4). The streams diverge by different routes, and the shape-biased one only when its input is achromatic and matured during training: the same network shown the same adult input from the first step, or given the ramp on its L−M channel instead, learns two texture-leaning streams. Only that conjunction divides the cortical hierarchy — the shape-biased stream carrying V1, the stronger, texture-leaning stream extrastriate cortex — and tracks which images human observers resolve. The integrated behaviour is shape-biased because a behavioural rate is a property of a readout over components, not a report of any one of them — a property that holds, in kind, in networks with no stream split at all (Fig. 6). Robustness, error consistency with human observers and representational dimensionality come apart from the rate, as the account predicts (Fig. 5).

From 2.1 The model, and what the developmental trajectory costs.

The network splits its input once, before any convolution (Fig. 1a): each image is converted to the DKL cone-opponent space, stream 1 receives the luminance channel and stream 2 the L−M channel, and each is a ResNet-18 trunk. Under the developmental diet the luminance channel passes through the acuity blur and the contrast-sensitivity cut of the developmental visual diet of Lu et al. (2026), both evaluated at one age that rises linearly with the training step from birth to 300 months (Methods 4.1). Every evaluation uses the endpoint, at which the acuity blur is inert and the contrast cut is an adult's, so at test the magno stream sees nearly clean luminance. The trunks are initialised bit-identically, so every stream difference below is attributable to the inputs, and a linear classifier reads out $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with $w = 0.5$ during training, under one cross-entropy on that output and no loss on either stream.

The decisive control holds the diet at its 300-month endpoint from the first step: the same architecture, recipe, seeds and test input, without the developmental trajectory. The originating study's network ("BA network") held a fixed mixture of ages instead — a $\sigma = 4$ px blur, about a 3-month-old's acuity, and a 48-month-old's contrast sensitivity — and is reported beside the model throughout, with its no-diet arm; trained with our recipe, the same fixed diet separates the recipe from the trajectory. Networks were trained on MiniEcoset (112 classes; Thorat & Kietzmann, 2024) for 50 epochs with AdamW, a cosine schedule, an exponential moving average of the weights and label smoothing, three seeds per condition (Methods 4.2); the BA network keeps the thesis's 150-epoch recipe.

The developmental trajectory costs accuracy, not stability (Fig. 1c): the model reaches 0.683 ±

0.002 validation top-1 at its final checkpoint, within 0.9 points of its best epoch in every seed, against 0.764 ± 0.003 for the control on the identical test input. The fixed BA diet reaches 0.665 ± 0.003 under the same recipe and 0.556 at its best saved checkpoint under the thesis's, after which it degrades (final 0.445). The trajectory therefore adds 1.8 points and 0.067 of shape bias (training-time evaluation) over the fixed diet under one recipe, and gives up 8.1 points against the network that learned the same input without the trajectory. A trajectory stopping at 48 months (one seed) is no better than the fixed diet (0.664 top-1, shape bias 0.637): the input has to reach adult acuity, not only start coarse. Every result for the model and its control below is from final checkpoints (Methods 4.7).

From 2.2 The two streams specialise in opposite directions, by different routes.

Read out separately with a linear probe on the cue-conflict images, the two streams end on opposite sides of the shape–texture boundary: on exemplars the probe was not trained on, shape − texture accuracy is +0.316 for the magno stream and −0.156 for the parvo stream, means of three seeds, every seed on its side and every seed's 5–95% interval over draws excluding zero (Fig. 2a; +0.578 and −0.454 with exemplars shared; Methods 4.3). This is the BA network's split (+0.337 and −0.168), at 0.683 top-1 against that network's 0.445 at the same, final, checkpoints. Behaviour agrees: read out alone, the magno stream's cue-conflict shape bias is 0.850 and the parvo stream's 0.477, against 0.682 for the integrated readout (Table 1).

Both components are built by learning, and the input's trajectory decides the sign of one (Fig. 2a, Table 1). Untrained, the magno stream on the model's test input sits where undieted luminance does (+0.087 against +0.090 on held-out exemplars), and the parvo stream begins nearly unbiased (−0.033). Training on the maturing input adds +0.229 to the magno stream — nearly twice what training added in the BA network (+0.127), whose fixed blur put +0.210 into the input before any learning — and −0.123 to the parvo stream. Training on the same adult input from the first step drives the magno stream the other way, across the sign flip to −0.177, below its own parvo stream (−0.069), and the integrated readout to 0.374, near the BA no-diet arm's 0.416. The two networks see identical images at test and differ only in whether the magno input matured during training: the magno stream's shape bias is a property not of the input it is tested on but of the input it learned from. The two magno streams sit on opposite sides of parity at every epoch evaluated during training (shape bias 0.63–0.92 against 0.32–0.495 over three seeds each; training-time evaluation), so the bias is present from the first evaluation, after one epoch at ages up to 6 months (top-1 0.09), and is kept as the input sharpens. Whether the coarse-to-fine direction matters, or only exposure to coarse input, these runs cannot say (Discussion, Limitations). It is not a filter on texture either: on held-out exemplars the model's magno stream decodes texture at 0.148 (chance 0.0625; 0.078 for the BA network's blurred stream) and shape at 0.464, so the stream holds texture information and leans to shape.

Which property of the input carries the effect? Crossing the carrier (achromatic luminance or chromatic L−M) with the developmental ramp gives four cells. They are achromatic with the ramp (+0.316), achromatic with the adult input from the start (−0.177), chromatic without the ramp (−0.156 in the model, −0.069 in the control) and chromatic with the ramp: the channel swap, which applies the same ramp to L−M with luminance untransformed (three seeds; −0.016). The ramp moves the achromatic stream by +0.49 and the chromatic stream by +0.05, which leaves it below parity; the carrier × ramp interaction is +0.44 on held-out exemplars (+0.73 with exemplars shared), and no seed of the conjunction cell (+0.30 to +0.33) overlaps a seed of any other cell (at most −0.011). Only the ramp on an achromatic carrier crosses zero, as only the BA network's fixed transform on an achromatic carrier did (interaction +0.36 on held-out exemplars; Supplementary Table S3). The swap reproduces its control on every other measure as well: 0.760 top-1, an integrated cue-conflict rate of 0.389 and both streams below parity (Table 1). Applied to the chromatic

channel, the trajectory buys neither the shape bias nor the accuracy cost.

Two qualifications belong here. First, cue-conflict shape bias is computed only over images a readout decides, a count that varies across readouts (426 to 580 of 1,200 in the model, 209 to 673 in the control; seed means); the probe, which classifies every image, and a forced two-way estimator over every image preserve every ordering this paper draws a conclusion from (forced estimator: magno 0.730, integrated 0.648, parvo 0.520; control 0.393, 0.383, 0.471), but the parvo stream sits within 0.03 of parity on both estimators and on opposite sides of it, so its side is read from the probe. Second, the BA study pre-registered its fatal outcome as a no-diet network at the diet's accuracy with a shape-biased magno stream, and closed it with an accuracy-matched retraining (Supplementary Note S4). The same worry applies here, since the model is the less accurate network (0.683 against 0.764), and the control's own training answers it: at the epoch where its top-1 matches the model's final 0.683 (epoch 7 of 50 in each of the three seeds), its magno stream's shape bias is 0.41–0.43, and at no epoch of any seed does it reach parity (highest 0.495; training-time evaluation).

From 2.3 In cortex, the magno stream carries V1 and the texture-biased stream the rest of the hierarchy.

The thesis's representational similarity analysis (Kriegeskorte et al., 2008) against seven NSD regions found that whichever stream was the more predictive correlated better with every region, a general factor that a correlation cannot separate from regional structure. We replaced it with voxel-wise banded-ridge encoding models for each of the eight NSD subjects, one band per stream, partitioning each voxel's cross-validated explained variance into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p) (Methods 4.5); Fig. 3a plots the per-voxel index $(\text{U_m} - \text{U_p})/(\text{U_m} + \text{U_p})$, median over reliable voxels per ROI, across V1v $\rightarrow$ hV4.

The profile crosses zero, and only with the maturing input. In the model the index runs from +0.55 in V1v, magno-explained, through zero at V2v to −0.76 in hV4, parvo-explained (Fig. 3a): the magno stream carries V1 and the parvo stream the ventral end of the retinotopic hierarchy (in V1, U_m 0.10–0.11 against U_p 0.03–0.04, U_m larger in 77–82% of voxels and in 8/8 subjects in every seed; in hV4, 0.015–0.020 against 0.13). The control, on the identical test input, is magno-explained everywhere and nearly flat (+0.81 in V1v, +0.73 in hV4), as are the channel swap (+0.85, +0.83) and the BA no-diet network (+0.91, +0.83); the BA network is parvo-explained from V1 on and descends without crossing (−0.00 to −0.84).

Quantified as the contrast hV4 − mean(V1v, V1d) of the per-ROI median (Fig. 3b; Extended Data Table 1), the gradient is −1.259 to −1.291 across the three seeds against −0.062 to −0.086 for the control's, each seed against the control seed of the same index at the exact Wilcoxon floor for eight subjects ($p = .0078$, 8/8), with the same separation on the raw statistic U_m − U_p (−0.186 to −0.193 against −0.003 to −0.007). It is steeper than the BA network's −0.745 to −0.833 in 8/8 subjects (mean Δ −0.48, $p = .0078$). The per-subject Spearman ρ between hierarchical rank and stream preference is −0.964 to −0.973 in every seed, against −0.29 to −0.41 for the control (Fig. 3c). The channel swap, the same ramp on the chromatic carrier, has no gradient either (−0.019 normalised, +0.002 raw, seed means; $\rho = -0.22$), and the model's lies below it in 8/8 subjects ($p = .0078$): in cortex as in behaviour, only the ramp on the achromatic carrier departs from the control.

The trajectory does not create the gradient; it decides what training does to the one the input supplies. An untrained network on the same test input already descends (−0.480 normalised, −0.195 raw, 8/8 subjects, $p = .0078$ on both; −0.544 and −0.479 for untrained networks on the BA network's two front ends). Training on the maturing input steepens it on the normalised index (to −1.27) and holds it on the raw difference (−0.19), so under the rule that a cortical claim needs both statistics (Methods 4.7) we claim only that training on the maturing input preserves it; training

on the adult input from the first step erases it on both (−0.070; −0.004, seed means).

Pushed through models fitted on natural scenes only, the cue-conflict images give a predicted magno − parvo shape-bias difference of +1.32 to +1.34 in the model against +0.07 for the control and −0.01 for the channel swap (8/8 subjects, seed means; Fig. 3d), a test that needs neither probe nor readout. It is smaller than the BA network's +1.53 to +1.57 (8/8), in the direction of the two magno streams' own rates (0.850 against 0.878).

The gradient continues past hV4 and does not turn. On NSD's full atlases (Extended Data Fig. 4), relative to V1, hV4 sits at −1.12 and every extrastriate group lower — ventral (VO, PHC) −1.25, lateral (LO, TO) −1.27, dorsal (V3A/B, IPS0–2) −1.22 — each in 8/8 subjects ($p = .0078$). The ventral and lateral groups sit a further 0.14–0.15 below hV4 ($p = .039$ and .016, uncorrected), and no contrast among the three extrastriate groups departs from zero (all $p \geq .08$). Beyond V3 the magno band contributes about 0.013 unique variance against 0.149 for the parvo band: the texture-biased stream carries extrastriate cortex as a whole, and the thesis's magno→dorsal mapping fails for want of any magno variance to map. Nor is the model's gradient an eccentricity or receptive-field-size contrast in disguise. Among voxels of matched eccentricity it is negative in each of NSD's five eccentricity bands (−0.58 in the foveal band to −1.33 beyond 4°), and among voxels of matched pRF size in each size tertile (−1.26 to −1.32), below the control's in every band and tertile (7/7 or 8/8 subjects; 6/6 in the smallest tertile), while the control and the channel swap stay near zero in each (Extended Data Fig. 3; Supplementary Note S3). Nor does the estimator manufacture it. Simulated with the model's fitted feature geometry, the measured noise ceilings and a true magno − parvo difference of zero everywhere, the recovered contrast is small and of the wrong sign (+0.018, $p = .11$; +0.066 on the control's geometry). Corrected by each arm's own floor, the model's gradient is −1.277 against −0.127 for the control, both non-zero in 8/8 subjects, and the model's is the steeper in every subject ($p = .0078$; Supplementary Note S3).

Three things this section does not settle. First, the trajectory buys the gradient at a small cost in BOLD prediction against its control (noise-ceiling-normalised median r^2 0.471 against 0.494, 8/8 subjects), although the model predicts BOLD better than the BA network does (0.395, 8/8; Fig. 3e). Second, the gradient was not pre-specified: it is the survivor of seven regional hypotheses tested on the BA network, six of which failed, and is read as sign and ordering rather than as a calibrated effect (Supplementary Notes S1 and S3). Third, "texture-biased" names the stream, not the reason cortex prefers it. Both streams receive near-full-bandwidth input at test, which removes the BA network's confound between texture bias and the magno stream's blur; but in the model the stream carrying extrastriate cortex is both the texture-leaning one and the chromatic one, so the design cannot separate texture bias from the carrier. It is also, in every network, the trunk that carries the task (Results 2.2), so task load is confounded with texture bias too; what only the model has is a V1 carried by the other trunk.

From 2.4 Behaviour reads out from both streams, and the rate is a property of the readout.

A readout of two oppositely biased components is nearly forced to land between them; what can be measured is how much of the rate it controls. Sweeping w while holding y_1 and y_2 exactly fixed (Fig. 4a) runs cue-conflict shape bias from 0.48 to 0.85 over 21 values of w and three seeds, and $w = 0.5$ reproduces each trained model exactly. The control's curve is flat (0.37–0.42), and so is the channel swap's (0.38–0.40): with both streams texture-leaning there is nothing for the weight to move. Because the sweep interpolates between two measured endpoints, we refitted a two-layer head on the frozen features at each w under the rule the BA study fixed in advance (Fig. 4b): its median span is 0.395 against 0.371 for the frozen affine head (106%; weakest seed 96%; threshold 80%), and 0.241 when restricted to readouts within 5 points of peak validation accuracy. The span is a property of two oppositely biased components under any readout, and the extreme

rates are bought with accuracy. Nothing here fits w to humans, and it could not: no observer's rate is reachable in any run (0 of 30 observer-runs), and a weight fitted to an observer's trial-by-trial choices barely cross-validates (held-out agreement 0.039 against 0.031 for an arbitrary weight, above it in 17 of 30 observer-runs).

The readout weight cortex prefers descends the hierarchy. For each w, the readout's input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ was the design of a ridge encoding model of every voxel in NSD's general visual mask, and the w that maximises an ROI's median r^2 is the weight that region prefers (Methods 4.5; Extended Data Fig. 5; Extended Data Table 2). In the model that weight falls from 0.65 in V1 to 0.30–0.35 in every extrastriate group, below V1 in 8/8 subjects ($\Delta = -0.323$ [−0.371, −0.269], exact $p = .0078$); at the extrastriate weight the network's own rate would be 0.60, at V1's 0.75. In the control the gradient reverses (0.43 in V1 to 0.62–0.78 beyond hV4; higher in 8/8 subjects, $p = .0078$), and so it does in the channel swap (0.38 in V1 to 0.62–1.00; higher in 7/8 subjects and tied in one, $p = .016$). The weight that best reproduces the observers' choices is not identified by these data (median 0.79 over observers, but per-fit bootstrap intervals average 0.70 wide on [0, 1] and only 10 of 30 fits beat a permutation null), so we do not claim that cortex and behaviour prefer different weights. The curves are shallow — every ROI is within 2% of its best r^2 at $w = 0.5$ — so this is an ordering of optima, not an r^2 difference (Supplementary Note S4).

Without a stream split, a gain on a network's own shape- and texture-predictive subspaces still moves the rate. The span of Fig. 4a is given by construction in a two-stream network, so we tested networks with no split, under plans committed before each analysis ran (Methods 4.8). In a frozen network's penultimate features, the fifteen discriminant directions of a 16-class shape and of a 16-class texture linear discriminant, made mutually orthogonal, span a shape- and a texture-predictive subspace; a gain of 2w on the first and $2(1 - w)$ on the second, the network's own head untouched, is the analogue of the mixing weight, and every w is gated on top-1 accuracy on held-out natural images (Fig. 6). The 1,200 cue-conflict images reuse 160 object and 48 texture exemplars, so the subspaces, and the bases the gate is measured on, are fitted on images sharing no exemplar with those the rate is scored on; the pre-registered split shared exemplars, and its values are given as exemplar-shared (Methods 4.8). In the six ImageNet networks of Fig. 6 the rate rises monotonically with w by 0.14–0.20, 37–55% of the model's two-stream span (0.37), so the pre-stated threshold (≥ 0.20 in at least four of the six), met in all six with exemplars shared, is met in one (Fig. 6a, b; Extended Data Tables 3 and 4).

Across the field's zoo the result holds and the ranking does not. In 38 torchvision ImageNet classifiers the gated span is 0.08–0.29 (median 0.18; 0.20–0.50 with exemplars shared) and unrelated to the trained rate ($\rho = +0.17$; Methods 4.8). Ranked by trained rate and by the rate at the shape end of the gate, the networks agree at Kendall $\tau = 0.63$, with 19% of the 703 pairs reversing. Re-estimated on resampled data, the trained ranking itself reverses 8.9% of pairs (images resampled), 9.1% (object exemplars) and 13.6% [8.3%, 20.9%] (texture exemplars), so against the strictest floor the all-pairs figure is not decisive; among the pairs the trained rate resolves, 39 of 490 reverse under image resampling (8.0%, against a floor of 0.33%) and 12 of 389 under texture-exemplar resampling (3.1%, against 0.47%). A linear probe's shape-minus-texture accuracy ranks the zoo differently again ($\tau = 0.43$; Extended Data Fig. 6; Extended Data Table 4). A gain on random 15-dimensional subspaces moves no network's rate by more than 0.08 (Supplementary Table S14).

The informative exception is a ResNet-50 trained on Stylized-ImageNet (Geirhos et al., 2018a): the highest trained rate of the 42 networks tested (0.81), the only one whose probe leans to shape on exemplars it was not trained on (+0.23 [+0.13, +0.34], against −0.52 to +0.01 for the ImageNet-trained networks), and the highest agreement with the observers (40% of the ceiling at its trained readout). Its bias sits in the representation, which the rate alone could not show, yet its span is not exceptional (0.11 on unseen exemplars, the 2nd smallest of 39); what sets it apart is which evidence

the gain acts on. In a post hoc decomposition under a fifth plan (Methods 4.8), the gain moves 37 of the 38 ImageNet networks toward shape mainly by removing texture evidence (median 89% of the effect) and the Stylized-ImageNet network mainly by adding shape evidence (texture share 11%; Supplementary Table S14).

Cortex loads on the texture-leaning units, not on the discriminant subspaces. With the two unit halves of the no-diet single trunk as encoding bands, the texture-leaning half carries every extrastriate group relative to V1 in 8/8 subjects ($\Delta = -0.32$, exact $p = .0078$), and the cortex-optimal weight of the lossy sum falls from 0.80 in V1 to 0.37–0.48 beyond hV4 (7/8, $p = .023$) while the behavioural rate rises toward $w = 1$ (Fig. 6c, d; Supplementary Table S12). In ImageNet networks the texture-leaning half carries early visual cortex as well, so the V1-relative prediction fails there, and the discriminant subspaces that move the rate are nearly silent as unique cortical predictors (at most 1.3% of normalised unique variance each), so the cortical claim is made under the unit rule only (Supplementary Note S4). The architecture is therefore not what makes the rate a readout property, nor, in the networks trained on our images, what makes extrastriate cortex load on the texture-biased component; it makes the components separable enough for one weight to move the rate from 0.48 to 0.85 (by 0.24 for a refitted head held within five points of peak accuracy), and for cortex to be carried almost entirely by one of them.

This is the mechanism we propose for the paradox: a system holding a shape-biased and a texture-biased component, whose behaviour reads out from both, will look shape-biased behaviourally and texture-biased wherever the texture-biased component dominates — and in this model of the subcortical split, the component that dominates beyond V1 is the texture-biased one, and the shape-biased component exists at all only if its input matured during training. It is an existence proof with a cortically grounded demonstration, not a measurement of the human readout.

From 2.5 Properties that do not follow from shape bias.

Robustness. The standard reading of Geirhos et al. (2018a, 2021) is that shape bias buys out-of-distribution robustness. The BA study pre-registered that prediction for its diet and it failed (Supplementary Table S5); the model repeats the failure against a control that sees the same test images (Methods 4.6; Fig. 5a; Extended Data Fig. 1). In relative-accuracy area under the curve, the model is less robust than its control on four of the five corruption families neither was trained on — uniform noise (-0.135), high-pass (-0.136), false colour (-0.084) and rotation (-0.040), every seed of the model below every seed of the control (exact Mann–Whitney $p = 0.10$, the floor for three seeds against three) — within 0.01 of it on phase scrambling ($+0.009$), and more robust only to low-pass ($+0.078$), the family its trajectory trains in, although at test its own input is no longer blurred; contrast, the other trained-in family, is level (-0.004, seeds interleaved). Robustness appears where it was trained in and nowhere else. The coupling belongs to the stylisation route of Geirhos et al., not to shape bias; against the source of the diet we reproduce its shape-based half and not its robust half (Supplementary Note S4; Methods 4.6).

Agreement with human decisions. Error consistency — Cohen's κ on trial-by-trial agreement — subtracts the agreement two decision-makers' marginal rates force (Geirhos et al., 2020); across the ten observers the human–human ceiling is $\kappa = 0.234$. In the BA network the most shape-biased readout, its magno stream at 0.878, had the lowest error consistency of seven models compared (3.5% of the ceiling; Supplementary Table S6). The model reverses that observation: its magno stream (0.850) is its most human-consistent readout, at 17.3% of the ceiling, above its integrated readout (13.2%), its parvo stream (10.3%) and the control (11.9% integrated, 12.6% magno), and the best of its 63 sweep readouts reaches 29.0%. Neither observation is about shape bias. Across the seven models now compared, κ rises with the rate by 0.019 [0.011, 0.027] (90% interval) over the 0.68 span of shape bias they cover, 8.1% of the ceiling; across the field's zoo agreement rises with

the rate to at most 37% of the ceiling, and that rise is the measure's own bound: Cohen's κ cannot exceed a value fixed by the two marginals, one quadratic in the rate accounts for the agreement ($R^2 = 0.90$), and at a matched rate of 0.40 agreement separates neither the zoo's representations nor the Stylized-ImageNet network (four of six predictions of a fourth plan failed; Supplementary Table S13; Supplementary Note S4). At a matched rate it does not separate our seeds either: the three magno streams sit at 0.871, 0.842 and 0.838 and reach 5.1%, 25.6% and 21.3% of the ceiling. The BA study's pre-registered prediction, that the diet would make the network more human-consistent than its control, is not supported: the model's integrated readout reaches 13.2% of the ceiling (seeds 9.4–17.7%) and the control's 11.9% (10.3–13.2%), with the seeds interleaved (exact Mann–Whitney $p = 1.0$; magno streams 17.3% against 12.6%, $p = 0.70$). The measure does detect a correspondence when one is there, and here the trajectory supplies it. Asked not which cue the observers pick on an image but whether it resolves to either cue at all, the model's readout recovers 22.3% of the split-half ceiling on held-out images (seeds 19.6–24.3%), against 9.0% for the control (6.6–10.4%) and 8.2% for the channel swap (7.1–10.4%), every model seed above every seed of both (exact Mann–Whitney $p = 0.10$, the floor); on which cue is picked the three are interleaved (13.3%, 10.4% and 9.5%; $p \geq 0.70$; Fig. 5c). The readout weight that recovers it is magno-heavy (0.91–0.98, chosen on held-out images). The BA network showed the same contrast against its no-diet arm (24.1% against 9.2%), but crossing weights and test input removed it in both directions (-0.6% and -3.7%; Fig. 5d): it measured whether a network is tested on the input it was trained on, not the diet. The model and its control are each tested on the input they finished training on, and on the same one, so the difference between them belongs to the trajectory.

Representational dimensionality. The magno/parvo ratio of the streams' participation ratios on NSD images ends above parity in every subject and seed of the model (24/24; seed means 1.88–1.98) and below it in 23 of its control's 24 (seed means 0.89–0.95), from the same untrained starting point (1.63) and the same test input (Extended Data Fig. 2). What the trajectory decides is which side of parity training ends on, as the BA diet did against its no-diet network (seed means 1.60–2.51 against 0.91–0.99). That dimensionality is not shape bias is shown twice. The channel swap also ends above parity (24/24; seed means 1.27–2.15) with neither stream shape-biased, and for the opposite reason: its ramped chromatic stream contracts (participation ratio 52–86 against 116–124 in the control, seed means) while its luminance stream stays where the control's is (109–113 against 108–110), whereas in the model the ramped luminance stream expands (272–281). And the BA network's `parvo_highsf` control's second stream becomes the more dimensional stream while remaining texture-leaning (probe -0.139 on held-out exemplars).

From 3. Discussion.

What the readout account licenses. A model matching the human shape-bias rate is routinely described as having human-like representations; here that inference fails twice. The same network holds components at $+0.32$ and -0.16 on held-out exemplars and behaves at 0.682, and one scalar over frozen representations moves the rate across most of the range the field's models occupy, so the rate does not identify a representation; and trial-by-trial agreement with the observers never exceeds 29% of the human–human ceiling in our networks and rises with the rate only as far as the bound on κ allows, so the rate is at best a weak proxy for the decision process. Prior work established that a representation can hold more shape than its behaviour reveals (Hermann et al., 2020; Islam et al., 2021), that the measured rate depends on the test design (Tartaglini et al., 2022) and on a prompt in vision–language models (Gavrikov et al., 2024), and that it tracks a network's spatial-frequency channel (Subramanian et al., 2023); new here is the form of the dissociation — both representations held fixed, the rate moved by the readout alone — and its cortical grounding: the same mixing weight, fitted to cortex, prefers the magno-heavy end in V1 and the parvo-heavy end beyond it (Extended Data Fig. 5). Nor does the account need the split: in networks with none

a gain on their own shape- and texture-predictive subspaces moves the rate on unseen exemplars and re-orders the zoo, and the texture-leaning units of our single trunks again carry extrastriate cortex (Fig. 6); the split amplifies the dissociation (a span of 0.37 between the streams against 0.20 for the same network's concatenated features with exemplars held out), it does not create it.

How to report shape bias. A shape-bias claim should carry three numbers, each answering a different question: the rate at the trained readout (an operating point); the span the same frozen features reach under a reciprocal gain on their own shape- and texture-predictive subspaces within an accuracy budget (how much of the rate the readout controls); and a probe's shape-minus-texture accuracy (what the representation holds) — the last two on held-out exemplars, since the standard set reuses 48 texture images across 1,200 stimuli and exemplar-shared estimates inflate both. The probe's preprocessing must be fixed too, because with a few hundred training images it sets the probe's size: we fix exemplar-balanced leave-one-exemplar-out folds with PCA and scaling fitted inside each fold and an untuned $C = 1$ (Methods 4.3), and read the probe for sign and order, since the ImageNet median runs from -0.14 to -0.39 across four other held-out preprocessings ($+0.03$ for one that standardises components fitted on all images) while every one puts the Stylized-ImageNet network above all 38 ImageNet networks. Across 38 ImageNet networks the three rank the zoo differently ($\tau = 0.63$, and 0.43 for the probe, 0.28–0.54 across preprocessings); error consistency, which at a network's own readout largely re-expresses the rate, should be read at a matched rate (Supplementary Table S13). None of these numbers alone licenses a claim about representations. One command in the repository prints all three, with the exemplar-shared values beside them, for any torchvision network (Methods 4.8).

What a developmental trajectory does, and what it costs. The intuitive account of low-pass developmental input is informational: coarse input shields the learner from texture, so it learns shape. Our results support a version of it in which the input's history matters and the information at test does not. The model and its control see the same images at test, and the untrained network on that input carries no more shape bias than undieted luminance; what differs is whether the magno stream's early learning happened on coarse, low-contrast input. Its route runs through the competition between the trunks: the coarse early input hands the task to the chromatic trunk, and the luminance trunk, learning beside it from a world that sharpens, ends the weaker and shape-biased; the swap shows that being the weaker trunk is not enough (Results 2.2). The trajectory is thus a constraint on the solution learning converges to, not a filter on what the network sees — the BA network's fixed 48-month transform conflated the two, putting most of its magno stream's shape bias into the input ($+0.210$ of $+0.337$ before training) at a cost in accuracy (0.556), stability and BOLD prediction (0.395). The trajectory costs less: 8.1 points of top-1 against its control, robustness to corruptions it was not trained on, and 0.024 of noise-ceiling-normalised r^2. Nothing here recommends it as a training method for accuracy; its interest is as a model of development, and its prediction is developmental: a visual system whose early input is coarse should show a stream-level shape bias that a system given adult input from the start, on the same images, does not.

Limitations. The load-bearing cortical result rests on eight subjects from one dataset and is read as sign and ordering only. The mapping between our chromatic stream and texture-selective cortex is by explanatory role, not content: nothing here asserts that the ventral stream is chromatic, the prediction that the chromatic stream should win in colour-biased cortex fails in the BA network (pooled VO $+0.005$, bounded within ± 0.017), and substituting the koniocellular axis makes it worse (Supplementary Note S2). In every network the trunk that carries extrastriate cortex is also the one that carries the task, so the cortical result cannot separate texture bias from task load; trunks matched in accuracy would. One trajectory shape was tested — linear in age from birth to 300 months over 50 epochs — together with one seed of a trajectory stopping at 48 months;

how the result depends on the shape and length of the trajectory is open, and so is whether its direction matters: the control removes both the coarse-to-fine order and the exposure to coarse input, and a reversed or age-shuffled schedule tested at the same endpoint would separate them. No single-stream network was trained with the trajectory, so whether its shape bias, like the BA diet's, survives only beside a second stream is untested. The model's recipe (AdamW, cosine schedule, weight averaging, label smoothing, 50 epochs) differs from the thesis's, so comparisons with the BA network mix diet and recipe; the BA diet trained with the model's recipe (0.665 top-1, shape bias 0.620, gradient −0.735) separates the two. The model of record was changed after its first seed was seen; the rule for the change, and the control that could have blocked it, were fixed before seeds 1–2 and the control were trained (Methods 4.2). Whether a two-stream architecture specialises on its own is untestable here, because the trunks are bit-identical at initialisation. Human comparisons are against ten published observers and of aggregate rate; the readout weight is a hyperparameter that nothing selects. The cue-conflict set reuses its exemplars, so spans and probes are reported with exemplars held out (about 60% and two thirds of their exemplar-shared values in the zoo); the unit rule's η^2 split, fitted on all images, has not been recalibrated this way. Seven plans were committed before their analyses ran, all on the BA network or the zoo, and every outcome is reported; their ledgers, failures included, are summarised in Supplementary Note S5. The sixth, re-estimating spans and probes with exemplars held out, is post hoc with respect to the five before it, as are the held-out accuracy gate, the reversal floors and the probe's preprocessing spread added to it after a later referee simulation. Of the main study's four pre-registered predictions, two failed on the BA network and are reported as failures; on the model, the robustness prediction fails again and the error-consistency one is not supported (model and control seeds interleaved). Claims deliberately not made are listed in Supplementary Note S1.

Supplementary Tables

Supplementary Table S1 | Monotonicity of the profile and the exact permutation null. Spearman ρ between hierarchical rank [1,1,2,2,3,3,4] and stream preference, mean over subjects; the null enumerates all 630 distinct labellings of the seven ROIs (one-sided in the pre-stated direction, floor $1/630 = .0016$). "per-voxel index" is the manuscript's statistic (Fig. 3c; Results 2.3): the median over reliable voxels of (U_m − U_p)/(U_m + U_p) per ROI, as in Extended Data Table 1; "ratio of medians" is median(U_m − U_p)/median(U_m + U_p) per ROI, the statistic of an earlier draft, kept for comparison. Source: `results/hierarchy_permutation_null.json` (which labels the two `raw` and `norm`).

arm	seed	statistic	observed mean ρ	exact p	most extreme of 630	null 2.5th pct
diet	0	per-voxel index	-0.964	0.0016	yes	-0.743
diet	0	ratio of medians	-0.964	0.0016	yes	-0.754
diet	1	per-voxel index	-0.973	0.0016	yes	-0.743
diet	1	ratio of medians	-0.973	0.0016	yes	-0.750
diet	2	per-voxel index	-0.966	0.0016	yes	-0.756
diet	2	ratio of medians	-0.966	0.0016	yes	-0.748

arm	seed	statistic	observed mean ρ	exact p	most extreme of 630	null 2.5th pct
no diet	0	per-voxel index	-0.571	0.0206	no	-0.559
no diet	0	ratio of medians	-0.502	0.0286	no	-0.502
no diet	1	per-voxel index	-0.661	0.0190	no	-0.594
no diet	1	ratio of medians	-0.606	0.0190	no	-0.546
no diet	2	per-voxel index	-0.463	0.0921	no	-0.594
no diet	2	ratio of medians	-0.580	0.0365	no	-0.616

Supplementary Table S2 | In-silico cue conflict through the fitted encoding models, per seed. Median over reliable voxels of the per-voxel predicted shape-bias index for each stream, mean over 8 subjects; the contrast is magno − parvo, paired within subject against the no-diet arm of the same seed. Source: `results/encoding_cc/`.

seed	arm	magno stream	parvo stream	magno − parvo	diet − no diet	sep.	p
0	diet	+0.791	−0.743	+1.534	+1.424	8/8	0.0078
0	no diet	−0.638	−0.748	+0.110			
1	diet	+0.796	−0.770	+1.566	+1.468	8/8	0.0078
1	no diet	−0.601	−0.699	+0.098			
2	diet	+0.799	−0.759	+1.558	+1.474	8/8	0.0078
2	no diet	−0.613	−0.696	+0.083			

Supplementary Table S3 | The carrier × transform 2×2 on the probe and on behaviour. Mean over training seeds with the seed range in brackets; "decided" is the mean number of the 1,200 cue-conflict images the readout classifies as the shape or the texture category (the shape-bias denominator). Cells are read out on stream 2 in three cases and on the diet's magno stream in the fourth. Source: `results/figures/fig1_data.npz` (from `results/shape_texture/`, `results/cue_conflict/`). "probe bias, exemplars held out" is the exemplar-balanced probe of Methods 4.3, scored on object and texture exemplars the probe was not trained on (three exemplars per category for both labels, leave one exemplar out, PCA inside the folds, mean of 20 draws; `docs/PLAN_EXEMPLAR_2026-09-21.md`, `results/figures_v2/exemplar_streams.json`); "probe bias" is the original stratified-fold probe, whose test images share exemplars with its training folds.

carrier	transform	condition, stream	probe bias	probe bias, exemplars held out	cue-conflict shape bias	decided / 1,200	seeds
chromatic (L−M)	none	no diet, parvo	−0.504 [0.035]	−0.118 [0.021]	0.364 [0.080]	270	3
chromatic (L−M)	yes	swap, parvo	−0.145 [0.043]	−0.002 [0.022]	0.495 [0.038]	165	3
achromatic (luminance)	none	parvo_highsf, stream 2	−0.271 [0.023]	−0.139 [0.012]	0.389 [0.028]	598	3

carrier	transform	condition, stream	probe bias	probe bias, exemplars held out	cue-conflict shape bias	decided / 1,200	seeds
achromatic (luminance)	yes	diet, magno	+0.819 [0.014]	+0.337 [0.007]	0.878 [0.022]	488	3

Supplementary Table S4 | Single-stream controls trained with the identical diet, and the readout ladder. Best saved checkpoint per seed, except the no-diet row, which is scored at the final checkpoint because its accuracy does not degrade (Fig. 2d; Methods 4.7); mean [range] over three seeds. Accuracy is validation top-1 at the scored checkpoint. Sources: `results/cue_conflict_bestep/`, `results/cue_conflict_single34_bestep/`, `results/cue_conflict_single_bestep/`, run histories, `results/param_matched/`.

model / readout	parameters	shape bias	decided / 1,200	accuracy
diet, magno stream alone	22,397,936	0.903 [0.888, 0.916]	504	0.556 [0.552, 0.564]
diet, both streams (w = 0.5)	22,397,936	0.655 [0.594, 0.718]	502	0.556 [0.552, 0.564]
diet, parvo stream alone	22,397,936	0.400 [0.372, 0.429]	388	0.556 [0.552, 0.564]
single stream, ResNet-34 (matched)	21,338,992	0.625 [0.604, 0.652]	537	0.623 [0.616, 0.634]
single stream, ResNet-18 (half)	11,230,832	0.567 [0.541, 0.590]	501	0.599 [0.589, 0.608]
no diet, both streams	22,397,936	0.416 [0.394, 0.433]	593	0.701 [0.698, 0.706]

Supplementary Table S5 | Out-of-distribution robustness: relative-accuracy AUC per corruption family and condition. Mean over three seeds [range]; the thesis's schedule-trained checkpoint is a single run. "sep." marks complete seed separation between the diet and no-diet arms (the exact 3 v 3 floor, p = 0.100). Families marked ‡ were declared trained-in before the run. Source: `results/ood_bestep/`.

family	diet	no diet	swap	parvo_highsf	thesis schedule	diet − no diet	sep.
uniform-noise	0.634 [0.619, 0.647]	0.729 [0.720, 0.735]	0.749 [0.744, 0.758]	0.767 [0.758, 0.773]	0.640	−0.095	yes
high-pass	0.678 [0.647, 0.696]	0.827 [0.820, 0.831]	0.831 [0.817, 0.841]	0.783 [0.775, 0.791]	0.680	−0.149	yes
false-colour	0.854 [0.845, 0.860]	0.938 [0.933, 0.943]	0.966 [0.960, 0.971]	0.874 [0.868, 0.880]	0.839	−0.084	yes
rotation	0.574 [0.567, 0.582]	0.615 [0.613, 0.616]	0.605 [0.597, 0.612]	0.611 [0.602, 0.616]	0.572	−0.041	yes

family	diet	no diet	swap	parvo_highsf	thesis schedule	diet − no diet	sep.
phase-scrambling	0.382 [0.365, 0.391]	0.373 [0.369, 0.377]	0.371 [0.365, 0.375]	0.382 [0.380, 0.383]	0.362	+0.009	no
contrast ‡	0.481 [0.460, 0.493]	0.549 [0.530, 0.578]	0.543 [0.539, 0.550]	0.511 [0.498, 0.526]	0.489	−0.068	yes
low-pass ‡	0.444 [0.434, 0.464]	0.356 [0.353, 0.359]	0.359 [0.353, 0.362]	0.385 [0.375, 0.392]	0.436	+0.089	yes

Supplementary Table S6 | Error consistency (Cohen's κ) with the ten human observers. The trained readout (w = 0.5), the magno stream alone (w = 1; the $\kappa = 0.0082$, 3.5% of Results 2.5 is the mean of the three final-checkpoint runs) and the best readout over the 21 values of w, per run; κ as a percentage of the human–human ceiling ($\kappa = 0.234$, 45 pairs). Published CNN values are from Geirhos et al. (2020). Source: `results/sweep_error_consistency/`.

run	shape bias at w = 0.5	κ at w = 0.5	% ceiling	κ at w = 1 (magno alone)	% ceiling	rate at w = 1	best κ over w	% ceiling	w at best	rate at best
dkl_base_seed0_final	0.421	+0.0309	13.2	+0.0404	17.3	0.438	+0.0404	17.3	1.00	0.438
dkl_base_seed1_final	0.431	+0.0333	14.3	+0.0390	16.7	0.431	+0.0421	18.0	0.75	0.429
dkl_base_seed2_final	0.392	+0.0372	15.9	+0.0349	15.0	0.400	+0.0412	17.6	0.35	0.410
dkl_bcs_seed0_best	0.594	+0.0096	4.1	+0.0178	7.6	0.890	+0.0178	7.6	1.00	0.890
dkl_bcs_seed0_final	0.692	+0.0234	10.0	+0.0361	15.5	0.872	+0.0405	17.3	0.90	0.857
dkl_bcs_seed1_best	0.718	+0.0269	11.5	+0.0058	2.5	0.913	+0.0425	18.2	0.35	0.614
dkl_bcs_seed1_final	0.686	+0.0319	13.7	+0.0028	1.2	0.894	+0.0437	18.7	0.45	0.651
dkl_bcs_seed2_best	0.652	+0.0372	15.9	-0.0055	-2.3	0.908	+0.0483	20.7	0.85	0.847
dkl_bcs_seed2_final	0.640	+0.0280	12.0	-0.0143	-6.1	0.871	+0.0405	17.3	0.75	0.784
AlexNet (published)	0.429	+0.0320	13.7							
GoogLeNet (published)	0.312	+0.0250	10.7							
VGG-16 (published)	0.172	+0.0170	7.3							

Supplementary Table S7 | The estimator's own bias, and the residual-misfit variants. Synthetic voxels were generated from each arm's fitted 1,000-component feature spaces with the measured per-ROI noise ceilings imposed and the true magno − parvo difference set to zero everywhere; the recovered contrast is the instrument's signed floor. "Displacement" is the real contrast minus the floor, paired per subject. The six variants differ in how the floor is paired (group or per-subject) and whether the residual misfit of the recovery simulation is also removed (before or after rescaling by the recovered slope). Sources: `results/vp_recovery_realnc/diet_floor_paired_2026-09-07.txt`, `floor_pairing_2026-09-07.txt`, `prediction_scored_2026-09-07.txt`.

quantity	diet	no diet
floor, normalised contrast (true = 0)	+0.0375 (8/8 > 0, p = .0078)	+0.0511 (8/8 > 0, p = .0078)
floor, raw contrast	−0.0021 (p = .15)	+0.0009 (p = .84)
real contrast, normalised (seed 0)	−0.8335	−0.0785
displacement real − floor	−0.871 (8/8, p = .0078)	−0.130 (8/8, p = .0078); n = 7: −0.103 (7/7, p = .0156)
recovered slope of estimate vs truth	1.210	1.281
variant: group floor, minimal	−0.720 (8/8, p = .0078)	−0.101 (8/8, p = .0078)
variant: per-subject floor, minimal	−0.720 (8/8, p = .0078)	−0.101 (8/8, p = .0078)
variant: group floor + residual before scaling	−0.626 (8/8, p = .0078)	−0.034 (6/8, p = .055)
variant: per-subject floor + residual before scaling	−0.626 (8/8, p = .0078)	−0.034 (5/8, p = .31)
variant: group floor + residual after scaling	−0.607 (8/8, p = .0078)	−0.015 (4/8, p = 1.00)
variant: per-subject floor + residual after scaling	−0.607 (8/8, p = .0078)	−0.015 (5/8, p = .84)
per-ROI normalised floor, V1v ... hV4 (no-diet geometry)		+0.025, +0.025, +0.026, +0.025, +0.027, +0.026, +0.026 (ρ vs rank +0.54, p = .22)

Supplementary Table S8 | Stream preference regressed on pRF eccentricity and pRF size within subject. Per-voxel stream-preference index regressed on log pRF eccentricity and log pRF size (NSD pRF estimates, voxels with pRF $R^2 > 5\%$, noise ceiling > 0.1); "ecc | size" is the partial slope for eccentricity controlling for size and vice versa; mean over 8 subjects, with the number of subjects on the stated side and the exact Wilcoxon p. Collinearity between the two predictors is modest (mean VIF 1.66). The arm interaction is the paired difference of partial slopes, seed-averaged within arm. Sources: `results/prf_continuous/dissociation_2026-09-08.txt`, `interaction_2026-09-08.txt`, `input_null_2026-09-08.txt`. The paired trained − untrained rows compare each subject's partial slope in the trained seed-0 network with the untrained network on the diet front end; training with the diet changes neither statistic significantly and the two disagree in sign, while training without the diet removes the dissociation on the raw statistic.

model	statistic	ecc alone	size alone	ecc | size	size | ecc
diet, seed 0	normalised	+0.066 (0/8 neg, .0078)	−0.013 (5/8 neg, .31)	+0.119 (0/8 neg, .0078)	−0.086 (8/8 neg, .0078)
diet, seed 1	normalised	+0.065 (0/8, .0078)	−0.029 (7/8, .039)	+0.132 (0/8, .0078)	−0.110 (8/8, .0078)
diet, seed 2	normalised	+0.064 (0/8, .0078)	−0.012 (5/8, .25)	+0.115 (0/8, .0078)	−0.083 (8/8, .0078)
no diet, seed 0	normalised	+0.018 (4/8, .64)	+0.033 (1/8, .039)	−0.004 (4/8, 1.00)	+0.035 (0/8, .0078)
no diet, seed 1	normalised	+0.004 (4/8, .95)	+0.015 (4/8, .38)	−0.009 (6/8, .46)	+0.021 (2/8, .11)
no diet, seed 2	normalised	−0.007 (5/8, .38)	+0.012 (3/8, .31)	−0.022 (6/8, .25)	+0.024 (1/8, .016)
diet, seed 0	raw	+0.103 (1/8, .039)	−0.158 (8/8, .0078)	+0.325 (0/8, .0078)	−0.356 (8/8, .0078)

model	statistic	ecc alone	size alone	ecc | size	size | ecc
no diet, seed 0	raw	+0.130 (1/8, .023)	+0.181 (0/8, .0078)	+0.028 (3/8, .84)	+0.163 (0/8, .0078)
Gabor bank, diet front end (input null)	normalised			+0.035 (6/8 pos, .15)	+0.002 (4/8 neg, .95)
untrained network, diet front end	normalised			+0.083 (7/8 pos, .016)	−0.038 (6/8 neg, .039)
untrained network, undieted front end	normalised			+0.089 (7/8 pos, .023)	−0.038 (8/8 neg, .0078)
trained diet − untrained (diet front end), paired	normalised			+0.036 (6/8 pos, .11)	
trained diet − untrained (diet front end), paired	raw			−0.077 (2/8 pos, .11)	
trained no diet − untrained (diet front end), paired	normalised			−0.087 (2/8 pos, .15)	
trained no diet − untrained (diet front end), paired	raw			−0.374 (0/8 pos, .0078)	
arm interaction, diet − no diet (seed-averaged)	normalised			+0.134 (8/8, .0078)	−0.119 (0/8 pos, .0078)
arm interaction, diet − no diet (seed-averaged)	raw			+0.305 (8/8, .0078)	−0.502 (0/8 pos, .0078)

Supplementary Table S9 | Cue-conflict shape bias under the forced two-way estimator. The primary estimator (Methods 4.3) restricts the logits to the 16 categories, takes the argmax, and scores shape against texture over the images decided as one of the two, so images decided as a third category drop out of the denominator ("decided"). The forced estimator keeps all 1,200 images and scores, per image, whether the logit of the shape category exceeds the logit of the texture category. Mean over seeds [range]. Source: the stored per-image logits in `results/cue_conflict*/`.

model / readout	restricted (primary)	decided / 1,200	forced two-way	seeds
diet, magno stream alone (best ckpt)	0.903 [0.888, 0.916]	504	0.775 [0.767, 0.788]	3

model / readout	restricted (primary)	decided / 1,200	forced two-way	seeds
diet, both streams w = 0.5 (best ckpt)	0.655 [0.594, 0.718]	502	0.588 [0.574, 0.609]	3
diet, parvo stream alone (best ckpt)	0.400 [0.372, 0.429]	388	0.414 [0.394, 0.428]	3
diet, magno stream alone (final ckpt)	0.878 [0.870, 0.892]	488	0.761 [0.757, 0.764]	3
diet, both streams w = 0.5 (final ckpt)	0.675 [0.646, 0.692]	475	0.608 [0.596, 0.618]	3
diet, parvo stream alone (final ckpt)	0.418 [0.406, 0.438]	353	0.449 [0.411, 0.472]	3
no diet, both streams	0.416 [0.394, 0.433]	593	0.375 [0.357, 0.388]	3
no diet, magno stream alone	0.421 [0.398, 0.434]	602	0.378 [0.352, 0.398]	3
no diet, parvo stream alone	0.364 [0.311, 0.391]	270	0.442 [0.410, 0.473]	3
swap, both streams	0.408 [0.403, 0.412]	586	0.365 [0.362, 0.367]	3
swap, parvo stream alone (L−M + transform)	0.495 [0.478, 0.516]	165	0.511 [0.491, 0.531]	3
parvo_highsf, both streams	0.486 [0.450, 0.516]	618	0.450 [0.438, 0.457]	3
parvo_highsf, stream 2 alone (L−M + luminance)	0.389 [0.374, 0.402]	598	0.346 [0.324, 0.358]	3
single stream, ResNet-34 (matched)	0.625 [0.604, 0.652]	537	0.561 [0.552, 0.573]	3
single stream, ResNet-18 (half)	0.567 [0.541, 0.590]	501	0.523 [0.506, 0.538]	3
no diet, accuracy-matched (epochs 10, 10, 11), magno alone	0.450 [0.432, 0.465]	533	0.364 [0.357, 0.370]	3
thesis checkpoint (progressive schedule), both streams	0.697	620	0.636	1
untrained, diet front end, magno alone	0.513	154	0.510	1
untrained, no-diet front end, magno alone	0.568	169	0.508	1

Supplementary Table S10 | Stream preference beyond hV4, on every NSD atlas. Median over reliable voxels of the per-voxel index (Um − Up)/(Um + Up) (noise ceiling > 0.1, Um + Up >

10^{-3}, $\geq$ 20 voxels per ROI per subject), mean over the eight subjects (n = subjects with enough voxels); the no-diet, diet, swap and parvo + high-SF arms are each the mean over three training seeds per subject, and the thesis column is the thesis's one checkpoint (Methods 4.5). Positive = magno band, negative = parvo band. Atlases: Wang et al. (2015) Kastner2015 ROIs beyond hV4 (the V1–hV4 rows are Extended Data Table 1 and Fig. 3), NSD's `streams` atlas, and the fLoc category-selective ROIs. Regional contrasts and the exact tests are in Extended Data Fig. 4c and `results/figures_v2/ext_gradient.md`. Source: the stored voxelwise fits in `results/encoding_cc/` and `results/encoding/`, relabelled; no refitting.

atlas	ROI	untrained (diet front end)	no diet (3 seeds)	diet (3 seeds)	swap (3 seeds)	parvo + high-SF (3 seeds)	thesis ckpt (n = 1)	voxels (median, diet)
Kastner2015	VO1	+0.00 (n=8)	+0.92 (n=8)	−0.87 (n=8)	+0.98 (n=8)	−0.99 (n=8)	−0.94 (n=8)	196
Kastner2015	VO2	+0.12 (n=8)	+0.97 (n=8)	−0.90 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	220
Kastner2015	PHC1	+0.41 (n=8)	+0.98 (n=8)	−0.89 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	148
Kastner2015	PHC2	+0.46 (n=8)	+0.99 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.93 (n=8)	150
Kastner2015	LO1	+0.13 (n=8)	+0.92 (n=8)	−0.83 (n=8)	+0.98 (n=8)	−0.98 (n=8)	−0.90 (n=8)	230
Kastner2015	LO2	−0.14 (n=8)	+0.92 (n=8)	−0.90 (n=8)	+0.98 (n=8)	−1.00 (n=8)	−0.94 (n=8)	166
Kastner2015	TO1	+0.25 (n=8)	+0.96 (n=8)	−0.86 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.91 (n=8)	280
Kastner2015	TO2	+0.20 (n=8)	+0.96 (n=8)	−0.89 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.90 (n=8)	36
Kastner2015	V3A	+0.39 (n=8)	+0.91 (n=8)	−0.81 (n=8)	+0.98 (n=8)	−0.98 (n=8)	−0.84 (n=8)	278
Kastner2015	V3B	+0.24 (n=8)	+0.96 (n=8)	−0.87 (n=8)	+0.99 (n=8)	−0.99 (n=8)	−0.94 (n=8)	254
Kastner2015	IPS0	+0.15 (n=8)	+0.98 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.92 (n=8)	317
Kastner2015	IPS1	−0.06 (n=8)	+0.98 (n=8)	−0.86 (n=8)	+1.00 (n=8)	−0.99 (n=8)	−0.92 (n=8)	239
Kastner2015	IPS2	+0.02 (n=8)	+1.00 (n=8)	−0.92 (n=8)	+1.01 (n=8)	−1.00 (n=8)	−0.94 (n=8)	47
streams	early	+0.27 (n=8)	+0.86 (n=8)	−0.42 (n=8)	+0.96 (n=8)	−0.93 (n=8)	−0.45 (n=8)	3133
streams	midventral	−0.15 (n=8)	+0.87 (n=8)	−0.84 (n=8)	+0.97 (n=8)	−0.99 (n=8)	−0.86 (n=8)	708
streams	midlateral	−0.04 (n=8)	+0.91 (n=8)	−0.86 (n=8)	+0.97 (n=8)	−0.99 (n=8)	−0.90 (n=8)	571
streams	midparietal	+0.32 (n=8)	+0.94 (n=8)	−0.84 (n=8)	+0.99 (n=8)	−0.99 (n=8)	−0.91 (n=8)	512
streams	ventral	+0.08 (n=8)	+0.96 (n=8)	−0.91 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.95 (n=8)	2724
streams	lateral	+0.25 (n=8)	+0.97 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.91 (n=8)	1842
streams	parietal	+0.12 (n=8)	+0.99 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.93 (n=8)	1082

atlas	ROI	untrained (diet front end)	no diet (3 seeds)	diet (3 seeds)	swap (3 seeds)	parvo + high-SF (3 seeds)	thesis ckpt (n = 1)	voxels (median, diet)
floc-faces	OFA	−0.07 (n=8)	+0.90 (n=8)	−0.89 (n=8)	+0.98 (n=8)	−0.99 (n=8)	−0.92 (n=8)	367
floc-faces	FFA-1	+0.01 (n=8)	+0.95 (n=8)	−0.91 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.94 (n=8)	399
floc-faces	FFA-2	+0.22 (n=7)	+0.98 (n=7)	−0.92 (n=7)	+1.00 (n=7)	−1.00 (n=7)	−0.95 (n=7)	260
floc-places	OPA	+0.34 (n=8)	+0.98 (n=8)	−0.86 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.93 (n=8)	1070
floc-places	PPA	+0.44 (n=8)	+0.99 (n=8)	−0.88 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	620
floc-bodies	EBA	+0.17 (n=8)	+0.97 (n=8)	−0.88 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.92 (n=8)	2174
floc-bodies	FBA-1	−0.07 (n=6)	+0.96 (n=6)	−0.91 (n=6)	+1.00 (n=6)	−1.00 (n=6)	−0.95 (n=6)	314
floc-bodies	FBA-2	+0.13 (n=8)	+0.98 (n=8)	−0.92 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	252
floc-words	OWFA	−0.29 (n=8)	+0.88 (n=8)	−0.85 (n=8)	+0.96 (n=8)	−0.99 (n=8)	−0.88 (n=8)	392
floc-words	VWFA-1	−0.16 (n=8)	+0.96 (n=8)	−0.91 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.96 (n=8)	454
floc-words	VWFA-2	−0.02 (n=5)	+0.95 (n=6)	−0.94 (n=6)	+0.99 (n=6)	−1.00 (n=6)	−0.96 (n=6)	150
floc-words	mfs-words	−0.08 (n=8)	+0.97 (n=8)	−0.93 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.96 (n=8)	121

Supplementary Table S11 | The readout span without a stream split under the pre-specified unit-level rule: fixed-head gain sweep on the shape- and texture-leaning halves of the units. The special case of Extended Data Table 3 in which the two subspaces are spanned by disjoint sets of units. Each unit of a network's penultimate layer is scored by the fraction of its variance between shape categories minus that between texture categories on the 1,200 cue-conflict images (one-way η^2); the median split gives a shape-leaning half S and a texture-leaning half T. logits(w) = head(g(w) ⊙ y) with g = 2w on S and 2(1 − w) on T, the network's own head untouched; w = 0.5 is the trained network. The split is defined on one stratified half of the images and the rate scored on the other (10 random splits, both directions, the same splits as Extended Data Table 3; mean, with the range over splits and seeds in brackets). Rates by the paper's primary estimator (112-class, decided images) for our networks and the Geirhos 16-class probability aggregation for the ImageNet networks; "forced-estimator span" is the same sweep on the forced two-way estimator. "shape-leaning units" is the fraction of units with η^2_shape > η^2_texture on all images; for the two-stream concatenation the fraction of S that belongs to stream 1 is given in parentheses (P3). "w within 5 pts" is the w-range over which top-1 on held-out natural images (MiniEcoset validation; a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks) stays within 5 points of the trained value. "refit span" is the lossy-sum protocol z(w) = w·y_S + (1 − w)·y_T with a linear and a two-layer head refitted at every w (Methods 4.3 budget), our networks only. The last two columns are a sensitivity analysis added after the pre-specified analysis, at a referee's request: the same held-out sweep with S defined as the absolute shape-leaning set (units with d_j > 0) or as the top decile of units by d_j, and T as the remainder; given are the held-out span with its range over splits and seeds, the minimum Spearman ρ, the mean size of S and the w-range within 5 points of the trained top-1 (no gate exists for the two-stream concatena-

tion, which has no held-out natural-image features). Pre-specification: `docs/PLAN_GENERALITY_2026-09-14.md`. Source: `results/figures_v2/generality_behaviour.json`, aggregated by `tmp_analysis/figs_v2/generality_behaviour.py` from the per-network outputs of `src/analysis/generality/` (`slurm/generality_{cc,mini,inval,rules}.sh`); the per-network files are outside the repository.

network	units	trained rate	w = 0	w = 1	span (held-out) [range]	ρ(w, rate)	forced-estimate span	shape-learning units	w within 5 pts of trained top-1	refit span linear / two-layer	span, S = {d > 0}	span, top decile	seeds
two-stream, diet, streams concatenated (validation)	1024	0.675	0.420	0.879	0.461 [0.39, 0.57]	1.00	0.307	0.49 (S from stream 1: 0.99)	—	—	0.46 [0.39, 0.59] ($\rho \geq$ 0.98;	S	= 504)
single ResNet-34, diet	512	0.625	0.565	0.688	0.136 [0.03, 0.24]	0.71	0.102	0.17	0.20–0.78 (top-1 0.626)	0.06 / 0.05	0.12 [0.03, 0.23] ($\rho \geq$ -0.78;	S	= 89; gate 0.00–0.73)
single ResNet-18, diet	512	0.557	0.467	0.652	0.194 [0.11, 0.31]	0.97	0.131	0.09	0.28–0.72 (top-1 0.547)	0.09 / 0.06	0.11 [0.04, 0.18] ($\rho \geq$ -0.63;	S	= 44; gate 0.00–0.73)
single ResNet-18, no diet (`rgb_tsn`)	512	0.445	0.374	0.566	0.195 [0.11, 0.39]	0.96	0.164	0.05	0.27–0.75 (top-1 0.678)	0.10 / 0.09	0.15 [0.03, 0.31] ($\rho \geq$ -0.57;	S	= 24; gate 0.00–0.78)
ImageNet ViT-B/16	768	0.399	0.371	0.455	0.086 [0.07, 0.11]	0.98	0.138	0.01	0.00–1.00 (top-1 0.815)	—	0.21 [0.10, 0.34] ($\rho \geq$ 0.53;	S	= 6; gate 0.00–0.95)
ImageNet ResNet-152	2048	0.293	0.227	0.440	0.214 [0.18, 0.25]	1.00	0.196	0.02	0.15–0.90 (top-1 0.785)	—	0.27 [0.22, 0.33] ($\rho \geq$ 0.86;	S	= 45; gate 0.00–0.85)
ImageNet ConvNeXt-B	1024	0.269	0.229	0.343	0.115 [0.09, 0.14]	0.99	0.173	0.00	0.00–1.00 (top-1 0.843)	—	0.28 [0.20, 0.35] ($\rho \geq$ 0.14;	S	= 2; gate 0.00–0.95)

network	units	trained rate	w = 0	w = 1	span (held-out) [range]	ρ(w, rate)	forced-estimate span	shape-leaning units	w within 5 pts of trained top-1	refit span linear / two-layer	span, S = {d > 0}	span, top decile	seeds
ImageNet ConvNeXt-S	768	0.248	0.211	0.335	0.124 [0.11, 0.15]	0.99	0.179	0.00	0.00–1.00 (top-1 0.840)	—	undefined (empty half)	0.17 [0.12, 0.20] ($\rho \geq$ 0.72;	S
ImageNet ResNet-50	2048	0.222	0.167	0.346	0.180 [0.14, 0.23]	0.99	0.202	0.02	0.15–0.90 (top-1 0.760)	—	0.27 [0.20, 0.32] ($\rho \geq$ 0.75;	S	= 34; gate 0.00–0.85)
ImageNet VGG-16	4096	0.110	0.096	0.189	0.095 [0.07, 0.11]	0.96	0.106	0.01	0.00–1.00 (top-1 0.711)	—	0.40 [0.34, 0.46] ($\rho \geq$ -0.38;	S	= 56; gate 0.00–0.95)

Supplementary Table S12 | Cortex without a stream split: the shape- and texture-predictive components as encoding bands. Upper block (subspace bands): the three bands are the shape-predictive subspace S = Y·B_S (15 dimensions), the texture-predictive subspace T = Y·B_T (15) and the residual R = Y·B_R of Extended Data Table 3 (bases fitted on all 1,200 images), with U_S = r²_{STR} − r²_{TR} and U_T = r²_{STR} − r²_{SR} from the three-band model and the two two-band models (random search, 30 and 20 samples); "shared" is r²_{STR} − U_S − U_T and "r²" the three-band model's noise-ceiling-normalised r². w*_cortex there is the weight of the 15-dimensional lossy sum z(w) = w·s + (1 − w)·t alone. Lower block (unit halves): for each single-trunk network the penultimate units were split as in Supplementary Table S11 (all 1,200 images) and the two halves entered as two bands of a banded ridge model of every `nsdgeneral` voxel (same split, folds and alpha grid as Methods 4.5); U_shape and U_texture are the noise-ceiling-normalised unique variances and the index is (U_shape − U_texture)/(U_shape + U_texture), median over reliable voxels (noise ceiling > 0.1, ≥ 20 voxels), mean over eight subjects (seeds averaged per subject). Regions: V1–hV4 from `prf-visualrois`; ventral (VO1/2, PHC1/2), lateral (LO1/2, TO1/2) and dorsal (V3A/B, IPS0–2) from Kastner2015. The two-stream diet row applies the same computation to the magno (shape-biased) and parvo (texture-biased) bands of Fig. 3. "shared" is the cross-validated variance the bands account for jointly (the full model's r² minus the unique variances) and "r²" the full model's noise-ceiling-normalised r², medians over the same voxels. w*_cortex is the mixing weight of the lossy sum z(w) = w·y_S + (1 − w)·y_T that maximises an ROI's median test R² (Extended Data Fig. 5 protocol). Exact paired Wilcoxon over subjects. The unit-rule rows for ImageNet ViT-B/16 and ConvNeXt-B were run after the pre-registered analysis, at the identical protocol, to ask whether the unit-rule cortical result depends on architecture; they are post hoc and are marked as such in `docs/PLAN_GENERALITY_2026-09-14.md`. Source: `results/figures_v2/generality_cortex.json` and `results/figures_v2/subspace_cortex.json`, aggregated by `tmp_analysis/figs_v2/generality_cortex.py` from the per-subject fits of `src/analysis/generality/cortex_bands.py` (`slurm/generality_{nsd,cortex}.sh`, `slurm/subspace_{nsd,cortex}.sh`), which are deposited alongside the encoding outputs of Fig. 3 (Code availability).

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
subspace bands S, T, R											
single ResNet-18, no diet	index	+0.12	-0.01	+0.01	+0.03	+0.35	+0.13	+0.32	-0.088 (5/8, 0.4609)	+0.151 (2/8, 0.0781)	-0.25
	U_shape	0.011	0.009	0.010	0.010	0.013	0.009	0.011			
	U_texture	0.008	0.009	0.009	0.009	0.006	0.007	0.006			
	shared	0.329	0.311	0.305	0.327	0.521	0.409	0.375			
	r^2 (full model, nc-normalised)	0.351	0.332	0.327	0.351	0.542	0.429	0.396			
	w*_cortex (median)	0.63	0.68	0.68	0.63	0.33	0.33	0.33		Δw* extrastriate − V1 -0.23 (6/8 neg, 0.0391)	
single ResNet-34, diet	index	-0.03	-0.13	-0.08	+0.13	+0.19	-0.02	+0.14	+0.153 (2/8, 0.1094)	+0.130 (2/8, 0.3828)	+0.25
	U_shape	0.007	0.007	0.007	0.010	0.011	0.007	0.010			
	U_texture	0.007	0.008	0.008	0.007	0.008	0.007	0.008			
	shared	0.325	0.298	0.293	0.316	0.503	0.379	0.358			
	r^2 (full model, nc-normalised)	0.341	0.316	0.313	0.337	0.524	0.397	0.378			
	w*_cortex (median)	0.62	0.57	0.55	0.45	0.55	0.68	0.48		Δw* extrastriate − V1 -0.01 (4/8 neg, 0.9453)	
ImageNet ResNet-50	index	-0.33	-0.19	-0.23	+0.07	-0.16	-0.19	-0.24	+0.395 (1/8, 0.0156)	+0.131 (2/8, 0.1484)	+0.60
	U_shape	0.004	0.005	0.005	0.008	0.005	0.004	0.004			

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
	U_texture	0.007	0.006	0.007	0.006	0.006	0.005	0.006			
	shared	0.333	0.311	0.307	0.365	0.635	0.496	0.462			
	r^2 (full model, nc-normalised)	0.348	0.325	0.324	0.384	0.649	0.508	0.479			
	w*_cortex (median)	0.45	0.40	0.45	0.40	0.40	0.45	0.40		Δw* extrastriate − V1 -0.02 (5/8 neg, 0.6406)	
ImageNet ViT-B/16	index	-0.45	-0.54	-0.45	-0.25	-0.50	-0.43	-0.70	+0.193 (1/8, 0.0781)	-0.097 (3/8, 0.8438)	+0.37
	U_shape	0.004	0.003	0.004	0.007	0.004	0.003	0.002			
	U_texture	0.006	0.006	0.007	0.009	0.010	0.005	0.008			
	shared	0.254	0.242	0.254	0.292	0.555	0.404	0.382			
	r^2 (full model, nc-normalised)	0.267	0.256	0.271	0.310	0.573	0.416	0.398			
	w*_cortex (median)	0.50	0.40	0.45	0.15	0.50	0.40	0.55		Δw* extrastriate − V1 +0.08 (3/8 neg, 0.3828)	
ImageNet ConvNeXt-B	index	+0.17	+0.01	-0.17	+0.38	+0.00	-0.30	-0.51	+0.212 (3/8, 0.1953)	-0.440 (7/8, 0.0156)	+0.08
	U_shape	0.006	0.005	0.005	0.008	0.006	0.004	0.003			
	U_texture	0.004	0.003	0.004	0.003	0.004	0.006	0.005			
	shared	0.256	0.240	0.233	0.285	0.540	0.408	0.369			
	r^2 (full model, nc-normalised)	0.267	0.253	0.246	0.301	0.554	0.422	0.385			

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
	w*_cortex (median)	0.45	0.55	0.55	0.00	0.75	0.70	0.75		Δw* extrastriate − V1 +0.20 (1/8 neg, 0.0391)	
unit halves S, T											
two-stream, diet (magno − parvo)	index	-0.06	-0.44	-0.70	-0.84	-0.88	-0.87	-0.87	—	—	—
	U_shape	0.087	0.045	0.026	0.016	0.014	0.014	0.013			
	U_texture	0.099	0.119	0.145	0.180	0.246	0.190	0.181			
	shared	0.289	0.286	0.261	0.219	0.230	0.184	0.168			
	r^2 (full model, nc-normalised)	0.485	0.457	0.439	0.428	0.496	0.390	0.373			
single ResNet-34, diet	index	+0.05	+0.02	+0.05	-0.03	-0.22	-0.09	-0.06	-0.080 (7/8, 0.0156)	-0.178 (7/8, 0.0156)	-0.25
	U_shape	0.032	0.030	0.028	0.025	0.017	0.016	0.020			
	U_texture	0.028	0.028	0.025	0.027	0.027	0.020	0.022			
	shared	0.263	0.245	0.245	0.274	0.458	0.353	0.320			
	r^2 (full model, nc-normalised)	0.327	0.304	0.300	0.329	0.506	0.392	0.366			
	w*_cortex (median)	0.53	0.60	0.57	0.53	0.53	0.55	0.57		Δw* extrastriate − V1 +0.01 (4/8 neg, 0.9453)	

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
single ResNet-18, no diet	index	+0.17	+0.13	+0.12	+0.17	-0.15	-0.14	-0.14	-0.005 (5/8, 0.9453)	-0.316 (8/8, 0.0078)	-0.15
	U_shape	0.038	0.035	0.030	0.031	0.019	0.016	0.018			
	U_texture	0.026	0.026	0.024	0.022	0.025	0.022	0.024			
	shared	0.272	0.258	0.262	0.288	0.483	0.371	0.335			
	r^2 (full model, nc-normalised)	0.340	0.321	0.319	0.347	0.530	0.415	0.381			
	w*_cortex (median)	0.80	0.80	0.68	0.68	0.42	0.48	0.37		Δw* extrastriate − V1 -0.32 (7/8 neg, 0.0234)	
ImageNet ResNet-50	index	-0.38	-0.38	-0.32	-0.50	-0.44	+0.06	-0.07	-0.120 (5/8, 0.3828)	+0.235 (1/8, 0.0391)	-0.13
	U_shape	0.008	0.008	0.008	0.006	0.006	0.010	0.009			
	U_texture	0.019	0.018	0.017	0.020	0.015	0.009	0.011			
	shared	0.312	0.292	0.297	0.365	0.606	0.489	0.446			
	r^2 (full model, nc-normalised)	0.343	0.322	0.327	0.397	0.633	0.512	0.471			
	w*_cortex (median)	0.10	0.10	0.00	0.05	0.00	0.85	0.70		Δw* extrastriate − V1 +0.07 (3/8 neg, 0.8125)	
ImageNet ViT-B/16	index	-0.19	-0.13	-0.20	-0.23	-0.31	-0.23	-0.27	-0.039 (5/8, 0.6406)	-0.081 (6/8, 0.2500)	-0.35
	U_shape	0.036	0.036	0.032	0.037	0.047	0.042	0.038			
	U_texture	0.053	0.047	0.050	0.062	0.092	0.070	0.069			
	shared	0.162	0.154	0.161	0.189	0.381	0.292	0.258			

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
	r^2 (full model, nc-normalised)	0.252	0.239	0.245	0.290	0.527	0.409	0.366			
	w*_cortex (median)	0.25	0.20	0.10	0.15	0.25	0.15	0.15		Δw* extrastriate − V1 -0.18 (5/8 neg, 0.5469)	
ImageNet-ConvNeXt-B	index	-0.45	-0.40	-0.37	-0.31	-0.30	-0.32	-0.09	+0.140 (2/8, 0.1094)	+0.218 (2/8, 0.0547)	+0.42
	U_shape	0.012	0.013	0.013	0.017	0.022	0.020	0.025			
	U_texture	0.030	0.029	0.029	0.035	0.039	0.035	0.029			
	shared	0.214	0.200	0.203	0.248	0.476	0.357	0.319			
	r^2 (full model, nc-normalised)	0.259	0.246	0.249	0.304	0.540	0.420	0.374			
	w*_cortex (median)	0.00	0.00	0.00	0.00	0.00	0.00	0.00		Δw* extrastriate − V1 +0.10 (0/8 neg, 0.5000)	

Supplementary Table S13 | Error consistency with the ten observers at a matched cue-conflict rate. Every frozen network of Extended Data Table 4 under the same gain (Methods 4.8), swept over w ∈ {0, 0.05, ..., 1}; for each network the row is the **gated** readout whose rate is closest to 0.40 (tolerance ± 0.05; networks whose gate does not reach it are absent, 5 of 38: EfficientNet-V2-S, VGG-11-BN, VGG-16-BN, VGG-16, VGG-19-BN). κ is Cohen's κ on trial-by-trial agreement over the images both parties decided, mean over the ten observers, as a percentage of the human–human ceiling $\kappa = 0.234$ (Methods 4.3; Supplementary Table S6). "κ at the trained readout" is the same quantity at w = 0.5, the number Extended Data Table 4 reports and the form in which the field reports error consistency. "κ / κ_max" normalises by the largest κ attainable at the two marginals, κ_max = $(1 - |\text{p_model} - \text{p_human}| - \text{p_e})/(1 - \text{p_e})$, expressed as a percentage of the same ratio for human pairs (35% of their own bound); values above 100% mean a network attains a larger share of its much smaller attainable agreement than two observers do of theirs, not that it is more human-like than a human. Over the 798 gated readouts of the 38 ImageNet-

trained networks, raw κ is a function of the rate alone (one quadratic in the rate, $R^2 = 0.90$, RMSE 1.7 points of ceiling; per-network offsets −2.2 to 2.8 points, unrelated to the probe bias, $\rho = 0.04$), the bound itself running from 0.006 to 0.219. Normalised by the bound, the rank dependence on the rate disappears (pooled rank R^2 0.001 against 0.891; within-network median Spearman ρ 0.20 against 0.99). Ranking the 33 networks by κ at their trained readouts against by κ here gives Kendall $\tau = 0.33$ with 178 of 528 pairs (34%) reversing. The Stylized-ImageNet ResNet-50 cannot be gained below 0.65 inside its gate; at a common rate of 0.65 it is ResNet-50 SIN 135% [104, 165]; ViT-B/32 120% [64, 174]; AlexNet 79% [61, 96] on the normalised scale (95% percentile bootstrap over the ten observers). Pre-specification: `docs/PLAN_KAPPA_2026-09-17.md` (predictions K-1 to K-6: four failed and a fifth could not be tested as specified; the normalisation by κ_max is post hoc). Source: `results/figures_v2/kappa_rate.json`, aggregated by `tmp_analysis/figs_v2/kappa_rate.py` from the per-network sweeps of `src/analysis/generality/kappa_sweep.py` (`slurm/zoo_kappa.sh`); the per-network files are outside the repository. * not ImageNet-trained.

network	w at the matched rate	rate	κ, % of human–human ceiling	κ at the trained readout	κ / κ_max, % of the human-pair value	probe bias	trained rate
GoogLeNet	0.90	0.397	18.7	9.5	185.9	−0.52	0.251
ShuffleNet-V2 x1.0	0.80	0.401	18.1	9.2	158.7	−0.43	0.234
Inception-v3	0.80	0.406	17.8	11.0	160.2	−0.56	0.305
Swin-S	0.95	0.413	17.5	8.9	172.7	−0.55	0.236
MobileNet-V3-L	0.80	0.407	17.0	10.8	173.6	−0.47	0.265
Swin-B	1.00	0.418	17.0	7.7	169.1	−0.57	0.233
MaxViT-T	0.85	0.401	16.0	9.1	171.2	−0.57	0.295
Swin-T	0.90	0.407	15.4	8.4	163.6	−0.52	0.225
ViT-L/16	0.75	0.408	15.4	11.5	168.1	−0.43	0.319
EfficientNet-B3	0.75	0.396	15.1	12.5	138.4	−0.50	0.291
DenseNet-201	0.85	0.403	14.9	8.5	137.8	−0.49	0.273
MobileNet-V2	0.95	0.389	14.4	5.6	139.4	−0.52	0.184
Wide-ResNet-101-2	0.90	0.402	13.9	7.9	137.8	−0.60	0.279
EfficientNet-B0	0.85	0.407	13.9	7.1	130.2	−0.54	0.271
ResNet-101	0.95	0.405	13.8	8.7	150.6	−0.52	0.284
ResNet-152	0.90	0.402	13.7	9.1	126.5	−0.49	0.293
ResNet-50 SIN+IN, ft IN *	0.95	0.397	13.5	5.9	127.8	−0.58	0.227
ResNet-50 SIN+IN *	0.60	0.399	13.3	11.7	140.6	−0.40	0.363
RegNet-Y-16GF	0.90	0.409	13.0	7.7	131.8	−0.50	0.261
ConvNeXt-B	0.80	0.398	13.0	8.0	132.1	−0.47	0.269

network	w at the matched rate	rate	κ, % of human–human ceiling	κ at the trained readout	κ / κ_max, % of the human-pair value	probe bias	trained rate
ViT-B/32	0.00	0.418	12.9	23.0	130.4	−0.30	0.617
ResNeXt-50 32x4d	0.90	0.383	12.9	5.6	130.6	−0.58	0.219
ResNet-18	0.90	0.407	12.7	6.3	107.3	−0.53	0.233
ResNet-34	0.85	0.403	12.5	6.1	131.0	−0.53	0.259
DenseNet-121	0.90	0.402	12.5	5.7	114.0	−0.55	0.224
MNASNet-1.0	0.85	0.406	12.5	8.7	113.7	−0.53	0.226
ConvNeXt-S	0.85	0.399	12.4	8.1	122.7	−0.53	0.248
ConvNeXt-T	0.85	0.395	12.3	7.0	115.3	−0.50	0.241
ResNet-50	1.00	0.394	12.2	5.1	116.9	−0.56	0.222
AlexNet	0.75	0.393	12.0	7.1	124.8	−0.45	0.261
ConvNeXt-L	0.80	0.401	11.8	9.2	122.4	−0.49	0.293
ViT-B/16	0.50	0.399	11.6	11.6	126.2	−0.43	0.399
RegNet-Y-8GF	0.90	0.407	11.5	6.7	103.1	−0.50	0.244
Wide-ResNet-50-2	0.95	0.405	10.5	5.0	100.5	−0.62	0.230
ResNeXt-101 32x8d	0.80	0.400	10.0	7.9	107.5	−0.53	0.294
ResNet-50 (V2 recipe) *	1.00	0.356	7.8	0.5	70.8	−0.62	0.185

Supplementary Table S14 | What the gain acts on: the span decomposed. Every frozen network of Extended Data Table 4, at its trained readout. For each cue-conflict image the Geirhos rule gives a margin Δ = log p_shape − log p_texture (p_c the mean softmax probability over category c's ImageNet classes; the forced two-way estimator calls the image "shape" iff $\Delta > 0$). The gain of Methods 4.8 adds $(2w - 1)\cdot W(P_S - P_T)y$ to the logits exactly, so to first order it changes Δ by $(2w - 1)(e_S - e_T)$, with e_S and e_T the evidence the head receives through the shape and the texture subspace (u·W P_S y and u·W P_T y, u the gradient of Δ with respect to the logits). "Shape term" is the mean of e_S over the 1,200 images (evidence added at the shape end), "texture term" the mean of −e_T (texture evidence removed there), both in units of the margin; texture share = texture term / (shape term + texture term). M0 is the first-order forced-choice span from the per-image margins, M1 the two-number span $\Phi(z + \lambda) - \Phi(z - \lambda)$ with z = mean Δ / sd Δ and λ = mean(e_S − e_T) / sd Δ. "Reported" is the held-out span of Extended Data Table 4 (primary estimator, on halves that share exemplars); "in sample" is the same estimator and "forced" the forced two-way estimator, both with the bases fitted on all 1,200 images. Random-direction span: ten draws (seeds 0–9) of two mutually orthogonal random 15-dimensional subspaces inside each network's 256-dimensional PCA space of the cue-conflict features, same gain, estimator (in sample) and gate; every random sweep stays inside its gate. Pre-specification: `docs/PLAN_SPAN_2026-09-18.md`, predictions S-1 to S-7, written after Extended Data Table 4 and before this analysis; the whole analysis is post hoc with respect to the manuscript's first three plans. Met: S-1

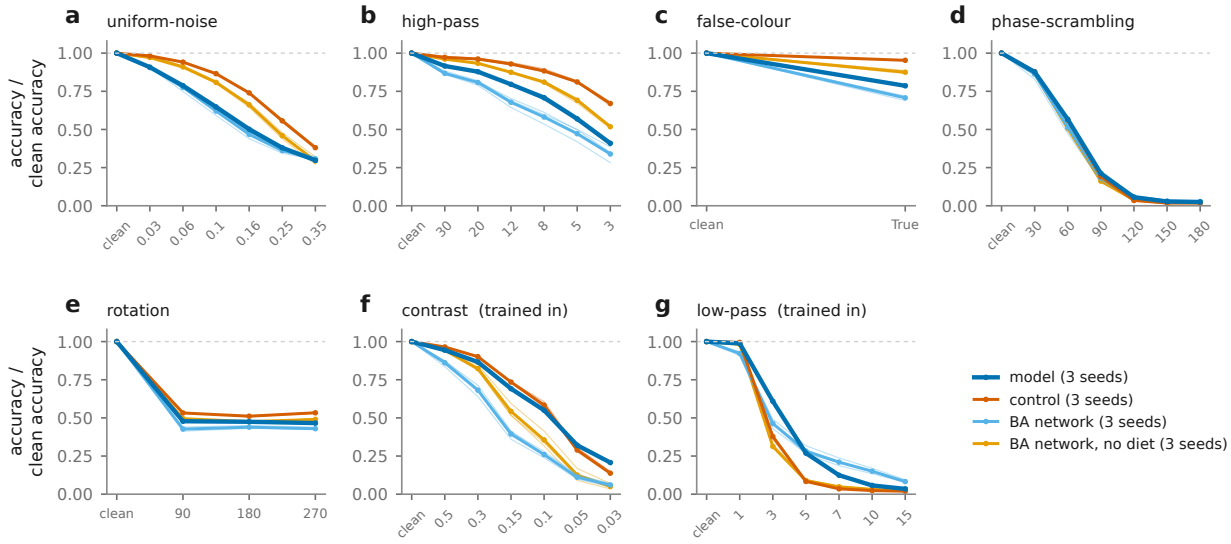
(M0 reproduces the forced-choice curve, median largest error 0.009; ρ(M0, forced span) = 0.99), S-5b (the Stylized-ImageNet texture term 0.31 is below every ImageNet network's, 0.93–3.95) and S-7 (median absolute mean signed random span 0.002; the reported span exceeds the largest random span in 38 of 38 ImageNet networks and 42 of 42). Failed: S-2 (M1 ranks the forced spans at $\rho = 0.95$ but the reported spans at 0.59, against 0.7), S-3 ($\rho(\lambda$, reported span) = 0.56, against 0.7), S-4 ($\rho(\lambda$, z) = −0.32, against $|\rho| \leq 0.3$), S-5a (the Stylized-ImageNet λ 0.53 lies inside the ImageNet range 0.43–0.85) and S-6 (a decision-free head-weighted subspace share ranks the reported spans at $\rho = 0.35$, against 0.5, no better than the unweighted variance share, 0.40). Post hoc: the reported and the forced spans rank the 38 ImageNet networks at $\rho = 0.58$ (in-sample primary 0.81 with the reported and 0.89 with the forced); the forced span rises with the trained forced rate ($\rho = 0.50$ [0.20, 0.73]) where the reported one does not (0.04); among the 39 networks (the 38 and itself) the Stylized-ImageNet span is the smallest reported, the 9th smallest in sample and the 16th smallest forced, and the 3rd smallest with exemplars held out. The texture term exceeds the shape term in 37 of 38 ImageNet networks (texture share median 0.89, range 0.45–0.95; the exception is ViT-B/32), and the texture share tracks the probe bias of Extended Data Table 4 (ρ = −0.81 [−0.91, −0.62]; −0.86 over all 42) more closely than the trained rate does (0.59 [0.29, 0.80]; difference in $|\rho|$ 0.22 [0.04, 0.44]); against the exemplar-balanced probe scored on held-out exemplars (Extended Data Table 4) it is −0.58 [−0.78, −0.29], no closer than the trained rate. Intervals are 95% percentile bootstraps over networks (10,000 resamples), descriptive of this fixed zoo. On the forced estimator the ranking by trained rate and by the shape-end rate agree at Kendall τ = 0.58 with 146 of 703 pairs (21%) reversing (in-sample primary $\tau = 0.50$, 25%; held-out primary 0.47, 27% with exemplars shared and 0.63, 18.6% with exemplars held out). Source: `results/figures_v2/span_model.json`, aggregated by `tmp_analysis/figs_v2/span_model.py` from `src/analysis/generality/span_model.py` (`slurm/zoo_span.sh`); the per-network files are outside the repository.

network	trained rate: primary / forced	span: reported / in sample / forced	forced span, M0 / M1	z / λ	shape term / texture term (texture share)	random-direction span, absolute: median / largest
ImageNet-trained (torchvision IMAGENET1K_V1)						
VGG-11-BN	0.105 / 0.117	0.235 / 0.153 / 0.197	0.206 / 0.300	−1.11 / 0.69	0.17 / 3.24 (0.95)	0.007 / 0.018
VGG-16-BN	0.115 / 0.117	0.226 / 0.157 / 0.177	0.174 / 0.289	−1.07 / 0.64	0.17 / 3.01 (0.95)	0.012 / 0.025
Wide-ResNet-50-2	0.230 / 0.225	0.261 / 0.274 / 0.306	0.312 / 0.348	−0.61 / 0.54	0.16 / 2.71 (0.94)	0.012 / 0.036
VGG-16	0.110 / 0.115	0.268 / 0.216 / 0.253	0.253 / 0.336	−1.04 / 0.72	0.25 / 3.95 (0.94)	0.006 / 0.020
VGG-19-BN	0.150 / 0.149	0.241 / 0.177 / 0.190	0.195 / 0.296	−0.90 / 0.56	0.19 / 2.88 (0.94)	0.009 / 0.023
Wide-ResNet-101-2	0.279 / 0.253	0.231 / 0.271 / 0.287	0.292 / 0.309	−0.50 / 0.45	0.19 / 2.42 (0.93)	0.008 / 0.022
EfficientNet-V2-S	0.184 / 0.183	0.198 / 0.189 / 0.258	0.273 / 0.393	−0.76 / 0.68	0.16 / 1.87 (0.92)	0.006 / 0.016

network	trained rate: primary / forced	span: reported / in sample / forced	forced span, M0 / M1	z / λ	shape term / texture term (texture share)	random-direction span, absolute: median / largest
ResNeXt-50 32x4d	0.219 / 0.219	0.316 / 0.298 / 0.305	0.307 / 0.361	−0.66 / 0.58	0.27 / 3.19 (0.92)	0.014 / 0.026
ResNet-50	0.222 / 0.210	0.290 / 0.254 / 0.264	0.276 / 0.328	−0.74 / 0.55	0.28 / 2.72 (0.91)	0.005 / 0.019
Swin-S	0.236 / 0.238	0.287 / 0.322 / 0.362	0.375 / 0.445	−0.59 / 0.70	0.23 / 2.15 (0.90)	0.011 / 0.022
GoogLeNet	0.251 / 0.247	0.296 / 0.278 / 0.278	0.286 / 0.387	−0.58 / 0.60	0.21 / 1.95 (0.90)	0.014 / 0.033
ResNet-18	0.233 / 0.232	0.316 / 0.326 / 0.320	0.330 / 0.390	−0.67 / 0.64	0.39 / 3.46 (0.90)	0.014 / 0.031
Swin-B	0.233 / 0.240	0.229 / 0.263 / 0.333	0.349 / 0.416	−0.60 / 0.65	0.24 / 2.15 (0.90)	0.012 / 0.021
ResNet-34	0.259 / 0.242	0.353 / 0.344 / 0.351	0.349 / 0.434	−0.66 / 0.71	0.38 / 3.36 (0.90)	0.003 / 0.032
ResNet-101	0.284 / 0.258	0.276 / 0.219 / 0.203	0.215 / 0.292	−0.58 / 0.44	0.26 / 2.21 (0.89)	0.006 / 0.019
DenseNet-121	0.224 / 0.202	0.347 / 0.363 / 0.355	0.349 / 0.432	−0.70 / 0.72	0.45 / 3.78 (0.89)	0.014 / 0.038
MobileNet-V2	0.184 / 0.185	0.346 / 0.330 / 0.306	0.318 / 0.405	−0.82 / 0.73	0.41 / 3.41 (0.89)	0.009 / 0.034
Swin-T	0.225 / 0.224	0.309 / 0.354 / 0.408	0.428 / 0.480	−0.63 / 0.78	0.29 / 2.34 (0.89)	0.011 / 0.017
ConvNeXt-T	0.241 / 0.246	0.353 / 0.379 / 0.393	0.415 / 0.502	−0.65 / 0.83	0.28 / 2.26 (0.89)	0.011 / 0.021
ConvNeXt-S	0.248 / 0.234	0.287 / 0.361 / 0.413	0.424 / 0.497	−0.62 / 0.81	0.28 / 2.23 (0.89)	0.006 / 0.019
RegNet-Y-8GF	0.244 / 0.227	0.365 / 0.324 / 0.337	0.332 / 0.382	−0.64 / 0.61	0.44 / 3.21 (0.88)	0.008 / 0.036
Inception-v3	0.305 / 0.283	0.246 / 0.305 / 0.302	0.300 / 0.360	−0.41 / 0.51	0.24 / 1.68 (0.88)	0.013 / 0.039
ConvNeXt-B	0.269 / 0.261	0.334 / 0.410 / 0.425	0.437 / 0.502	−0.56 / 0.79	0.31 / 2.18 (0.88)	0.009 / 0.019
DenseNet-201	0.273 / 0.228	0.334 / 0.325 / 0.338	0.333 / 0.419	−0.60 / 0.66	0.51 / 3.50 (0.87)	0.010 / 0.025
MNASNet-1.0	0.226 / 0.210	0.395 / 0.391 / 0.349	0.352 / 0.436	−0.73 / 0.75	0.55 / 3.62 (0.87)	0.009 / 0.022
ResNeXt-101 32x8d	0.294 / 0.265	0.283 / 0.303 / 0.322	0.326 / 0.363	−0.48 / 0.53	0.44 / 2.89 (0.87)	0.017 / 0.036
ConvNeXt-L	0.293 / 0.289	0.332 / 0.360 / 0.372	0.388 / 0.476	−0.51 / 0.72	0.35 / 2.19 (0.86)	0.009 / 0.023
MaxViT-T	0.295 / 0.297	0.239 / 0.272 / 0.338	0.353 / 0.406	−0.43 / 0.58	0.33 / 1.88 (0.85)	0.005 / 0.017
ResNet-152	0.293 / 0.255	0.246 / 0.235 / 0.219	0.226 / 0.286	−0.55 / 0.43	0.38 / 2.12 (0.85)	0.013 / 0.030
ShuffleNet-V2 x1.0	0.234 / 0.248	0.381 / 0.414 / 0.350	0.352 / 0.414	−0.62 / 0.66	0.48 / 2.71 (0.85)	0.010 / 0.069
RegNet-Y-16GF	0.261 / 0.242	0.300 / 0.291 / 0.317	0.318 / 0.352	−0.56 / 0.53	0.53 / 2.93 (0.85)	0.021 / 0.076

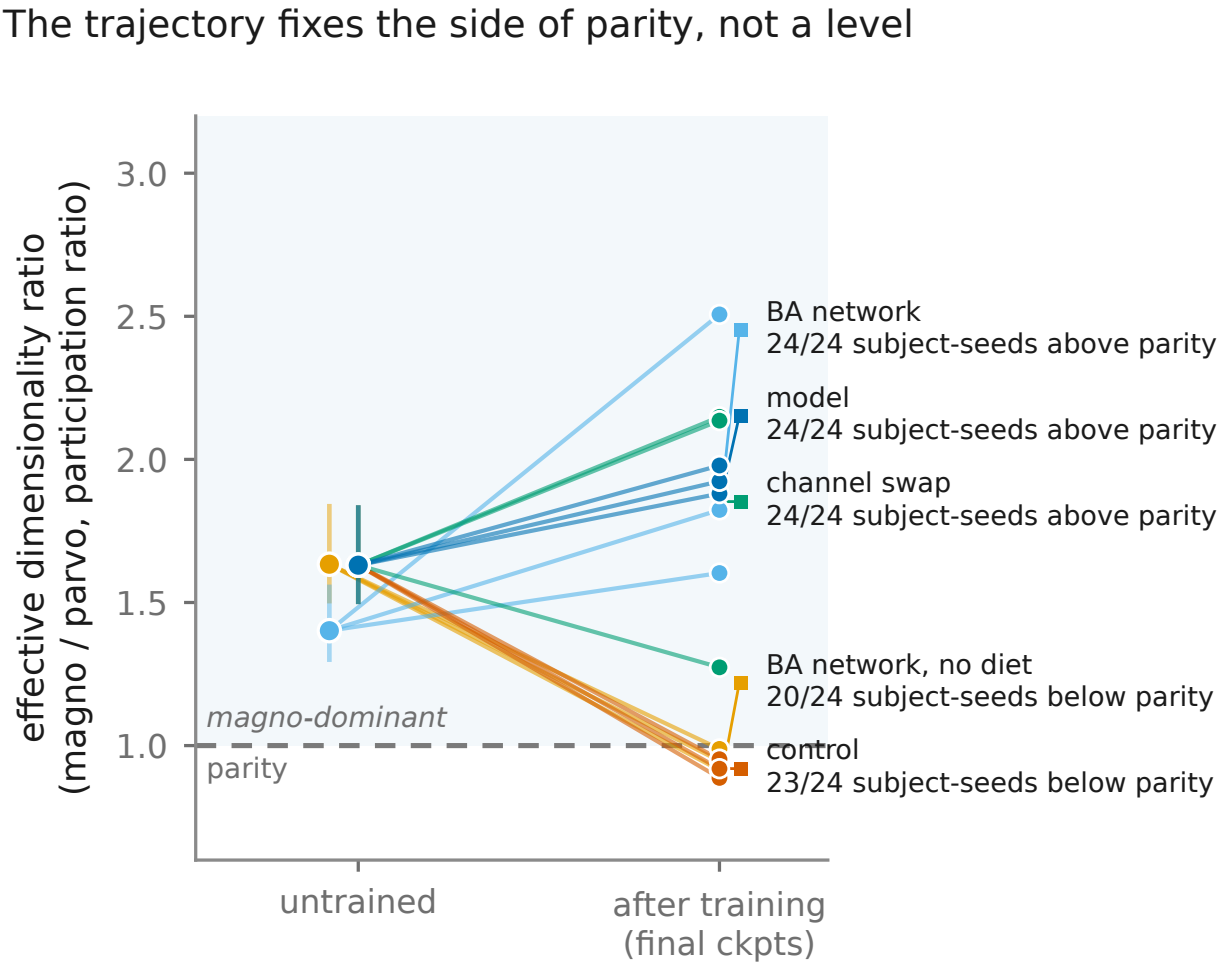
network	trained rate: primary / forced	span: reported / in sample / forced	forced span, M0 / M1	z / λ	shape term / texture term (texture share)	random-direction span, absolute: median / largest
AlexNet	0.261 / 0.244	0.504 / 0.497 / 0.447	0.435 / 0.502	−0.68 / 0.85	0.77 / 3.50 (0.82)	0.010 / 0.020
EfficientNet-B0	0.271 / 0.278	0.327 / 0.361 / 0.361	0.373 / 0.450	−0.49 / 0.67	0.43 / 1.94 (0.82)	0.010 / 0.016
EfficientNet-B3	0.291 / 0.275	0.310 / 0.361 / 0.386	0.395 / 0.465	−0.42 / 0.68	0.46 / 1.88 (0.80)	0.010 / 0.022
MobileNet-V3-L	0.265 / 0.268	0.435 / 0.450 / 0.434	0.437 / 0.508	−0.57 / 0.81	0.92 / 3.50 (0.79)	0.009 / 0.020
ViT-L/16	0.319 / 0.310	0.337 / 0.363 / 0.408	0.421 / 0.481	−0.40 / 0.70	0.59 / 2.03 (0.77)	0.008 / 0.017
ViT-B/16	0.399 / 0.362	0.282 / 0.357 / 0.376	0.376 / 0.422	−0.24 / 0.57	0.61 / 1.48 (0.71)	0.005 / 0.015
ViT-B/32	0.617 / 0.520	0.239 / 0.356 / 0.415	0.418 / 0.453	0.14 / 0.61	1.15 / 0.93 (0.45)	0.008 / 0.023
same architecture, different training recipe						
ResNet-50 (V2 recipe)	0.185 / 0.191	0.334 / 0.234 / 0.288	0.304 / 0.389	−0.72 / 0.66	0.12 / 1.48 (0.92)	0.013 / 0.024
Stylized-ImageNet controls (Geirhos et al., 2018a)						
ResNet-50 SIN+IN, ft IN	0.227 / 0.223	0.277 / 0.265 / 0.273	0.267 / 0.341	−0.66 / 0.55	0.26 / 2.75 (0.91)	0.011 / 0.029
ResNet-50 SIN+IN	0.363 / 0.327	0.350 / 0.358 / 0.370	0.373 / 0.428	−0.37 / 0.60	0.73 / 2.20 (0.75)	0.019 / 0.026
ResNet-50 SIN	0.810 / 0.703	0.166 / 0.255 / 0.318	0.328 / 0.347	0.58 / 0.53	2.43 / 0.31 (0.11)	0.015 / 0.028

Extended Data



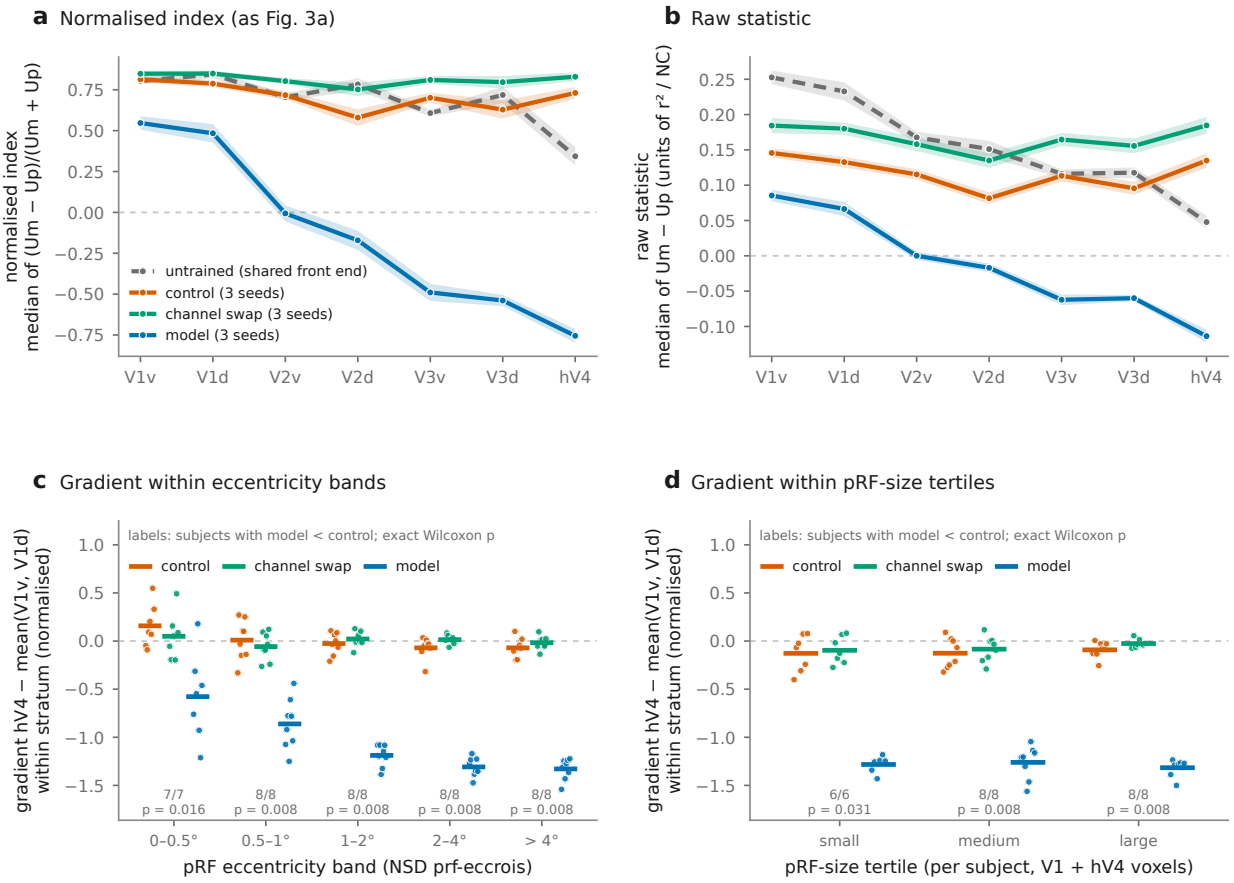
Extended Data Figure 1 | The out-of-distribution degradation curves behind Fig. 5a.** Accuracy relative to clean accuracy against corruption severity for the seven families, for the model and its control (final checkpoints; identical test input) and the BA network's two arms (best saved checkpoints); thin lines: seeds, thick: seed mean. Contrast and low-pass are trained in by the diets. The model is less robust than its control on the four untrained families other than phase scrambling, and more robust only to low-pass.**

Sources: `results/ood/`, `results/ood_bestep/`; `tmp_analysis/figs_v2/figE1_ood_curves.py`.



Extended Data Figure 2 | The trajectory decides which side of parity training ends on, not a level.** Magno/parvo ratio of the participation ratios of the two streams' features on 1,000 NSD images per subject (n = 8). Left: the untrained network (vertical bar: subject range); right: each training seed's mean over subjects, fanned from where it started. The model, its control and the channel swap start from the same untrained network on the same test input (1.63). The model ends above parity in every subject and seed (24/24; seed means 1.88–1.98), the control below it in 23 of 24 (seed means 0.89–0.95). The channel swap, the ramp on L−M, also ends above parity (24/24; seed means 1.27–2.15) with neither stream shape-biased, and for the opposite reason: its ramped chromatic stream contracts (participation ratio 52–86 against 116–124 in the control) while in the model the ramped luminance stream expands (272–281 against 108–110). The BA network's two arms start from their own front ends and end on opposite sides (seed means 1.60–2.51 with the diet, 0.91–0.99 without). No level is labelled, because the within-arm seed range exceeds the between-arm gaps.**

Sources: `results/figures_v2/fs2_pr_ratio.json` (`tmp_analysis/fs2_pr_ratio.py`), `results/figures/fig1_data.npz` (BA arms); `tmp_analysis/figs_v2/figE2_dimensionality.py`.



Extended Data Figure 3 | The cortical gradient against its nuisance axes: the raw statistic, eccentricity and receptive-field size.

(**a, b**) The stream-preference profile of Fig. 3a on both statistics: (a) the normalised per-voxel index (U_m − U_p)/(U_m + U_p), median over reliable voxels per ROI, and (b) the raw per-voxel difference U_m − U_p in units of noise-ceiling-normalised r². Lines are means over 8 subjects, shading ± SEM; the model, the control and the channel swap average three seeds per subject, and the untrained network on their shared front end is a single network. The untrained network already descends on both statistics (gradient −0.480 normalised, −0.195 raw). The model keeps that descent and steepens it on the normalised index (−1.27; −0.19 raw), whereas the control (−0.070; −0.004) and the channel swap (−0.019; +0.002) end flat. The two-statistics rule of Methods 4.7 therefore licenses "preserves", not "steepens".

(**c**) The gradient hV4 − mean(V1v, V1d) on the normalised index, computed within each of NSD's five eccentricity bands (`prf-eccrois`), so that V1 and hV4 voxels are compared only with voxels of matched eccentricity. One point per subject per band (seed mean), bars = mean. A cell needs at least 20 voxels per ROI or the subject is dropped from that band; in the foveal band this drops subject 02 in every arm. - The model's gradient is negative in every band: 6/7 subjects in the foveal band, where subject 05 is positive, and 8/8 in the other four. - It grows with eccentricity, from −0.58 in the foveal band to −1.33 beyond 4°. - The control stays between −0.07 and +0.16 in every band, and the channel swap between −0.06 and +0.05.

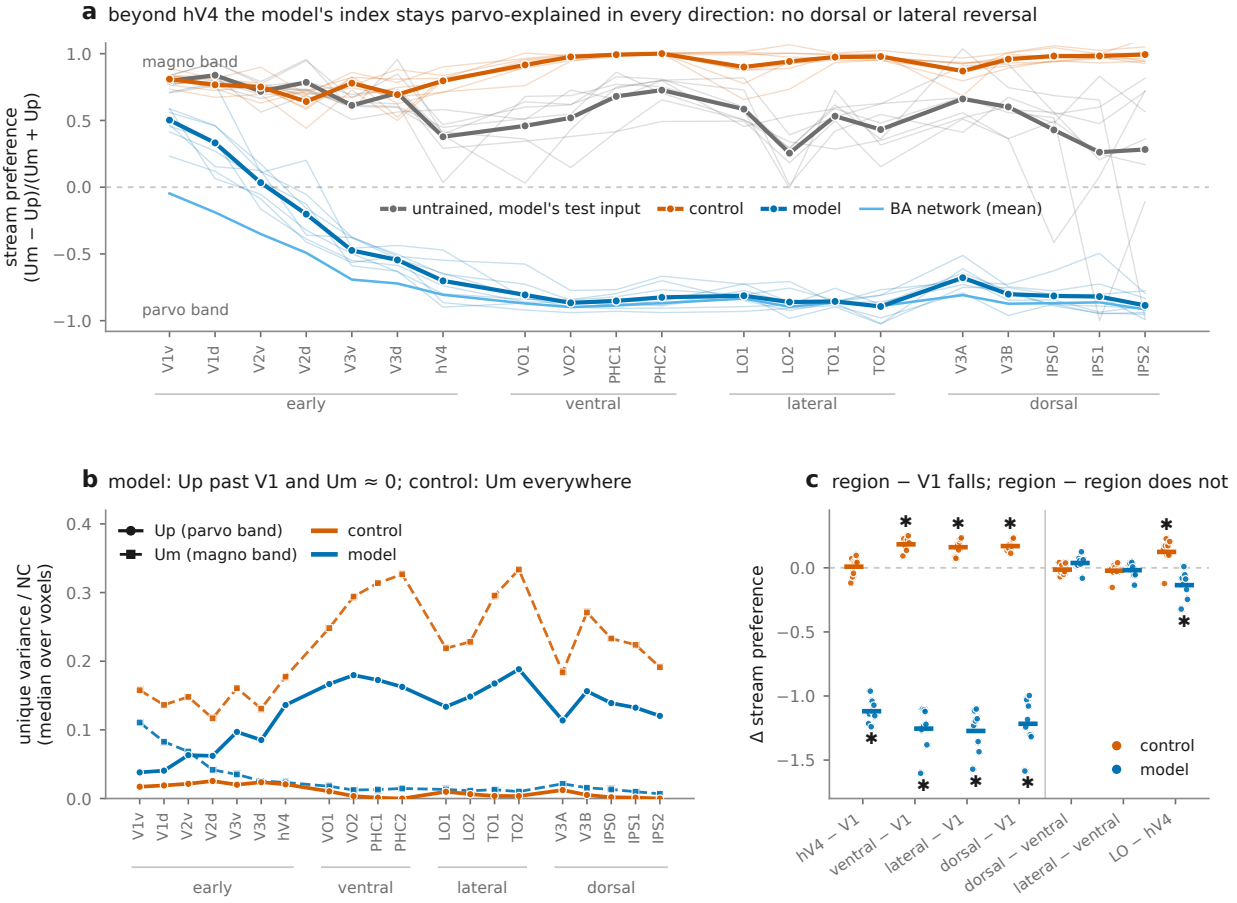
Labels give the number of subjects in which the model's gradient is below the control's, and the exact Wilcoxon p for that band (7/7, p = .016, in the foveal band; 8/8, p = .0078, in the other four). Eccentricity modulates the model's gradient but does not produce it.

(**d**) The same gradient within pRF-size tertiles. Tertile boundaries are set per subject on the pooled V1 + hV4 voxels, using NSD's pRF size estimates (voxels with pRF $R^2 > 5\%$). - The model's gradient is negative in every tertile (−1.28, −1.26, −1.32; every subject). - The control (−0.13, −0.13, −0.09) and the channel swap (−0.10, −0.08, −0.03) stay near zero. - In the small tertile, subjects 03 and 08 have too few voxels in the model, so its comparison with the control rests on 6 subjects (6/6, p = .031); the other two tertiles give 8/8, p = .0078.

The hierarchical contrast is therefore not a receptive-field-size contrast in disguise. The within-subject re-

gressions on the continuous predictors (Supplementary Note S3) give the same picture.

Source: `results/figures_v3/cortex_controls.json` (`tmp_analysis/fs2_cortex_controls.py`). The BA network's version of this figure is PAPER_v2 Extended Data Fig. 3.



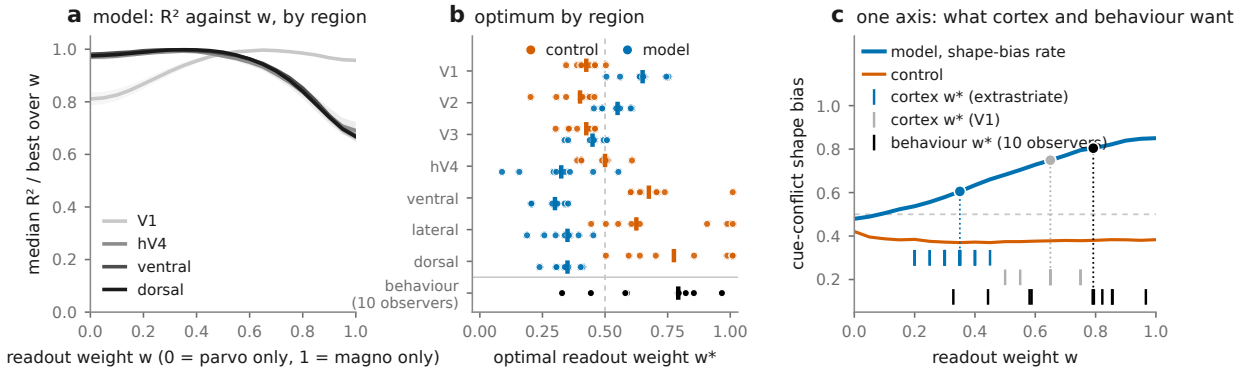
Extended Data Figure 4 | The gradient continues past hV4 and does not turn.** The stored banded-ridge fits of Fig. 3, relabelled with NSD's Kastner2015 atlas (the 22 ROIs that pass the ≥ 20-voxel rule), grouped as early (V1–hV4), ventral (VO1/2, PHC1/2), lateral (LO1/2, TO1/2) and dorsal (V3A/B, IPS0–2).**

(**a**) Stream preference (U_m − U_p)/(U_m + U_p) per ROI: the untrained network on the model's test input, the control and the model (thick: mean over subjects of each subject's seed mean; thin: subjects), and the BA network's mean.

(**b**) The two components, noise-ceiling-normalised unique variance of the parvo band (U_p, solid) and the magno band (U_m, dashed), median over reliable voxels, for the model and its control. In the model U_p carries every region past V1 and U_m is near zero; in the control U_m carries every region and U_p is near zero.

(**c**) Regional contrasts per subject: each region's mean index minus the V1 mean, and region minus region. Relative to V1 the model's hV4 sits at −1.12 and every extrastriate group lower — ventral −1.26, lateral −1.27, dorsal −1.22 — each in 8/8 subjects (exact Wilcoxon p = .0078); dorsal − ventral and lateral − ventral do not depart from zero. ∗: exact Wilcoxon $p < .05$, n = 8.

Sources: `results/figures_v3/ext_gradient.json` (`tmp_analysis/figs_v2/ext_gradient.py --set v3`); `tmp_analysis/figs_v2/figE4_ext_gradient.py`.



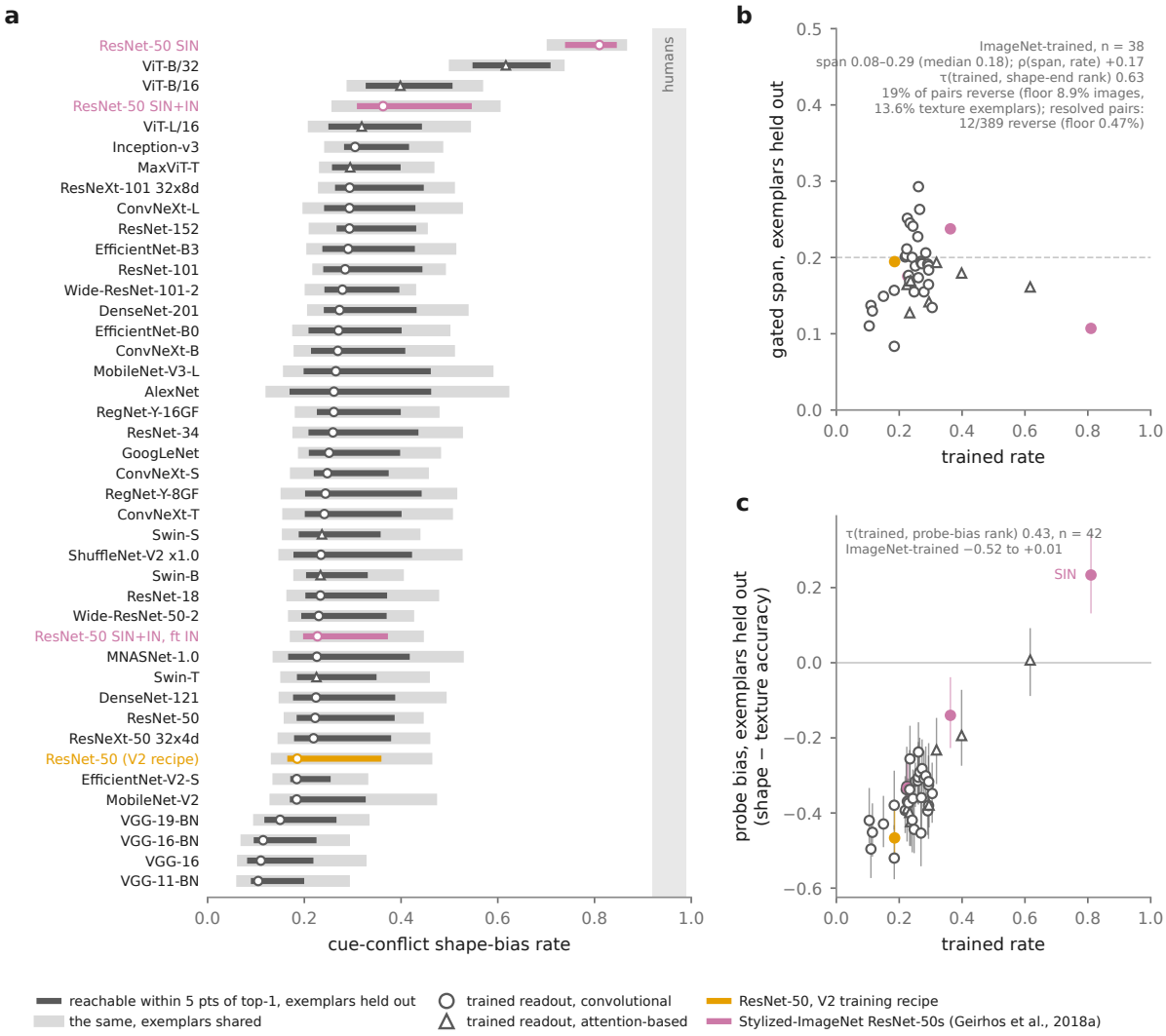
Extended Data Figure 5 | The readout weight cortex prefers descends the hierarchy.** For each mixing weight w the classifier's own input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ is the design of a ridge encoding model of every nsdgeneral voxel (Methods 4.5); a region's w* is the w that maximises its median test r^2 over reliable voxels (mean over seeds per subject).**

(**a**) The model's median r^2 against w, normalised to each region's best, for V1, hV4 and the ventral and dorsal groups (mean ± s.e.m. over subjects). Every ROI is within 2% of its best r^2 at $w = 0.5$: the panel shows an ordering of optima, not an r^2 difference.

(**b**) w* per region and subject for the model and its control (bar: median). In the model w* falls from 0.65 in V1 to 0.30–0.35 in every extrastriate group, below V1 in 8/8 subjects ($\Delta = -0.323$ [−0.371, −0.269], exact $p = .0078$); in the control the gradient reverses (0.43 in V1 to 0.62–0.78 beyond hV4; higher in 8/8 subjects, $\Delta = +0.308$ [+0.194, +0.425], $p = .0078$; three seeds). Bottom row: the weight a trial-by-trial fit to each of the ten observers prefers (median 0.79; not identified by these data, see Results 2.4).

(**c**) The model's shape-bias rate against w (the control's beneath it), with each region's and each observer's w* marked: at the extrastriate weight the model's rate would be 0.60, at V1's 0.75.

Sources: `results/figures_v3/readout_w_cortex.json` and Extended Data Table 2 in `results/figures_v3/ed_tables.md` (`tmp_analysis/figs_v2/readout_w_cortex.py --set v3`, from `/scratch/byrong/mpk/readout_w/results/2026-10-16_fs2_w21`).



Extended Data Figure 6 | The readout account across the field's model zoo, on exemplars the subspaces were not fitted on.

Forty-two frozen ImageNet networks under the rule of Fig. 6a–b: y is the input of each network's last linear layer, the 16-class shape and texture linear discriminants of y (PCA to 256 dimensions) give orthogonal shape- and texture-predictive subspaces, and logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b with the head untouched; the rate is scored by the Geirhos 16-class rule. Because the 1,200 cue-conflict images reuse 160 object and 48 texture exemplars, the drawn estimator splits both into three folds, fits the subspaces on the images whose object and texture exemplars both lie outside one fold and scores the rate on the images whose exemplars both lie inside it (ten repetitions, mean curve; Methods 4.8). The pale bars are the pre-registered estimator of `docs/PLAN_ZOO_2026-09-15.md`, fitted on one stratified half of the images and scored on the other, which shares 98% of its object exemplars and all of its texture exemplars with the fitting half. Thirty-eight networks are torchvision `IMAGENET1K_V1` classifiers (grey), one is the same ResNet-50 under torchvision's 2021 training recipe (orange), and three are the ResNet-50s of Geirhos et al. (2018a) trained on Stylized-ImageNet (SIN), on SIN + ImageNet, and on SIN + ImageNet then fine-tuned on ImageNet (purple).

(**a**) Networks sorted by their trained rate (symbol; circles convolutional, triangles attention-based), which is scored on all 1,200 images; the held-out curves are scored on the images of the held-out folds only and pass within 0.02 of it at w = 0.5. Bars: the rates reachable within 5 points of the trained top-1 on a stratified 10-per-class ILSVRC-2012 validation subset (10,000 images), with exemplars held out (dark) and shared (pale). The gate is measured on the same held-out fold bases as the curves (mean top-1 over 30 bases; Methods 4.8): the largest top-1 change is a median 1.4 points (at most 6.3), and the gate narrows in three networks (ResNet-18 to w = 0.10–0.95, MobileNet-V2 and SIN to 0.05–0.95); on bases fitted on all images the whole

sweep of every network lies inside it (at most 4.4 points). With exemplars held out the reachable span of the 38 ImageNet-trained networks is 0.08–0.29 (median 0.18; 11 of the 38 reach the pre-registered 0.20), against 0.20–0.50 (median 0.30) with exemplars shared, and the held-out mean curve rises monotonically with w in all 42 networks ($\rho \geq 0.99$). Grey band, the ten human observers (0.92–0.99). The SIN-trained network has the highest trained rate of all 42 (0.81) and a held-out span of 0.11 ($0.74 \rightarrow 0.85$), the 2nd smallest of 39 (0.12 and the 3rd smallest on the all-image gate); with exemplars shared its span is 0.17 ($0.70 \rightarrow 0.87$), and on the forced two-way estimator 0.32, the 16th smallest (Extended Data Table 4 and Supplementary Table S14).

(b) Held-out gated span against trained rate over the 38 ImageNet-trained networks: the span is unrelated to the rate (Spearman $\rho = +0.17$; +0.10 on the all-image gate, +0.04 with exemplars shared) and to top-1 (-0.19), and averages 0.19 in convolutional and 0.16 in attention-based networks. Ranking by the trained rate and by the rate at the shape end of each network's gate agree at Kendall $\tau = 0.63$; 131 of the 703 pairs (19%) reverse order, against 8.9% [6.0, 12.2] for the trained ranking re-estimated on two independent resamples of the images, 9.1% [6.0, 12.7] resampling object exemplars and 13.6% [8.3, 20.9] resampling texture exemplars; among the pairs the trained rate resolves, 39 of 490 reverse (8.0%, floor 0.33%) under image resampling and 12 of 389 (3.1%, floor 0.47%) under texture-exemplar resampling ($\tau = 0.47$ and 27% with exemplars shared; `results/figures_v2/exemplar_floors.json`). The dashed line is the pre-registered 0.20.

(c) The exemplar-balanced probe bias — shape minus texture accuracy of a linear probe on the same features with three exemplars per class for both labels, one exemplar left out per fold and PCA and scaling fitted inside each fold; mean of 20 draws, bars the 5–95% range over draws — against the trained rate. Thirty-seven of the 38 ImageNet-trained networks are texture-leaning (-0.52 to +0.01, the exception ViT-B/32 at +0.01 [-0.09, +0.09]), and the two rankings agree only at $\tau = 0.43$. The SIN network is the only shape-leaning network of the 42 (+0.23 [+0.13, +0.34]): its bias is in the representation, a distinction the rate alone cannot make, and the gain moves it mainly by adding shape evidence where it moves 37 of the 38 ImageNet-trained networks mainly by removing texture evidence (Supplementary Table S14). The paper's original stratified-fold probe, whose test images share exemplars with its training folds, gives -0.30 to -0.66 for every ImageNet-trained network, -0.11 for SIN and $\tau = 0.44$ (Extended Data Table 4). Error consistency with the observers (Extended Data Table 4, not drawn) is 2–23% of the human–human ceiling at the trained readouts, rises with the rate across networks ($\rho = 0.80$) and within every gate (38 of 38), and reaches at most 37% at any gated readout of an ImageNet-trained network; the SIN network reaches 40% trained and 53% at its shape end.

Pre-specification and outcomes: `docs/PLAN_ZOO_2026-09-15.md` (P-Z1, P-Z2 and P-Z4 met; P-Z3 met in parts a and c, not in part b, the SIN span falling short of 0.20) and `docs/PLAN_EXEMPLAR_2026-09-21.md` (the held-out estimators; P-X1–P-X10, of which P-X7 and P-X8 failed). Per-network values: Extended Data Table 4 and `results/figures_v2/exemplar_zoo.json`. Preprocessing Resize 256 / CenterCrop 224 except Inception-v3 (342/299), EfficientNet-B3 (320/300) and EfficientNet-V2-S (384/384).